



**GREEN
CLIMATE
FUND**

Meeting of the Board
26 – 28 February 2019
Songdo, Incheon, Republic of Korea
Provisional agenda item 34

GCF/B.22/10/Add.04

1 February 2019

Consideration of funding proposals - Addendum IV

Funding proposal package for FP102

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled “Mali solar rural electrification project”;
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat’s assessment;
- e) Independent Technical Advisory Panel’s assessment;
- f) Response of the accredited entity to the independent Technical Advisory Panel’s assessment; and
- g) Gender documentation.

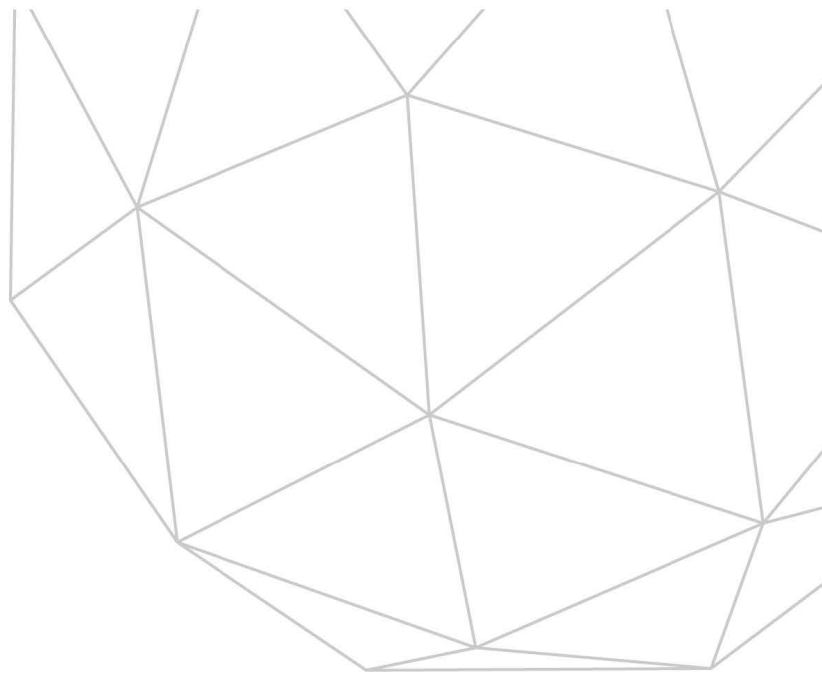
The funding proposal package for FP102 is being submitted for the first time for the Board's consideration at its twenty-second meeting.

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Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: **Mali solar rural electrification project**

Country/Region: **Mali (West Africa)**

Accredited Entity: **Banque Ouest Africaine de Développement (BOAD)**

Date of Submission: **24th June, 2018. Update : 9th, January 2019**

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Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“[FP]-[Agency Short Name]-[Date]-[Serial Number]”

ACRONYMS

() Some institutions' names and acronyms are kept in French*

AER-Mali	: Agence des Energies Renouvelables – Mali
AFDB	: African Development Bank
AEDD	: Agence pour l'Environnement et le Développement Durable
AMADER:	: Agence Malienne pour le Développement de l'Energie Domestique et de l'Electrification Rural
API-Mali	: Investment Promotion Council of Mali
AML/CFT	: Anti-money laundering - combating finance of terrorism
APR	: Annual Performance Report
BCEAO	: Central Bank of West African States
BE	: Baseline Emissions
BOAD	: Banque Ouest Africaine de Développement
BOT	: Build Operate Transfert
CoP	: Conférence of Parties
CREE	: Commission de Régulation de l'Électricité et de l'Eau
ER	: Emission Reduction
ESMP	: Environmental and Social Measures project
EDF	: Energy Development Fund
ESMF	: Environmental and social management Framework
DNE	: Direction Nationale de l'Energie
EDM-SA	: Energie du Mali- Société Anonyme
DREP	: Decade Rural Electrification Program
ESIA	: Environmental and Social Impact Assessment
FER	: Fonds d'Electrification Rurale or Rural Electrification Fund
FSR	: Feasibility studies report
GCF	: Green Climate Fund
GDP	: Gross Domestic Product
HDI	: Human Development Index
IRENA	: International Renewable Energy Agency
IPP	: Independent Power Producers
ISPS	: Isolated Solar Photovoltaic System
IRF	: Investment Renewal Fund
IFA	: International Financial Action
IRENA	: International Renewable Energy Agency
LFI	: Local financial institutions
MEE	: Ministère de l'Energie et de l'Eau
MPPT	: Maximum Power Point Tracking
MFI	: Microfinance Institutions
MEF	: Ministry of Economy and Finance
MSME	: Micro Small Medium Enterprises
M&E	: Monitoring and Evaluation
NDCs	: Nationally Determined Contributions (NDCs)
NGO	: Non-Governmental Organization
NEP	: National Energy Policy
NCSERE	: National Center for Solar Energy and Renewable Energies
NDSPC	: National Directorate for Sanitation and Pollution Control
O&M	: Operation and Maintenance
OECD	: Organization for Economic Cooperation and Development
OECD DAC	: Organization for Economic Co-operation, and Development Assistance Committee
PE	: Project Emissions
PANA	: National Adaptation Program of Action
PV	: Photovoltaïque
PMU	: Project Management Unit
PPPs	: Public Private Partnership
RE	: Renewable Energy
SDGs	: Sustainable Development Goals
TOR	: Terms of Reference
UNDP	: United Nations Development Program
UNFCCC	: United Nation Framework Convention on Climate Change
WEEE	: waste management for electrical and electronic equipment's
WAEMU	: West African economic and monetary union

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Mali solar rural electrification project
A.1.2. Project or programme		Project
A.1.3. Country (ies) / region		Mali
A.1.4. National designated authority (ies)		Agence pour l'Environnement et le Développement Durable (THE ENVIRONMENT AND SUSTAINABLE DEVELOPMENT AGENCY) (AEDD)
A.1.5. Accredited entity		Banque Ouest Africaine de Développement (BOAD)
A.1.5.a. Access modality		☒ Direct ☐ International
A.1.6. Executing entity / beneficiary		Executing Entity: Ministry of Energy and Water. Beneficiaries: a. 31 000 households (310,698 people) across 70 communities in 6 Regions (Kayes, Koulikoro, Sikasso, Segou, Mopti and Tombouctou) b. National institutions and policy makers (AER, AMADER and FER) responsible to create the enabling environment for RE investments
A.1.7. Project size category (Total investment, million USD)		☐ Micro (≤ 10) ☒ Medium ($50 < x \leq 250$) ☐ Small ($10 < x \leq 50$) ☐ Large (> 250)
A.1.8. Mitigation / adaptation focus		☒ Mitigation ☐ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		1 st submission: 24th June 2018
A.1.10. Project contact details	Contact person, position	- Yacoubou BIO-SAWÉ, Director for Environment and Climate Finance - Ibrahim TRAORE, Head of Climate Finance Division
	Organization	Banque Ouest Africaine de Développement (BOAD)
	Email address	itraore@boad.org ; ybiosawe@boad.org ; (cc eamoussou@boad.org)
	Telephone number	+228 22 23 26 92
	Mailing address	Avenue de la Libération, Lomé (Togo), BP. 1172

A.1.11. Results areas (mark all that apply)	
<u>Reduced emissions from:</u>	
☒	Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.)
☐	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☐	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
<u>Increased resilience of:</u>	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☐	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)
☐	Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.)
☐	Ecosystem and ecosystem services (E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Project / Programme Executive Summary

The main objective of this project is to promote rural electrification through isolated solar photovoltaic systems as a low-carbon and resilient solution to the effects of climate change in the energy sector in Mali.

The project will: (i) strengthen the capacity of public institutions (AMADER, AER and FER) engaged in rural electrification to accelerate solar-powered electrification; and (ii) increase the rural population's access to electricity in 70 communities through installation of isolated solar photovoltaic minigrid systems for a total capacity of 4.83 MW. The project will thus catalyze the development of an efficient solar market that will enable Mali to meet its renewable energy and greenhouse gas (GHG) reduction targets. In this case, it will help avoid 41 049 tCO₂ / year, and 1 027 227 tCO₂ for the entire life of the equipment (25 years). Direct beneficiaries are estimated at 310 698 inhabitants, 50% of whom are women in the project area. Around 31,000 new households will have access to electrification in rural areas (28,378 for Tranche 1). The project is tranché at the request of GCF Secretariat. First Tranche will achieve 80% of the impact (821 782 tCO₂ eq) and will target 50 localities (283 777 inhabitants).

The project has significant potential for paradigm shift in the rural electrification sector in Mali. Traditionally, the growing energy access demand has been met through fossil fuels. This project however provides a total shift in energy production i.e. replacing fossil fuels additions with renewable energy solutions. The project will positively contribute to the regulatory framework through capacity buildings, demonstration of technologies in areas where not available, putting in place an institutional framework for FER to strengthen its capacity to further promote the deployment of renewable energy.

The paradigm shifting potential lies in the joint efforts to create a conducive environment by putting in place an enabling regulatory framework and policies, that will incentivize the private sector participation in long term concessions under a sustainable and innovative structural arrangement. It is anticipated that the project model can be replicated across the whole country and in the wider West African subregion, notably to support different African countries to meet their commitment under the Paris Agreement i.e. Nationally Determined Contributions (NDCs), and to achieve their 2063 Agenda as well as Sustainable Development Goals (SDGs).

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	Q1 March, 2019
Expected financial close (if applicable)	Q1 March, 2019
Estimated implementation start and end date	Start: <u>01/07/2019</u> End: <u>30/06/2023</u>
Project/programme lifespan	25 years

B.1. Description of Financial Elements of the Project / Programme

Financial Model

A detailed financial model is attached in the annexes of this funding proposal. The financial model includes the breakdown of cost estimates for total project costs and GCF financing by sub-component. The project will be financed in EUR. As aforementioned, the project is tranced, and the first Tranche will target 50 communities, then BOAD will submit a request for Tranche 2 funding.

Table 1: Breakdown of cost estimates for total project (70 communities)

Component	Indicative cost	GCF funding	Co-financing	
	EUR	EUR	Amount	Institution
			EUR	
1. Capacity building of rural electrification institutions and Technical Assistance	487 349	487 349		
2. Detailed Engineering and Installation of solar powered minigrids	43 890 497	33 354 043	10 536 454	BOAD
3. Support to Productive Use of Electricity	1 500 000	1 500 000		
Project Management Cost	404 269	307	97 694	BOAD
Indicative total cost	46 282 115	35 341 699	10 634 148	

The total amount of the project is EUR 46,282,115, including PMC.

Table 2: Breakdown of cost estimates for Tranche 1 (50 communities)

Component	Tranche 1					Tranche 2		
	Indicative cost	GCF funding		BOAD Co-financing		Indicative cost	GCF funding	Co-financing (Loan)
	EUR	EUR	Financial instrument	EUR	Financial instrument	EUR	EUR	EUR
1. Capacity building of rural electrification institutions and Technical Assistance	487 349	487 349	Grant		-	-	-	
2. Detailed Engineering and Installation of solar powered minigrids	853 214	853 214	Grant		-	341 285	341 285	
	31 007 376	22 821 475	Loan	8 185 901	Loan	11 688 622	9 338 069	2 350 553
3. Support to Productive Use of Electricity	1 500 000	1 500 000	Loan		-	-	-	
Project Management Cost	404 269	306 575	Grant	97 694	Loan	-	-	-
Indicative total cost	34 252 208	25 968 612		8 283 595		12 029 908	9 679 355	2 350 553

For Tranche 1, the amount is EUR 34,252,208, including PMC, and the expected GCF funding is EUR 25,968,612. BOAD will provide a co-financing in loan for EUR 8,283 595.

Tranche 2 will cover the remainder of the total project funding and will be submitted after the mid-term evaluation.

The private sector will provide additional EUR 8.5 M at the whole project level and EUR 6.2 M for Tranche 1 as parallel financing, which corresponds to 20% of the investment for the installation of the mini-grids.

B.2. Project Financing Information

	Financial Instrument	Amount	Currency	Tenor	Pricing		
(a) Total project financing	(a) = (b) + (c)	34.25	<u>million euro</u> (€)				
(b) GCF financing to recipient	(i) Senior Loans	24.32	<u>million euro</u> (€)	40 years	0,75 %		
	(ii) Grants	1.65	<u>million euro</u> (€)	N/A	N/A		
	* Please provide economic and financial justification in section F.1 for the concessionality that GCF is expected to provide, particularly in the case of grants. Please specify difference in tenor and price between GCF financing and that of accredited entities. Please note that the level of concessionality should correspond to the level of the project/programme's expected performance against the investment criteria indicated in section E .						
	Total requested (i+ii)	25,97	<u>million euro</u> (€)				
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
	<u>Senior Loans</u>	8.3	<u>million euro</u> (€)	BOAD	20 years	5.97 %	<u>senior</u>
	Lead financing institution: BOAD (co-financing) in the amount of EUR 8.3 M						
	The private sector will provide additional EUR 8.5 M at the whole project level and EUR 6.2 M for Tranche 1 as parallel financing, which corresponds to 20% of the investment for the installation of the minigrids.						
Find in annex letter of commitment issued by BOAD.							

Financial instrument 1: Senior Loan: The loan will be used for installation of minigrid systems and needs a lower interest rate, for concessional loans are needed to finance projects as this one in a least developed country such as Mali, where national debt ratio is over 98% of GDP since 2016. It is important to notice that with a lower concessionality, the indebtedness level of the country will increase and IMF controls will be triggered, resulting in the impossibility for Mali to take the loan. Therefore, the project would fall apart.

Regarding BOAD's loan, it is made in local currency, which limits foreign exchange risks and favors a sustained debt. The interest rate is indicative and will be negotiated with the government at the time of project appraisal. Actually, BOAD is the common development financial institution for the 8 west African economic and monetary union (WAEMU) countries, and though it mobilizes resources at a market rate, its financing conditions are quite the lower one in the subregion's financial market. Every year, BOAD makes available to each country an amount of € 16 million of concessional financing at rates of up to 3%. The government decides in a sovereign way which project will benefit from those resources. So, when BOAD will appraise this project for consideration to its Board, negotiations with the government will help to agree on the appropriate rate to be applied.

The loan is directly given to the Malian Government. The beneficiaries will benefit from the concessionality of GCF through electricity tariffs that will be capped at 0.26 EUR/kWh (174 XOF/kWh) including an electricity price of 0.24 EUR/kWh, which is within communities' ability to pay, as informed by the feasibility study. According to the baseline scenario, the average tariff generally applied in rural areas for minigrid operated by private sector is 0.35 EUR/kWh (235 XOF/kWh) up to 0.53 EUR/kWh (350 XOF/kWh) whilst in urban areas and for grid connected systems, a subsidized tariff of 0.26 EUR/kWh (174 XOF/kWh) is applied by the government.

The Mali Government is the borrower and the owner of the assets. Ownership will remain with the public sector while concession agreements will be signed with private sector for Operation and Maintenance (O&M). The treasury of the Government of Mali will be the source of repayment of the loan. The loan will not be repaid from the cash flows to be generated by the project.

Financing instrument 2: grant

Mali is a highly indebted country with a high poverty rate. The GCF grant is specially needed for capacity building, Technical assistance and sensitization activities under component 1, with the aim to incentivize the private sector participation in long term concessions under a sustainable and innovative structural arrangement in clean energy sector, and to support the replication of the project's model accross the whole country.

B.3. Financial Markets Overview (if applicable)

How market price or expected commercial rate return was (non-concessional) determined?

Commercial banks throughout West Africa including Mali would offer too high rates (12 – 15% is typical in this region) and they usually require shorter terms of maturity (5 – 7 years). Mini-grid operators are new to the Region and commercial banks consider it too high risk to lend money to them. The prevailing high interest rate offered by commercial banks are one of the greatest barriers prohibiting private sector participant from investing in rural electrification projects because they cannot offer cost-reflective tariffs while as the same time having to bear high capital costs. The consequence is that the private sector operators would always resume to the need of subsidies, otherwise they will not be able to pay back the loan.

We have never observed a single case where a commercial debt transaction took place in our subregion to implement a mini-grid system project. All of the observed debt to mini-grid projects have been made possible based on concessional terms through peculiar grant for projects that are considered innovative enough and energy efficient. The cost of technology has not reach yet the lows where commercial debt alone would suffice to do mini-grid projects.

Blending concessional resources with commercial debt is detrimental a first stage to building the foundation of the market that would attract local and regional financial institutions. In this Mali project, it is anticipated that a positive shift in paradigm will commence through:

- a) Meeting the rapidly growing electricity access demand by renewable energy instead of continuous addition of fossil fuels generating units;
- b) Establishing a conducive regulation framework in order to attract private sector participants to come into the rural electrification business; and
- c) Establishing a sustainable rural electricity tariff model that will make it financially viable for private minigrid operators to self regenerate, renew and expand their operation, so as to build a further momentum to shift the paradigm, which shall set up an example that can be easily and quickly be duplicated throughout the West Africa region and beyond.

Please fill out applicable sub-sections and provide additional information if necessary, as these requirements may vary depending on the nature of the project / programme.

C.1. Strategic Context

This is a greenfield rural electrification project with the aim to meet the growing demand for power from the non-electrified communities. Greening the current energy provision would require extensive measures and gigantic investments that are not available to the country. However, it is anticipated that the successful implementation of this project will spur and foster further diffusion of renewable energy. It has been observed that in the past few years there has been continuous growth in fossil fuel based generating capacity. Therefore, it is of critical importance to meet future demand through renewable energy instead of continuously adding fossil fuel based power generation units.

Geography and climate

The Republic of Mali is a landlocked country in the Sahel belt of West Africa located between 11 ° and 25 ° North latitude. It covers an area of 1,246,814 km² of which 51% consists of desert land and 4% of cultivated land. This geographical situation, combined with the fact that Mali does not have fossil fuels on its territory, forces the Government to import all the fossil fuels it needs and makes it highly dependent on price volatility.

The country has an intertropical climatic regime of Sudano-Sahelian character, with high temperatures (average between 26 and 30 ° C), influenced by its continentality and is dominated by an alternation of dry season whose duration varies between 6 to 9 months of the south to north and a rainy season from 6 to 3 months.

Rainfall is characterized by high inter-annual variation and poor spatial distribution. There is a steady decrease in rainfall from the South (1000 mm / yr) to the North (<200 mm / yr) and the duration of the rainy season (MEA, 2009a). While historical records show a rise in average temperatures of about 0.7 ° C since 1960, projections project a continuation and acceleration of this rise in average temperatures as well as a decrease in overall rainfall amounts (with extreme events - floods - more frequent).

Socio-economic context

The population of Mali is estimated at about 18,957,258 inhabitants in 2017. The population growth rate is estimated at 3.4% per year with 62.5% of the population residing in rural areas. This population growth is a major concern for the future of the country, as it leads to an increase in needs to be met, which will inevitably lead to additional pressure on electricity production. The average population density is 15.2 inhabitants per square kilometer.

The administrative division of Mali comprises ten (10) administrative regions and one urban district (Bamako). These are the regions of Kayes, Koulikoro, Sikasso, Segou, Mopti, Gao, Timbuktu, Kidal, Taoudenit and Menaka. The regions are subdivided into 49 circles and 703 communes (including 684 rural) with more than 12,000 villages and fractions.

Gross Domestic Product (GDP) per capita was US \$ 781 in 2016. Dominated by agriculture and the mining sector, the Malian economy recorded a GDP growth rate of 5.3% in 2017. The economic outlook remains positive with projected real GDP growth rates of 5% for 2018 and 4.9% for 2019.

Energy situation and state of play of the subsectors of electricity and renewable energies

Energy balance of Mali

Mali's energy situation is characterized by a biomass-dominated balance sheet that accounts for 76% of primary energy, followed by 20% for hydrocarbon imports and 4% for electricity.

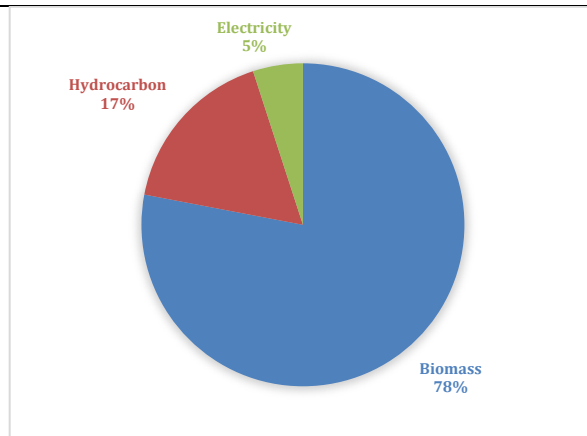


Figure 1: Energy balance of Mali, 2014

Source : Malian Energy Information System, Direction Nationale de l'Energie (DNE), 2017

Energy consumption in Mali is dominated by the residential sector which basically relies on biomass. Wood and charcoal are mainly used as cooking or heating fuels, a situation which results in strong pressure on the country's forest resources. In a country exposed to the harmful effects of climatic variations and where the degradation of land and natural resources is a real problem, fuelwood as a resource cannot adequately meet the increasing energy needs. Nevertheless, since Malian households will not do away with the use of wood fuels in the short term, primary energy in Mali will continue to depend on biomass (particularly fuelwood).

Apart from biomass, the sector is, therefore, characterized by a strong dependence on petroleum products whose importation is on a constant increase as a result of economic growth and the increasing demand of the population. Between 2000 and 2015, the quantity of imported hydrocarbons doubled from 0.5 million tonnes of oil equivalent (Mtoe) to 1.05 Mtoe and maintained a net increase until 2016 when it reached 1.37 Mtoe.

Mali imports 100 % of its consumption basically through road transport, representing about 26% of the country's total imports, a situation which, as the hydrocarbon monitoring indicators show, consequently makes it highly sensitive to the trends of hydrocarbon prices on the international market as well as the stability of the neighbouring countries. The main countries from which petroleum products come to Mali (2016) are Benin (45%), Senegal (30%) and Togo (15%)¹, followed by other countries like Cote d'Ivoire, Ghana or Mauritania.

This situation exposes the economy in its entirety to the volatility of petroleum prices and makes it vulnerable to foreign exchange reserve pressures, including in the energy sector.

In fact, the cost of hydrocarbons required to run thermal plants accounts for a significant share of the total national import burden and is one of the factors responsible for the high cost of electricity. In spite of the fall in the pump prices of super petrol, diesel, and fuel oil 180 between 2014 and 2016, a slight increase was observed in 2017 before reaching FCFA 692/l for super petrol, FCFA 603/l for diesel and FCFA 415/l for fuel oil 180. Kerosene, which was cheaper than super petrol and diesel in 2014 and 2015 (FCFA 570/l), reached FCFA 684/l in 2016.

The electricity subsector

According to 2017 figures, the rate of access to electricity was estimated at 42% at the national level (19% in the rural areas and 94% in the urban areas), with an increase of about 4% per year (DNE, 2016). Electricity demand in Mali increased historically by 10% a year (an average of 7.8% a year between 2005 and 2015, due to the strong impact of the crisis period (2012)).

The electricity sub-sector can be divided into four segments: the interconnected network of the Malian National Electricity Company (EDM-SA), isolated EDM-SA centres, integrated electricity production of private producers (mining and manufacturing companies in particular) which can sell their potential production surplus to the network and rural

¹ ONAP, 2017

electrification. The transmission and distribution network of EDM SA is made up of two components: (i) the interconnected network covering a number of urban centres including the capital, and (ii) isolated centres covered mainly by thermal plants operated by EDM-SA.

On the interconnected network, electricity is produced by hydroelectric plants, particularly the Manantali dam (out of which 104 MW of the total of 200 MW belong to Mali) and the Sélingué dam (46 MW), and thermal plants. In the current situation, the Manantali hydroelectric plant covers 26% of Mali's demand, followed by imports from Cote d'Ivoire (baseload imports of 30 MW increased to 45 MW through a provisional amendment), following the commissioning of the 225 kV Ferkessedougou-Sikasso line in 2012, and leased plants.

According to EDM-SA data, energy security in Mali is suffering from a chronic gap characterized by:

- ✓ at least 300 MW of electricity demand to be covered immediately or in the short term on the network;
- ✓ about 200 MW of thermal power to be replaced by competitive sources of energy;
- ✓ at least 500 MW in the long term for the connection or supply of big industrial extractive and food-processing units, etc.;
- ✓ at least 150 MW to be rehabilitated on the network's production capacity, and
- ✓ a fast-expanding network to support the development of the import and/or export of electric energy.

It is very unlikely that over the next ten years, the expected expansion of the national electricity network can ensure that a significant proportion of the low-income population living in isolated areas is connected to the network. This situation is expected to create an important market for off-network electrification schemes.

Off-network, the distribution of electricity is mainly ensured by local energy service providers who operate in isolated rural areas. In 2015, the country had 160 isolated mini-networks. The production technologies used to this day basically include diesel-run units (in 2015, 96% of the electricity produced off-network was through diesel) as well as related transmission and distribution systems. The total installed capacity of the mini-networks is about 15 MW.

It is important to note that the energy mix underwent a significant change with the contribution of hydroelectric power dropping from 80% in 2005 to 45% in 2015 and the contribution of thermal plants doubling to 41% over the same period, a situation projected by the electricity recovery plan in 2014 (see figure).

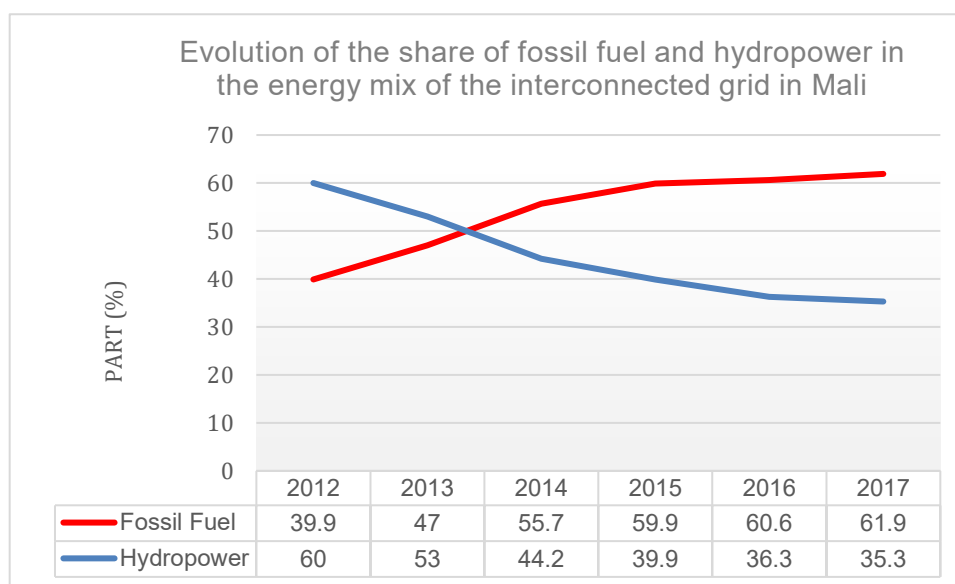


Figure 2: Energy mix in the interconnected network of Mali, including imports.
Source: Mali - Provisional Electricity Recovery Plan 2014

The tariffs within the mini-grids reflect the cost prices as well as the very high prices of the transportation of fuel to the hinterland; weighed against the purchasing power of many households, they prove to be inadequate.

The rural electrification sub-sector

Rural electrification in Mali is characterized as follows:

- ✓ about 70% of the Malian population live in the rural areas;
- ✓ the rate of electrification in the rural areas was 19% in 2016 against less than 1% in 2004;
- ✓ Mali has more than 10,000 rural towns;
- ✓ at least 255 towns are currently being supplied with electricity by private operators with the support of Donors and Technical and Financial Partners;
- ✓ the rural electrification systems developed by the operators are basically thermal;
- ✓ in 2015, about 131 towns were connected to diesel units, 32 towns were connected to hybrid PV-Diesel systems, 15 towns used biofuel units, 20 towns benefitted from the extension of the EDM-SA network, 43 towns were supplied by Multifunctional Platforms, 13 towns were supplied by solar kits and one town was supplied by a hydroelectric micro-plant;
- ✓ a total installed power of about 23 MW, with nearly 2 MWc PV solar, and
- ✓ operators enjoy subsidies from the Rural Electrification Fund managed by AMADER.

It should be noted that rural energy service providers operating mini-grids fueled by fossil fuel power plants are particularly affected by volatile oil prices, as well as exorbitant fuel transportation costs.

The renewable energy subsector

With the exception of medium-scale and large-scale hydroelectric production, the production of electricity from renewable sources connected to the network was 8.7 MW in 2016 (EDM S.A), while the off-network installed renewable energy capacity consisting mainly of small-scale solar energy reached 15 MWc in 2015. These rural energy production projects show the interest of the rural population and private operators in electricity supply.

The current development of renewable energy in Mali is characterized by a technological development of equipment which has significantly taken the concerns of Sahelian users into consideration as well as by an appreciable drop in the prices of equipment.

Many PV projects are being developed particularly at the Kati (65 MWc), Kita (50 MWc), Ségou (33 MWc), Sikasso (50 MWc) and Koutiala (25 MWc) sites. Plans are afoot for others, especially Fana (50 MWc), SREP Mali project (20 MWc) and Sélingué (Greenwish – 40 MWc), and the construction of several other plants hooked onto to the interconnected network for an accumulated power of 260 MW.

Solar PV is the subject of many applications:

- photovoltaic lighting (domestic needs, lighting of health centres, literacy centres and public areas);
- photovoltaic refrigeration used mainly in health centres to preserve pharmaceutical products;
- broadcasting, telecommunications and radio communication;
- solar pumping for irrigation and village water systems (water for domestic use) and grazing systems (watering places for animals), and
- solar fields and hybrid systems for the decentralized production of electricity (mini-networks) or connected to the network.

National Energy Policy

The sector is mainly governed by the National Energy Policy (NEP) adopted in 2006 and targeted at promoting the sustainable development of the country through the provision of affordable energy services in order to increase access to electricity and to promote socio-economic activities.

The specific objectives of the NEP are as follows: i) meet energy needs in terms of quality, quantity and cost; ii) protect persons, goods and the environment from the risk of a shortfall on the part of energy providers; iii) strengthen capacity in

the area of policies, management, monitoring and control of the energy sector, and iv) consolidate the gains from international cooperation in the area of energy.

The issue of renewable energy has been tackled in several regulatory enactments, particularly the NEP (2006) and the above-mentioned statement (2009). The most important policy document on this issue is the National Renewable Energy Development Strategy adopted in 2006 and with the objective of: i) promoting a vast use of renewable energy technologies and equipment to increase the share of renewable energy in national electricity production (up to 10% in 2015); ii) developing the biofuel sub-sector for various uses (electricity production, transport, agricultural mechanization, etc.); iii) creating better conditions to make renewable energy services sustainable; and iv) looking for sustainable financing mechanisms suited to renewable energy.

Policy Framework for Private Sector Engagement

Generally speaking, the policy environment is relatively favourable for energy investments. It is characterised by the following policy tools: (i) the Rural Electrification Fund; (ii) the National Industrial Development Policy (whose overall objective is to ensure an orderly, rapid, sustainable and balanced industrial development of the country while generating employment); (iii) the Public Private Partnership Framework (in the form of a concession contract based on a BOOT)² arrangement); (iv) the Investment Code (which is aimed at establishing a privileged customs and tax regime with a view to promoting investments); (v) the role of the Investment Promotion Council of Mali (API-Mali), offering a one-stop-shop for all procedures in setting up businesses, assisting investors and issuing permits in accordance with the Investment Code of Mali; and (vi) the directives of an *Ordonnance*³ guaranteeing the free exercise of competition and organising the public power service sector (including, among other things, the role and capacities of various players in the power sector, the conditions governing the exercise of the public electricity service contracting authority, the conditions governing the delegation and operation of the public electricity service, tariff and accounting principles in the sector)⁴.

United Nations Framework Convention on Climate Change

Apart from the strategies linked to the energy sector, the energy issue is taken into account in the commitments made by Mali to protect the environment and to combat climate change.

Mali has signed the United Nations Convention on Climate Change and the Kyoto Protocol as well as formulated its National Adaptation Programme of Action (PANA) on the harmful effects of climate change in 2007. The PANA consists of renewable energy development projects, some of which have been partly implemented.

Furthermore, in 2011, the Government of Mali formulated a national policy on climate change and a national climate change strategy (both of them factor RE into their thrusts of action).

At the COP 21 held in Paris in December, 2015, Mali committed to contribute to the reduction in GHGs by up to 31% for the energy sector. The Government of Mali intends to gradually reduce the carbon intensity of the electricity network by increasing investments and the use of renewable energy sources like photovoltaic solar (PV).

Moreover, Mali is fully committed to the protection of the environment and the fight against climate warming. The Government of Mali demands the systematic consideration of the assessment and reduction of environmental impacts in the design, manufacturing and use of energy infrastructure and equipment.

The country's solar potential and key constraints

The solar potential (PV and CSP) of the whole of Africa was analyzed by IRENA in 2014 (IRENA, 2014). According to this study, Mali is situated in a region with high solar potential and is considered as a country that is particularly conducive

² Under the BOOT contract, a company or a consortium of companies' finance, build, own and operate a new facility or a system, which is transferred back to the government after a predetermined period of time.

³ Ordonnance n. 00-019/P-RM du 15 mars 2000.

⁴ In this regard, many forms of partnerships aimed at enhancing a conducive environment for private investors (under various forms of contracts, such as concession contracts, leasing and management contracts).

to the development of solar technologies. The average solar radiation is estimated at 5-7 kWh/m²/day with a daily sunshine duration of 7 to 10 hours. Thus, the production potential from solar PV is estimated at 7906 TWh/year (IRENA, 2014).

According to the categorization made in relation to solar resources (GHI for PV and DNI for CSP) (see table below), this potential is considered adequate.

Categorization	Limited relevance	Adequate	Very adequate	Excellent
PV (GHI in [kWh/m ² /year])	<1000	1000-1500	1500-2500	2500-3000
CSP (DNI in [kWh/m ² /year])	<1800	1800-2000	2000-2500	2500-3000

The following figures show at the continental level the potential regions for the development of solar PV and CSP indicating that Mali is located in a region with high solar potential and presents itself as a country particularly conducive to the development of solar technologies. (IRENA, 2014).

Moreover, malian solar radiation map (see figure 5) shows that solar radiation is more important in the northern part of the country than in the southern part.

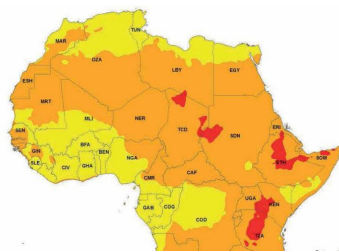


Figure 3: PV grading (based on GHI) ; from less to more adequate = yellow-orange-red (IRENA, 2014)

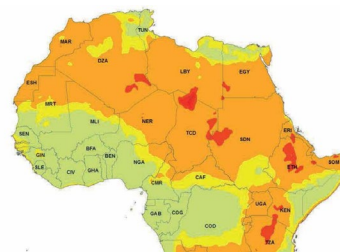


Figure 4: CSP grading (based on DNI) ; from less to more adequate = green-yellow-orange-red (IRENA, 2014)

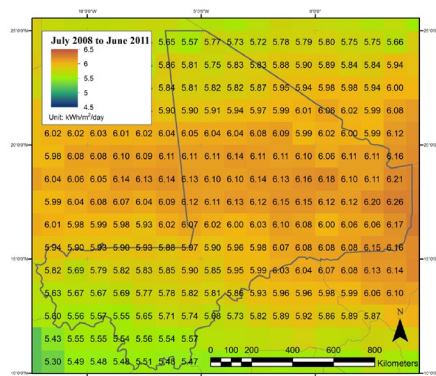


Fig 5. Solar Radiation in Mali

Despite the immense potential of the country, it represents only 3% of electricity production from renewable sources (excluding hydroelectric production). Its development remains slow and should reach only 10% of the energy mix by 2020 (national policy objective). Indeed, the widespread use of renewable energy technologies encounter significant barriers from a economic and financial, institutional, regulatory, sociocultural and environmental point of view. Mobilizing related investment is necessary to see a transformation of the sector.

Nonetheless, there are a number of major constraints to overcome for this purpose:

- **Economic and financial:** Weak domestic financial institutions; high up-front cost of RE technology; inadequate financial incentives to attract the private sector; weak canalization of finance from international sources for larger-scale RE development.
- **Institutional:** Coordination among proliferating agencies concerned with RE development; weak planning processes; incomplete framework for PPPs, in particular for utility scale independent power producers (IPPs).
- **Technical:** Limited capacity of human resources in the sector; limited studies; impact assessments on RE for electricity generation do not yet create adequate conditions for a robust, standardized and programmatic approach.
- **Social:** Inadequate provision of information and awareness of consumers on RE opportunities and challenges; poverty of Malian rural households impedes affordability, access and use of modern RE without subsidy schemes.

C.2. Project / Programme Objective against Baseline

1. Baseline Scenario

1.1. Climate change and energy challenges

1.1.1. Vulnerability of the energy sector to climate change

Climate change is one of the major challenges for the development of countries like Mali, which is ranked 175th out of 187 countries on the Human Development Index (HDI), as a result of their low incomes, weakness of their human capital and the weakness of their economies which basically depend on natural resources and climatic factors.

It has been observed that since the advent of the periods of drought in 1970, Mali has been experiencing a more arid climate throughout the country characterized by a 20% overall decline in useful rain. This situation leads to the worsening of the vulnerability of the main sectors like energy, particularly hydroelectric power which accounts for nearly 45% of cheap electricity supply by the national network from the Manantali dam (of which 104 MW out of a total capacity of 200 MW belong to Mali) and the Sélingué dam (46 MW).

With regard to energy production, one can find two hydroelectric plants in the River Niger basin – one at the Sélingué dam on the Sankarani and the other on a diversion of the river at Sotuba near Bamako. At the first one, the average annual production was 226,8 GWh, while the second one recorded 35,6 GWh in 2007 (EDM-SA report, 2007). The energy produced by the two plants met some 30% of the country's electric energy demand.

However, the rainfall situation, coupled with the technical constraints, affected the production of hydroelectric power by the said dams, where it was observed that there was a drop of more than 20% and 10% respectively over the last ten years.

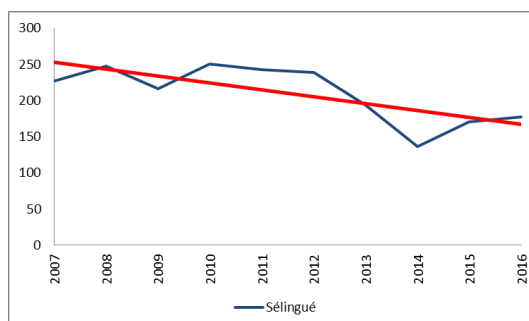


Figure 6: Change in hydropower production by the Sélingué dam (2007-2016)

1.1.2. Projections and future impacts on the energy sector

Climatic scenarios provided by the MAGICC and SCENGEN tools of the GIEC reckon a +3°C average increase in temperatures and a 22% decrease in rainfall throughout the country by 2100, with a risk of a reduction in water resource

recharge. Yet, it is acknowledged that a 1% decline in flows will bring about a reduction in electric power production in the region of 1.3 million kWh.

The consequence of this situation is poor access to affordable energy which limits the social opportunities of the poor segments of the population, particularly women and the youth in the rural areas, especially where the access rate is lower than 20%. Thus, energy is one of the sectors requiring an urgent roll-out of climate change adaptation measures in accordance with the National Adaptation Action Plan (PANA) on climate change in Mali.

1.2. Overview of Mali's GHG Emissions Profile due to the Energy Sector

The three national communications of Mali (1st in 1995, 2nd in 2000 and 3rd at the formulation stage) all established that the energy sector is one of the biggest emitters of greenhouse gases (GHGs), having increased from 945 kT to 2,782 kT, i.e. an increase of more than 190% over the 17-year 1995-2012 period. The basic scenario shows that GHG emissions will increase from 3,400 kTeq CO₂ in 2015 to 4,750 kTeq CO₂ in 2020, before reaching 9,269 kTeq CO₂ in 2030. This gives an average annual increase of 6.91% between 2015 and 2030.

This situation is correlated with a major deployment of thermal plants over the last few years by both EDM-SA, the national company, and private operators for the electrification of the country's isolated areas. Notwithstanding the high cost prices and the low profitability of this electrification option, this situation is highly unlikely to improve in the next few years without a sustainable alternative due to the ever-growing demand of the population (3.6% a year), economic growth and social pressure. In fact, it appears that it is simple and quick to deploy the thermal solution. Thus, to meet this demand, EDM-SA, the national company, is investing more and more in medium-sized thermal plant projects (fuel oil plants) and is also supplying more than twenty isolated centres solely with plants operating on diesel. Furthermore, the huge consumers of the mining and industrial sector exclusively use thermal energy for their needs (200 MW), while rural electrification is basically provided by local suppliers who operate diesel-powered isolated mini-networks.

Although the country's contribution to greenhouse gas emissions (GHGs) in the world is very low (0.06%), the government has agreed to participate in the collective ambition to attain the ultimate objective of the United Nations Framework Convention on Climate Change (less than 2°C) by reducing GHG emissions in the energy sector by 31% and introduced an investment plan in this regard in order to attain its objectives.

The National Environmental Protection Policy is being updated under the supervision of the Ministry of Environment and Sanitation. Achieving the political objectives related to environmental sustainability requires the prioritization of new energy investments for energy infrastructure that is environmentally friendly and resilient to climate change.

Constraints and challenges to the development of solar-based electrification in the context of the fight against climate change

- At COP 21 held in Paris in December 2015, Mali committed to contributing to the 31% GHG reduction mitigation scenario for the energy sector and included it in its document on defined NDCs and then in the Investment Plan for implementation of the NDCs. During consultations with the various actors for drawing up the Plan, it was noted that a good number of actors lacked knowledge of the NDCs document. Thus, it was identified that there was a priority need for awareness on the NDCs in each of the various areas, including the energy sector. Such an approach could enable the main players in the energy sector, particularly public institutions (National Energy Directorate -NED-, AMADER, AER-Mali), the private sector (investors and operators of existing rural electrification services) and the end users of basic energy services to work towards achieving the objectives set in terms of commitment by the country.
- The high initial investment cost in PV plants is a major obstacle to private sector involvement in rural electrification, as private investors are called upon to mobilize resources from the local banking system, at a cost which is expensive for them. The second main obstacle concerns the fixing of an appropriate tariff that ensures the financial viability of the system, while taking into account the capacity to pay in rural areas. The extent to which cost and risk factors apply varies with the technology used and the geographical location, and project promoters expect some form of financial risk sharing to offset them. To date, this risk has been covered by external donors through the Rural Electrification Fund (FER). The FER has therefore succeeded in encouraging more than 60 energy supply companies to intervene in Mali. They are involved in nearly 200 mini-grids. The initial

investment subsidies granted by the FER aim at making tariffs acceptable for consumers in rural areas and thereby guarantee financial profitability for private operators. Investment subsidies are limited to 70-80% of the investment cost and are based on a results-based approach (determined by the number of consumers expected to be connected to the grid during the first two years, the average tariff and the investment cost for off-grid consumers). No subsidy is granted for electricity consumption. However, the Rural Electrification Fund (FER) is, in practice, a bank account managed by AMADER which was first created to manage direct grants or an onlending of concessional loans. This fund is still not a real financial institution capable of borrowing, issuing guarantees and equity, setting up loans, towards rural electrification companies (for a fixed or unlimited period). It goes without saying that this situation is not sustainable, due to the decrease in the financial resources available from external sources to continue providing such subsidies. The government is therefore considering strategic actions such as building capacity of the FER to co-ordinate the mobilization of the financial resources on the international market which are needed to finance investments by the agencies in charge of promoting the State's renewable energy programmes.

- Minigrid companies focus primarily on supply of power and often do not pay enough attention to demand stimulation. Too often, new customers have been slow to connect and only use a small amount of electricity, leading to lost revenue and insufficient volume to spread fixed cost. Productive-use appliances, like grain mills, carpentry, irrigation, and cold storage, are particularly important because they are often coincident with cheaper generation from solar and have higher consumption than residential loads. Commercial and productive use customers are thus critical for long-term viability of minigrids/microgrids - as they have higher demand per connection, their loads are more coincident with solar generation, and they can provide immediate new income for the community to help afford the new electricity service. Furthermore, this is particularly important for rural areas where energy sources are needed in synergy with potentials for productive energy uses and employment creation. However, the upfront cost to connect or for purchasing electric appliances may be too high, requiring financing or subsidy whilst rural population in Mali faces several difficulties, including the lack of an appropriate financial service to support Productive Users of Energy (PUE) and back them up. Indeed, despite the significant demand for financial services in rural areas, institutions offering financial services – such as banks, credit unions, microfinance institutions (MFI) or insurance companies – are typically reluctant to serve rural areas. The key issue is often related to guarantee. Indeed, surveys revealed that although some producers' cooperatives are eligible to credit from financial institutions, they have trouble accessing credits due to issues of guarantee.

The removal of these barriers, supported by this project, will help the country to strengthen the resilience of the energy sector and hence the resilience of local populations to the effects of climate change, while satisfying the growing need for electricity, and reduce GHG emissions related to the energy sector.

C.3. Project / Programme Description

Describe the main activities and the planned measures of the project/programme according to each of its components.

1. Components of the project

The main components of the project are: (i) Capacity building of rural electrification institutions and Technical Assistance; (ii) Detailed Engineering and Installation of 70 solar powered minigrids (50 mini-grids at Tranche 1) to achieve access for 31,000 households and businesses (28,378 at Tranche 1); and (iii) Support to Productive Use of Electricity for solar based mini-grids.

Component 1: Capacity building of rural electrification institutions and Technical Assistance

This component takes into account activities related to the establishment of an enabling environment for the promotion of rural electrification from solar photovoltaics. It focuses on capacity building for structures such as AMADER, AER, and the Rural Electrification Fund as well as actions to raise investor awareness and disseminate project activities and results.

1.1. Training, raising awareness of stakeholders and dissemination of project activities/results

The project will contribute, on the one hand, to raising awareness among stakeholders on the country's commitments through the NDCs in the field of energy and related implications, by:

- organizing dissemination workshops on potential actions to be implemented in order to achieve the objectives of the CDN in the field of energy;
- organizing specialized training sessions for designated executives of NED, AMADER and AER-Mali on climate change and issues related to the energy sector;
- developing and disseminating public awareness material on climate change and CDN in the energy sector.

These activities will be carried out with support from the Environment and Sustainable Development Agency (AEDD) which is in charge of the national policy on climate change drawn up in 2011 together with a strategy and an action plan.

1.2. Communicating project results and lessons learned

Communication will be made towards various stakeholders on project activities, results achieved and best practices/lessons learned from the project, for replication across the country and the West African sub-region.

The expected results from this activity include particularly:

- Developing a communication and raising awareness/promotion plan for national (and international) investors and production of materials (including videos) will be published on the experience/best practices and lessons learned from the project. These materials will be made available electronically and they will be widely disseminated throughout the region and in countries considering similar renewable energy systems for rural electrification. In addition, they will be posted on the project website;
- Organizing information sessions with investors who are already active in the energy/renewable energy sector in the country, local businessmen who wish to develop their activities to include energy for rural areas and, possibly, organize a roadshow to attract foreign investors in order to create consortia with local companies to provide rural areas with modern quality energy services.
- Organizing mid-term and end-of-project workshops for institutions concerned (DNE, AMADER, AER-Mali, FER) to ensure monitoring and document the project experience. Training will be provided by international/local consultants for stakeholders on how to monitor, record and document the project experience.

1.3. Capacity building of the Rural Electrification Fund (Fonds d'Electrification Rural -FER)

This project will work to achieve this objective by supporting the setting up of technical assistance to develop the Rural Electrification Fund into a financial institution with its own legal status, through:

- (i) drafting of organisational and operational legal provisions as part of a reform of the FER;
- (ii) recruitment of a dedicated staff; and
- (iii) capacity building of the FER staff.

1.4. Technical Assistance on Regulatory Framework and on Procurement and Contracting of minigrid construction and O&M companies

Technical assistance will further be provided for the tender process and selection of O&M companies that will operate the mini-grids. This will include the writing of standard contract for 15 years O&M and the procedures that set the relations between AMADER and O&M companies.

A detailed implementation plan is provided on Annex 10.

Component 2: Detailed Engineering and Installation of solar powered minigrids

For each of the selected localities, this component includes specific engineering studies including site

identification/selection and E&S analysis, acquisition, installation and commissioning of the following equipment: (i) 50, 80 and 150 kWp stand-alone solar power plants with storage systems; (ii) low and medium voltage grids with their respective lengths of 8, 10 and 12 km; (iii) 14,189 prepaid meters; (iv) household in-door electrical wiring; (v) civil engineering works of buildings; and (vi) 10 stand-alone street lighting solar systems.

The project will set up a certain amount of metering equipment and grid connection line, then in the implementation of the project a deployment will be made according to the demand thanks to the income from the sale of electricity. The operators will charge an additional fee for households' connections according to the national regulation and within the limits of predetermined fees agreed with the rural electrification regulator (AMADER).

The equipment and their auxiliaries will be supplied with the essential spare parts and the specific tools for their maintenance as well as the required documentation.

AMADER is the entity which on behalf of Mali government will have the ownership of the assets. A concession agreement for 15 years (renewable) will be awarded by AMADER to eligible operators. The financial eligibility criteria is the ability to invest 20% of the total cost per minigrid. Detailed technical selection criteria will be developed as part of the Technical Assistance in component 1.

2.1 Preparation of detailed engineering studies

This component includes activities such as site selection, engineering studies and specific environmental and social analysis for each site.

2.2. Supply installation and commissioning of equipment

The 14 localities in category 1 (population less than or equal to 2,000 inhabitants) will each be supplied by a 30 kWp photovoltaic solar power plant and a 5 km-long three-phase LV grid.

The 21 localities in category 2 (population between 2,000 and 4,000 inhabitants) will each be supplied by a 50 kWp photovoltaic solar power plant and a 8 km long three-phase LV grid.

The 27 localities in category 3 (population between 4,000 and 7,000 inhabitants) will each be supplied by an 80 kWp photovoltaic solar power plant and a 4.5 km-long three-phase LV grid.

The 8 localities in category 3 (population of 7,000 inhabitants or more) will each be supplied by a 150 kWp photovoltaic solar power plant and a 12 km-long three-phase LV grid.

The total installed power will be: 0,42 MW + 1,05 MW + 2,16 MW + 1,2 MW = **4,83 MW**.

For Tranche 1 the total installed power will be: 0,42 MW + 2,16 MW + 1,2 MW = **3.78 MW**

2.3. Implementation of Environmental and social measures

This subcomponent takes into account the environmental and social actions for the purpose of mitigating the negative impacts and risks of the project on the environment and the project area. These include:

(i) institutional strengthening measures (strengthening the environmental expertise of the AER, AMADER and private operators); (ii) Technical studies, audits and procedures (environmental and social impact notices to be prepared; facilities maintenance manual); (iii) Training of the stakeholders involved in project implementation, particularly the Environmental and Social Safeguard Unit/AER, Information and sensitization of communities and populations in the project area; (iv) Monitoring, supervision and evaluation of project activities.

Component 3: Support to Productive Use of Electricity

This component is in line with the need to improve the profitability of electrification in rural areas and encourage investment, through support for energy demand, particularly with regard to the "productive use of energy".

Indeed, by supporting the acquisition of equipment for productive use by rural populations, the project will contribute to increasing the use of energy for income-generating activities, which will stimulate the sale of electricity and economic development of the beneficiary localities.

For this purpose, local financial institutions (microfinances) are willing to put in place "equipment or investment credit" for productive users. Types of energy equipment for productive use include: electric tools for carpentry, woodworking and small industry; sewing machines; welding machines; battery charging equipment; refrigeration and ice cube production; seed oil presses, grain mills, crop drying equipment and other equipment for value-added processing of local agricultural produce (honey, mango, cashew nuts, etc.).

However, in order to succeed in this operation, there is a need for risk guarantee in order to encourage banks and non-bank financial institutions to provide financial services to productive energy users. This will also make it possible to control the interest rates of the loans granted to the populations to support the users of productive energy.

Surveys conducted in local communities showed that the terms for credit dedicated to equipment for productive use would be as follows:

- loan amount between XOF 250,000 and 5 Million (Euro 381 to 7622) estimated at XOF 500,000 (Euro 762) on average;
- down payment requirements of 20 to 30% of the equipment cost;
- Tenor of 6-24 months, usually 12 months on average;
- interest rates between 18 and 20% per year, with set-up costs;

This component aims at promoting increased access to financial services for productive energy users in rural areas through setting up a financial guarantee fund toward financial institutions, for derisking purposes.

Upon BOAD's experience, this financial guarantee will be a loan proceed committed for covering part of the risk taken by the microfinance institutions (maximum 40% of receivables deemed irrecoverable). With hindsight, the following conditions are required:

- when all avenues have been exhausted in recovering overdue credit in accordance with the institution's credit policy;
- where the delay period exceeds 24 months, which is the legal period for downgrading credits as a loss.

The loan proceeds will be exclusively used for the guarantee itself and will be effectively in place upon completion of minigrids. The resources will be managed by BOAD via FER. FER is part of government of Mali who is the borrower. No new legal entity will be created since FER is already existing. The mechanism and the operationalisation of the guarantee fund are covered in component 1 (Capacity building and Technical Assistance to the Rural Electrification Fund – FER). Once FER has the capacity in place, it will use the proceeds to guarantee micro loans given by the local microfinance institutions for productive use. The productive use micro loans will be given to productive use consumers in the areas where the minigrids will be installed.

Microfinance Institutions which will be partners for this project will comply to Malian regulations for microfinance institutions such as:

- approval given by the Ministry of Economy and Finance (MEF)
- losses must not exceed 5% of the institution's total loan portfolio;
- interest rates are capped at 27%;
- loans to a client may not exceed 10% of total portfolio;
- 50% of the profits must be reserved and reinvested in the MFI;
- 15% of the loan portfolio must remain on deposit
- Annual financial reports and returns must be submitted to the Ministry of Economy and Finance and must indicate the following:
 - o amount of deposits taken from customers;
 - o total loan portfolio;
 - o number of customers;
 - o Interest rate

Mali is a member of the Central Bank of West African States (BCEAO), and therefore its banking system is regulated by this regional monetary institution which has drawn up a set of regulations covering MFIs. A copy of the above reports will therefore be shared with BCEAO for registration under the decentralized financial systems.

A partnership and performance agreement will be signed between the FER and the eligible MFIs.

Eligibility of the LFIs will be proven through the following: (i) approval given by the Ministry of Economy and Finance (MEF); (ii) number of years of experience, and (iii) effective and continuous implementation of activities in the field; (iv) a reassuring financial situation over the last three (03) years through the following indicators: (i) solvency (equity capital/total net assets is greater than or equal to 15%); (ii) portfolio quality (90-day overdue/portfolio outstanding less than 3%); and (iii) internal control quality.

In the context of this project, concerning ALM/FT, eligible LFIs will:

- ensure that the funds financing the Project are not of illicit origin and in particular are not related to fraud against Mali's financial interests, corruption, organized criminal activities, terrorism or drug trafficking; and
- forward to the Bank without delay any information raising suspicions as to the unlawfulness of the sums invested in the company and in the Project;
- notify the Bank without delay if it has known at any time of any information indicating the illicit origin of all or part of the funds of the structure;
- will not enter into a business relationship, directly or indirectly, with persons or entities on the lists established by the United Nations Security Council or its committees pursuant to Security Council resolutions, by the Council of the European Union in application of its Common Positions and / or by the African Union as well as any other relative or complementary resolution and any act of implementation thereof in connection with the fight against the Laundering of Capital and the financing of terrorism.

Under its due diligence, BOAD will assess civil / criminal and regulatory antecedents and sanctions lists. The Bank conducts also administrative investigations into corruption, fraud, coercion, collisions and inconvenient practices, and make use of the relevant national authorities for the necessary criminal investigations.

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

Describe the quality of the management team, overall strategy and financial profile of the Sponsor (Executing Entity) and how it will support the project/programme in terms of equity investment, management, operations, production and marketing.

Main entities involved in the project are:

- The Government of Mali will be the borrower and will take the credit risk through a sovereign guarantee.
- The Ministry of Energy and Water as the executing entity who delegates execution to the Renewable Energy Agency (Agence des Energies Renouvelables – AER) for component 2. (see Letter of the Ministry of energy dated 11 July 2018 in Annex)
- BOAD will provide direct support (selection of consultants) to AER in executing component 1 (capacity building and technical assistance), and the Rural Electrification Fund (FER) will execute component 3. BOAD has conducted technical and financial capacity assessments of the entities involved in the project as per requirements of the Accreditation Master Agreement (AMA) in line with BOAD's policies and procedures. It is BOAD's reasonable conclusion that the national entities involved therein are capable of carrying out the execution and operation of the project component activities under BOAD's assistance and support.

The Renewable Energies Agency (AER)

The Renewable Energies Agency in Mali (AER) is a Public Establishment of Scientific and Technological Character

endowed with legal personality and financial autonomy (inherited from the National Center for Solar Energy and Renewable Energies (CNESOLER)). Its mission is to: i) assess and inventory the country's potential for renewable energy resources; ii) contribute to the definition of national renewable energy strategies; (iii) conduct research / development activities in the field of renewable energies; (iv) conduct studies and monitor the implementation of renewable energy programs and projects for the benefit of industry stakeholders; (v) contribute to the development and capacity building of craftsmen, agents of State technical structures, local and regional authorities and private individuals, as well as those of renewable energy institutions in other countries; (vi) contribute to the information and awareness of promoters and users of renewable energy equipment; vii) perform testing, quality control and labeling of renewable energy equipment for the benefit of developers; viii) research and set up sustainable and adapted financing mechanisms for renewable energy projects and programs within a Public Private Partnership framework; (ix) to participate in international cooperation actions in the field of renewable energies.

AER has a staff of 47 agents including 14 senior managers (05 researchers (including a Doctor, Research Director, a Director of Memory and 03 Research Assistants) 04 Energy Engineers, 02 Electrical Engineers, 01 Statistician-Planner Engineer, 01 Engineer in Industry and Mine and 01 administrative and financial manager. (See AER's oragnizational flowchart below)

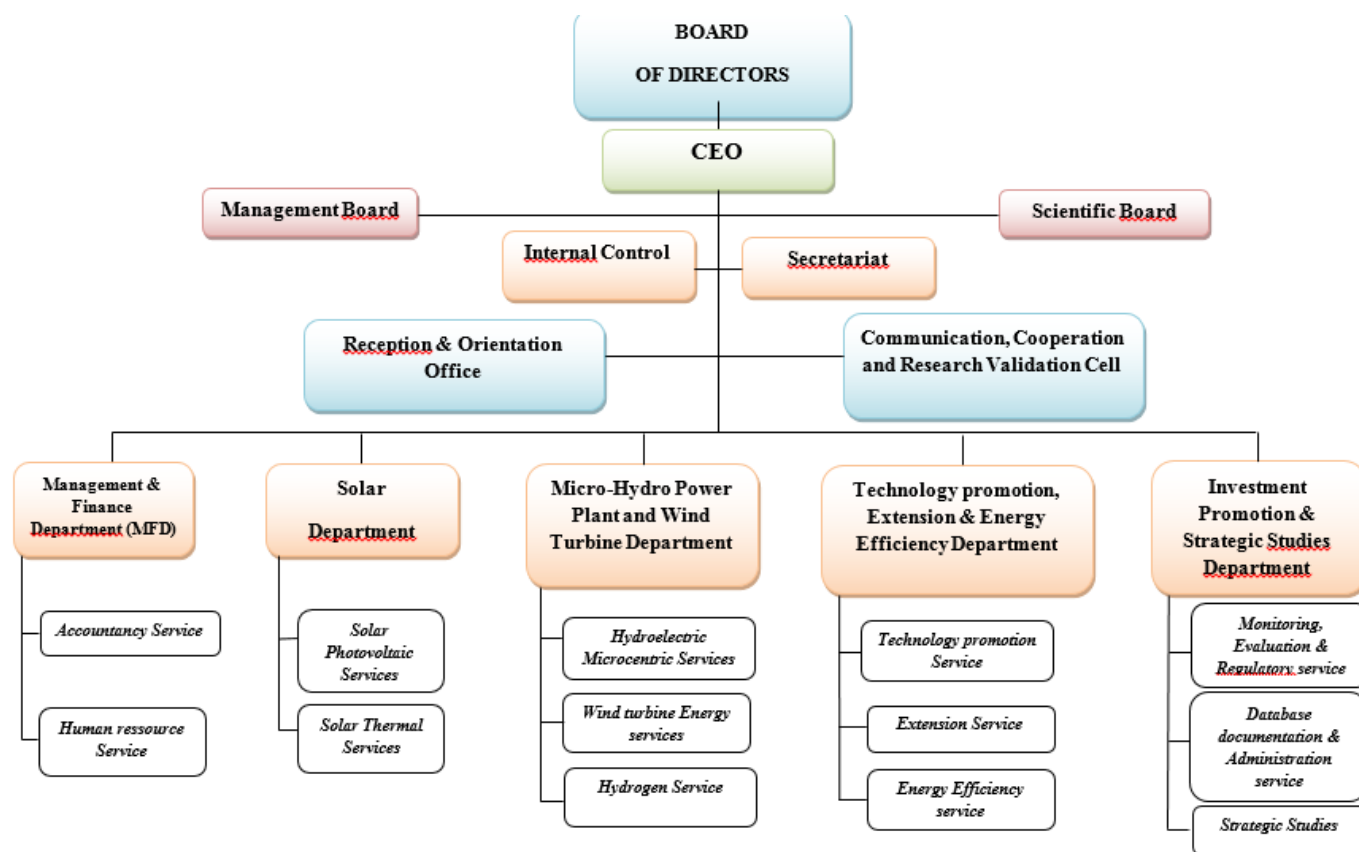


Figure 7: AER's organizational flowchart

Since its establishment, AER has conducted 7 solar based projects with the support of various partners such as UNDP, Indian Government, Chinese cooperation, European Union, AfDB, etc. A summarized list of the projects is presented below:

PROJECT TITLE	INSTALLED CAPACITY	BUDGET	FUNDING SOURCE	PROJECT DURATION

PROMOTING SUSTAINABLE ELECTRICITY PRODUCTION IN RURAL AREAS THROUGH HYBRID TECHNOLOGIES	147 kWp	12 585 568 500 XOF	UNDP MALI Gov.	2017-2020
SOLAR ENERGY VILLAGE ELECTRIFICATION PROJECT (PEVES III)	1,287 MWp	4.676.100.000 XOF	INDIAN Gov. MALI Gov. & BENEFICIARIES	
SUPPORT PROJECT FOR THE PROMOTION OF RENEWABLE ENERGY IN MALI (PAPERM)		1. 272.000 XOF	AFDB	
SINO-MALIAN COOPERATION SOLAR ENERGY BASED PILOT PROJECT FOR KIRINA	28, 520 kWp	400 000 000 XOF	CHINESE COOPERATION	2012- 2013
RENEWABLE ENERGIES PROMOTION FOR WOMEN EMPOWERMENT (PENRAF)		1 598 000 000 F XOF	UNDP, MALI Gov.	
"ENHANCING THE LIVES OF CHILDREN AND THEIR FAMILIES THROUGH ACCESS TO MODERN, CLEAN AND AFFORDABLE ENERGY SERVICES IN 30 DEPRIVED COMMUNITIES IN THE KITA CIRCLE": PASER-K	74,18 kWp	1 725 157 000 XOF	EUROPEAN UNION, PLAN ESPAGNE	2015-2018
ACCESS TO MODERN ENERGY THROUGH 36,123 PUBLIC LIGHTING PHOTOVOLTAIC SOLAR SYSTEMS IN 311 VILLAGES AND CITIES IN MALI	11,32 MWp	36.000.000.00 0 XOF	MALI GOVERNMENT	

West African Development Bank (BOAD)

The West African Development Bank (BOAD), is a regional financial institution with a mandate to promote development in West Africa, and foster economic integration within the sub region. For over 40 years, BOAD acquired strong experience in financial resources mobilization and management from its member countries⁵ and partners, including AfDB, AFD, etc. and on international financial market.

The Bank's technical expertise and fiduciary capacity are deemed complied with international standards by several international funds, such as the Green Climate Fund (GCF), the Global Environment Facility (GEF) and the Adaptation Fund (AF) to which BOAD is accredited as project entity.

BOAD has already financed solar based rural electrification projects in Senegal, Niger and Togo over the last three years.

BOAD has significant experience in the region using loans instruments with government to de-risk operations with a guarantee.

As the implementing entity (the Accredited Entity) BOAD has a team of energy engineers, rural engineers, environmentalists and various experienced specialists to accompany the project in its implementation phases. Moreover, the Bank could also consider the recruitment of specialists to support the project teams if necessary.

BOAD has relevant experience in energy sector through the management of a fund so-called "**Energy Development Fund (FDE)**". Established in September 2009, the Fund aims to contribute to improved access to energy in the WAEMU member countries and promote the development of sustainable energy under concessional conditions in order to finance eligible projects. The Fund started its operations in the first quarter of 2010 and focuses on the following areas: emergency plans and investment projects, audits, studies, engineering and control of investment projects, capacity building and technical assistance, development and execution of business plans, reform of the legal and regulatory framework, accompaniment to the restructuring and consolidation of the energy sector. The initial allocation of the Fund is FCFA 250 billion (~USD 125 million). As at 30 September 2016, 14 projects were financed in the form of loans involving a total amount of FCFA 229.7 billion (~USD 459 million).

⁵ Benin, Burkina Faso, Côte d'Ivoire, Guinea Bissau, Mali, Niger, Senegal and Togo.

BOAD's experience in Mali as for energy sector projects is as follows.

- ***EDM-SA Interconnected Network Strengthening Project***

The project involves the construction of a national dispatching center in Bamako and a 107-km long 63-kV power line between the towns of Ségou and Niono, as well as the renewal and extension of the distribution network. from the city of Niono.

With a total cost of M XOF 12,766 excluding taxes, the project is financed by BOAD and EDM-SA respectively for MXOF 11,000 and MXOF 1,766. The BOAD loan of MXOF 11,000 was granted by its Board of Directors in March 2008.

- ***Interconnection project of the electricity networks of Mali and Ivory Coast***

The project involves the construction of a 225-kV electrical interconnection line, covering a distance of 519 km, from Ferkessédougou in Ivory Coast to Ségou in Mali via Sikasso and Koutiala. The main objective of the project is to increase supply and strengthen the security of electricity supply in Mali.

The total cost before tax of the project is estimated at MXOF 74,177 and its financing plan is as follows: i) BOAD: MXOF 12,500; (ii) EXIM BANK from India: MXOF 33,750; (iii) EBID: MXOF 13,500; iv) Ivorian State: 976 MFCFA; v) EDM-SA: MXOF 1,527; and (vi) Government of Malian: MXOF 11,894. It should be noted that in the context of its financing of its counterpart, the Malian Government obtained from Exim Bank of India, an additional financing of US \$ 36 million.

- ***Rehabilitation of Sélingué and Sotuba hydroelectric facilities Project***

The purpose of the project is the rehabilitation of the Sélingué and Sotuba hydroelectric power stations, including the Sélingué energy evacuation stations and lines, as well as the connection to the interconnected network of Kangaba and Ouéléssébougou to the interconnected EDM-SA network with the electrification of fifteen (15) localities. The overall objective of the project is to contribute to the regular supply of electric power at lower cost.

The total cost of the project is MXOF 28 441 and its financing plan is as follows: i) BOAD: MXOF 25 000 (88%); (ii) Mali Gouvernement: MXOF 1,332 (5%); and iii) EDM SA: M XOF 2,109 (7%).

BOAD will monitor the project, on the basis of the Borrower's quarterly reports and field supervision missions.

An implementation team for this project will be constituted within BOAD, and be in charge of the due diligence. It is an internal bank team set up to coordinate the preparation, design, financing and implementation as well as monitoring and evaluation aspect of the project. The same team will be responsible for the liaison with the GCF and the subsequent reporting. The team at the bank will consist of financial analyst, environment social and gender specialist, renewable energy expert, monitoring and evaluation expert, climate change specialist, procurement specialist.

The organizational flowchart and the flow of funds are presented below:

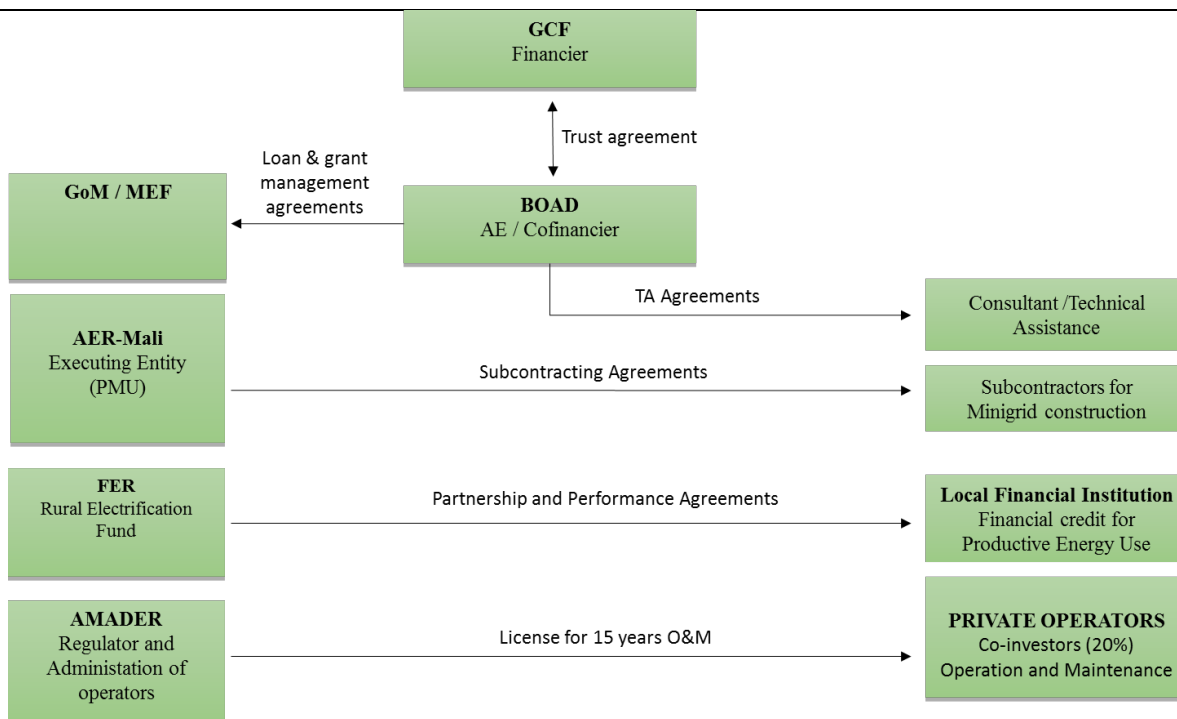


Figure 8: Organizational arrangement

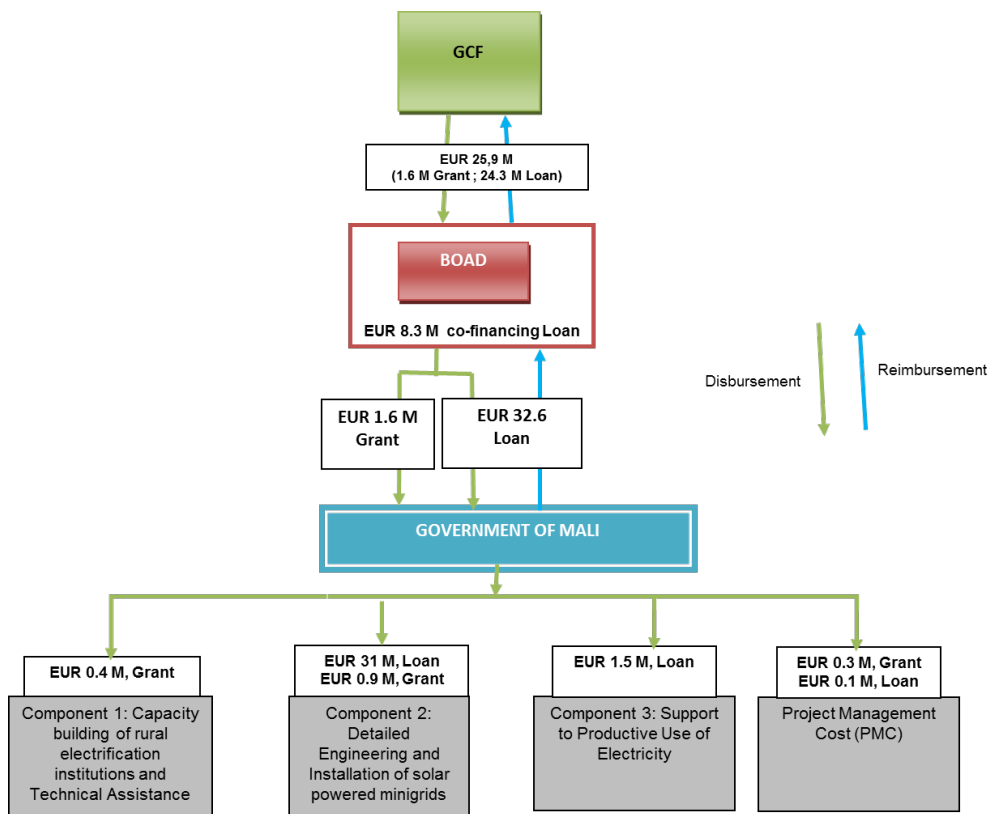


Figure 9: Global Flow of funds

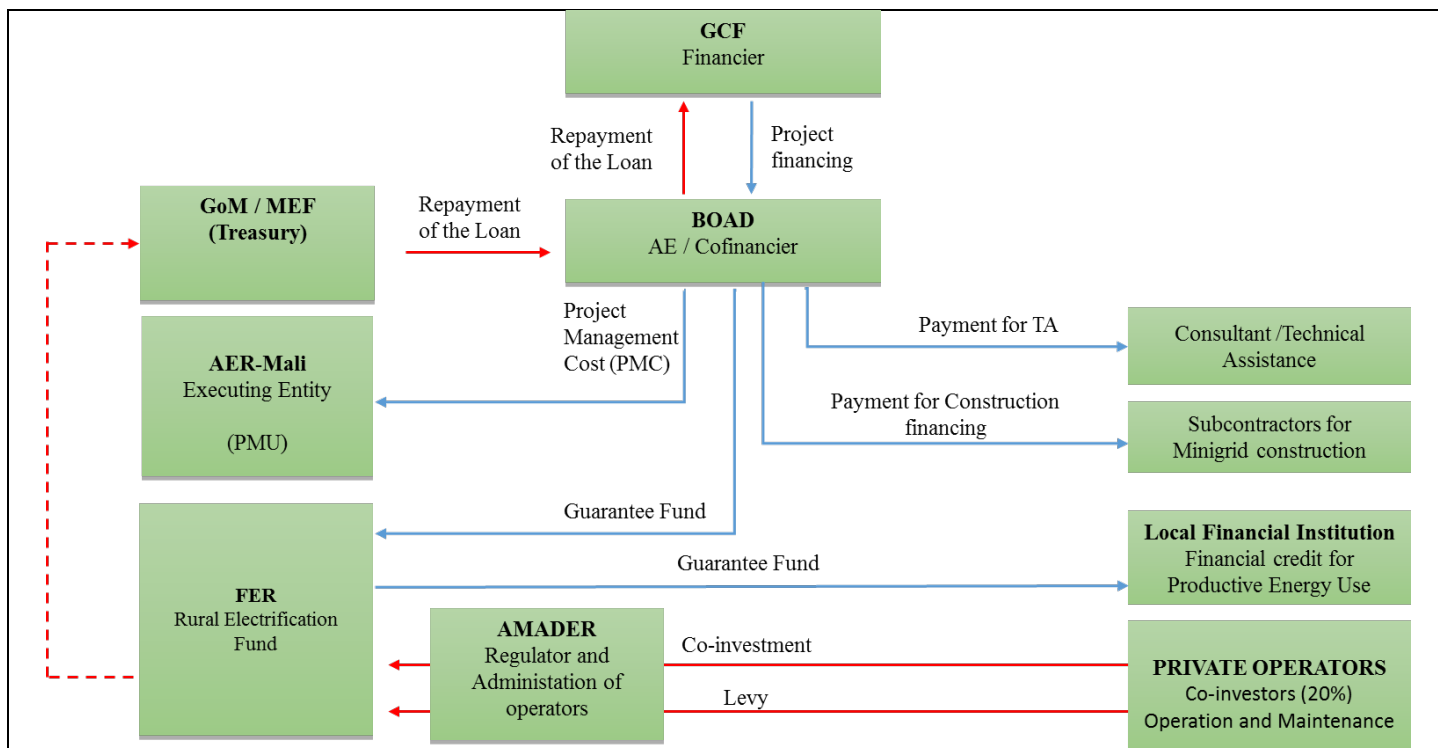


Figure 10. Detailed Flow of funds showing execution of payment and entities involved

Organizational Arrangement for Component 3

The proceeds will be managed by BOAD as implementing agency and the Rural Electrification Fund (FER) will be the executive institution.

A guarantee agreement will be signed with local financial institutions (LFIs).

A partnership and performance agreement will be signed between the FER and the eligible LFIs.

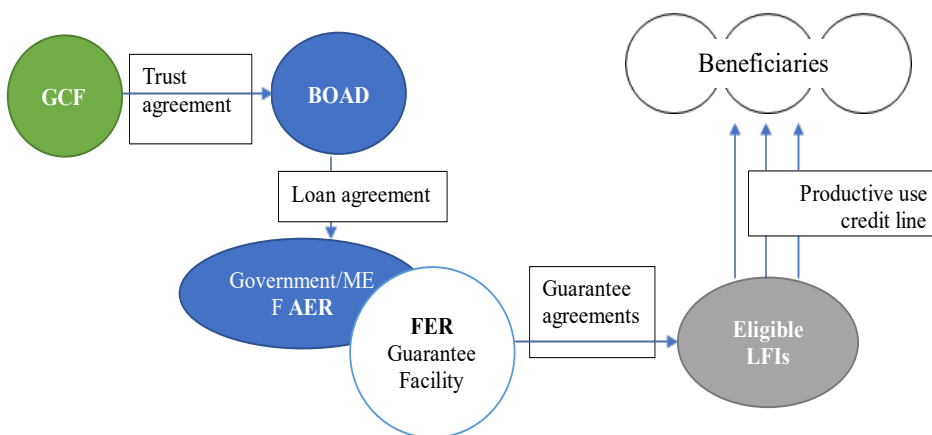


Figure 11: Organizational Chart

C.5. Market Overview (if applicable)

Describe the market for the product(s) or services including the historical data and forecasts.

According to EDM-SA data, energy security in Mali is suffering from a chronic gap characterized by:

- ✓ at least 300 MW of electricity demand to be covered immediately or in the short term on the network;
- ✓ about 200 MW of thermal power to be replaced by competitive sources of energy;
- ✓ at least 500 MW in the long term for the connection or supply of big industrial extractive and food-processing units, etc.;
- ✓ at least 150 MW to be rehabilitated on the network's production capacity, and
- ✓ a fast-expanding network to support the development of the import and/or export of electric energy.

It is very unlikely that over the next ten years, the expected expansion of the national electricity network can ensure that a significant proportion of the low-income population living in isolated areas is connected to the network. This situation is expected to create an important market for off-network electrification schemes.

Off-network, the distribution of electricity is mainly ensured by local energy service providers who operate in isolated rural areas. In 2015, the country had 160 isolated mini-networks. The production technologies used to this day basically include diesel-run units (in 2015, 96% of the electricity produced off-network was through diesel) as well as related transmission and distribution systems. The total installed capacity of the mini-networks is about 15 MW.

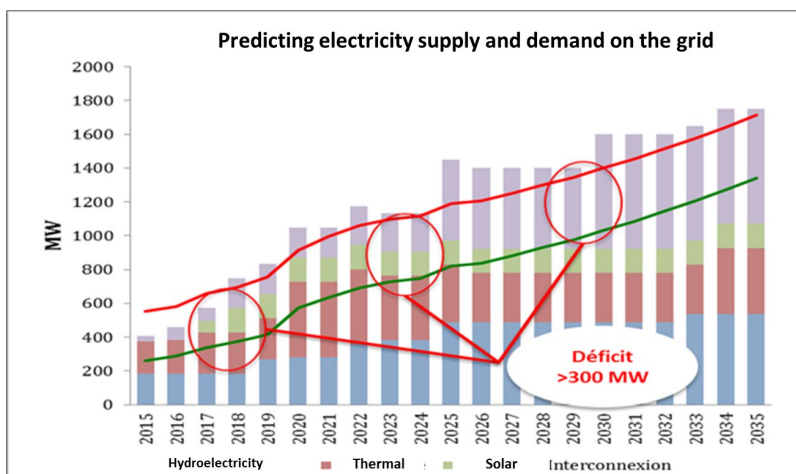


Figure 12: power generation deficit in Mali

Rural electrification in Mali is characterized as follows:

- ✓ about 70% of the Malian population live in the rural areas;
- ✓ the rate of electrification in the rural areas was 19% in 2016 against less than 1% in 2004;
- ✓ Mali has more than 10,000 rural towns;
- ✓ at least 255 towns are currently being supplied with electricity by private operators with the support of Donors and Technical and Financial Partners;
- ✓ the rural electrification systems developed by the operators are basically thermal;
- ✓ in 2015, about 131 towns were connected to diesel units, 32 towns were connected to hybrid PV-Diesel systems, 15 towns used biofuel units, 20 towns benefitted from the extension of the EDM-SA network, 43 towns were supplied by Multifunctional Platforms, 13 towns were supplied by solar kits and one town was supplied by a hydroelectric micro-plant;
- ✓ a total installed power of about 23 MW, with nearly 2 MWc PV solar

Among public sector actors involved in the energy sector, some national entities and technical services are also involved in the development of renewable energy:

Énergie du Mali SA (EDM SA), the national state-owned utility

Agence Malienne pour le Développement de l'Énergie Domestique et de l'Électrification Rurale (AMADER) aims at increasing energy efficiency and at expanding access to electricity in rural and sub-urban areas. It is also responsible for implementing the *Programme Décennal d'Électrification Rurale (PRODER)* through PPPs.

Private local energy companies and local initiatives (i.e., communities and women associations), with support from AMADER, are the drivers of Mali's rural energy access agenda. The *Fonds pour l'Électrification Rurale* has attracted almost 50 local energy service companies to the rural energy service delivery market in Mali. Most of them operate diesel-based mini-grid systems.

Autonomous producers are consumers who generate electricity for their own needs, either to cope with EDM SA failures to provide them with sufficient power or because they are outside the grid. They use diesel-powered generators.

The demand/supply projection and the cost of production show that in Mali, the cost of electricity production is high and unsustainable.

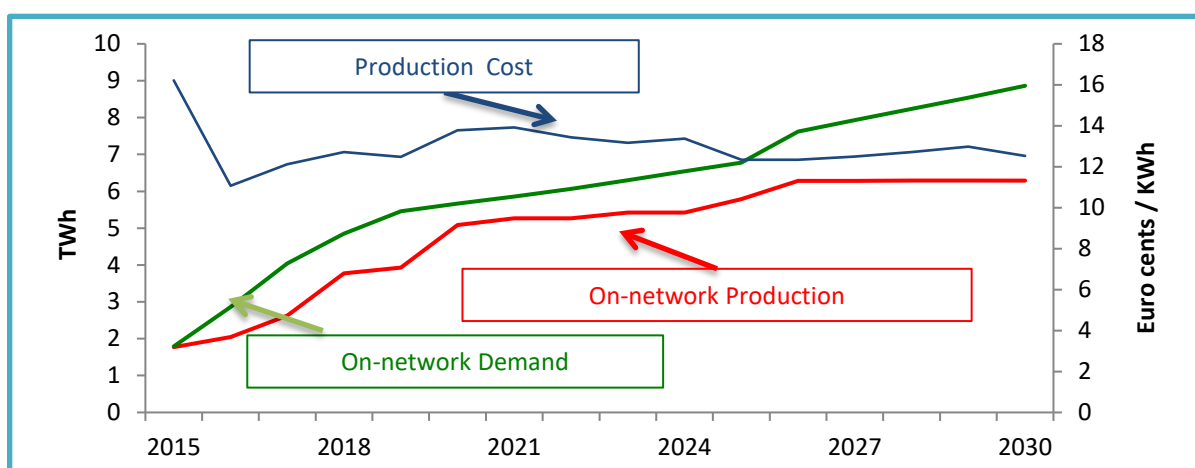


Figure 13: Demand/Supply Projection and Cost of Production in 2015

Power Tariff

General Tariff Principles

Electricity is considered to be commercial commodity, and therefore no supply to a third party may be made free of charge. Accessing such service gives rise to subscribing to a contract, which must be approved by CREE.

Equal treatment is guaranteed to all users with same consumption characteristics within the same Concession or among customers under the same lease.

Tariff regulations

- principle:** Pricing freedom for authorisation and declarations (and regulatory tariff for concessions) in compliance with provisions of contracts signed with customers.
- Sales subject to tariff regulation:**

The sales of Concessions benefiting from an exclusivity or natural monopoly are subjected to tariff regulation by a Directive from the CREE.

Regulated tariffs are price ceilings defined on the basis of the following principles:

- ✓ They are based on budgeted costs and allow the operator to cover all expenses and charges justified by operational needs
- ✓ They include an adequate rate of return that enables the Concessionaire to break even and get return on investment;
- ✓ They include price adjustment coefficients for the main cost components, making it possible to offset the effect of changes in the main economic parameters over which the operator has no control;
- ✓ They incorporate a downward adjustment term that forces the operator to increase productivity;
- ✓ They are in line with the Government's energy policy and aim to stimulate efficiency in the use of electricity and optimise the use of generation, transmission and distribution capacities;
- ✓ They shall be no discriminatory and transparent for same category of operator

Giving variations on characteristics of demand, regulated pricing scheme are defined by tariff quota periods and are reviewed by CREE directives.

Within the scope of the EDM-SA concession, there are provisions for tariff equalisation.

Tariff schedule and revision

The following tables show EDM-SA's low and medium voltage tariff and variations between 2009 and 2015

Table : Tariffs between 2009 and 2015

Tariff categories	2009 Tariff 2009 without VAT	2015 Tariff without VAT
social tariff (2-phase meters, 5 amps)		
Proportional price (CFA Francs /kWh)		
- Window 1: 0 – 50 kWh/month	59	59
- Window 2: 51 – 100 kWh/month	91	105
- Window 3: 101 – 200 kWh/month	94	120
- Window 4: > 200 kWh/month	108	130
Normal Tariff (2-phase meters > 5 A and meters 4)		
Proportional price (CFA Francs /kWh)		
- Window 1: 0 – 200 kWh/month	106	109
- Window 2: > 200 kWh/month	124	130

Public Lighting Tariff		
For the first 120 hours of the subscribed voltage	107	114
For more	74	79

It should be noted that this EDM tariff does not apply to rural mini-grids that have another tariff regime.

Despite the low level of incomes in the area, households expressed their willingness to pay the monthly electricity service charge. This willingness to pay the monthly electricity service charge by households has also been confirmed by local authorities

C.6. Regulation, Taxation and Insurance (if applicable)

Regulation related to the electricity and rural electrification sector

In Mali, AMADER is the only government regulatory authority in charge of rural electrification, setting of policies under the guidance of the Ministry of Energy.

Operating Regimes for Electricity Facilities

The Government entrusts public power service to operators under either a Public Service Permit or a Concession Agreement, issued under the conditions provided for by Ordinance No. 019/P-RM of 15 March 2000.

Any person or entity wishing to carry out Power Production activities for third parties must first obtain either a Permit or a Concession, depending on the level of installed capacity envisaged.

Other permits and authorizations are required as presented below. They will be obtained by the operators once selected via BOAD tender processes.

Table 1: Permits and Issuing Authorities

Permits and authorizations	Issuing authority
– Community authorization	Local Authority (Town Hall)
– Construction permit	Ministry of Urban Planning and Housing
– Environmental Permit	Ministry of the Environment, Sanitation and Sustainable Development
– Preliminary permit of 6 months	AMADER
– Agreement between AMADER and the Operator	AMADER
– Authorization from the Minister of Energy for a period of 15 years	Minister in charge of Energy
– Specifications for the operation and maintenance	AMADER

Import of equipment will be exempted from taxes.

Currency regime

The currency regime is the XOF, which is locked against the EUR. The project funds will be provided in EUR.

Privileges and Immunities

The GCF resources will enjoy an equal treatment with the BOAD resources in terms of privileges and immunities. This includes especially exemption from taxation.

C.7. Institutional / Implementation Arrangements

Organization of the execution

The project owner is the Government of Mali represented by the Ministry of Energy and Water Resources. The project will be executed by the Renewable Energy Agency (AER-Mali). In this regard, AER-Mali shall see to the programming, coordination, monitoring and supervision of all Project activities.

A Project Management Unit (PMU) shall be set up within AER-Mali and managed by a senior manager Project Coordinator, who must be an Electrical Engineer or other profile with experience. In addition to the coordinator, the project unit will include nine (09) other experts namely: (02 Executive Electrical Engineers or other executives, 02 Technicians, 02 Senior Construction Technicians, 01 Procurement Specialist, an Environmentalist and an Administrative and Finance manager) as well as support staff: 02 drivers, 01 Secretary, 01 courier and one (01) responsible for monitoring-evaluation. The project team will be established by a decision of the Minister of Energy and Water Resources. The PMU Coordinator's CV must first receive a no objection notice from the BOAD.

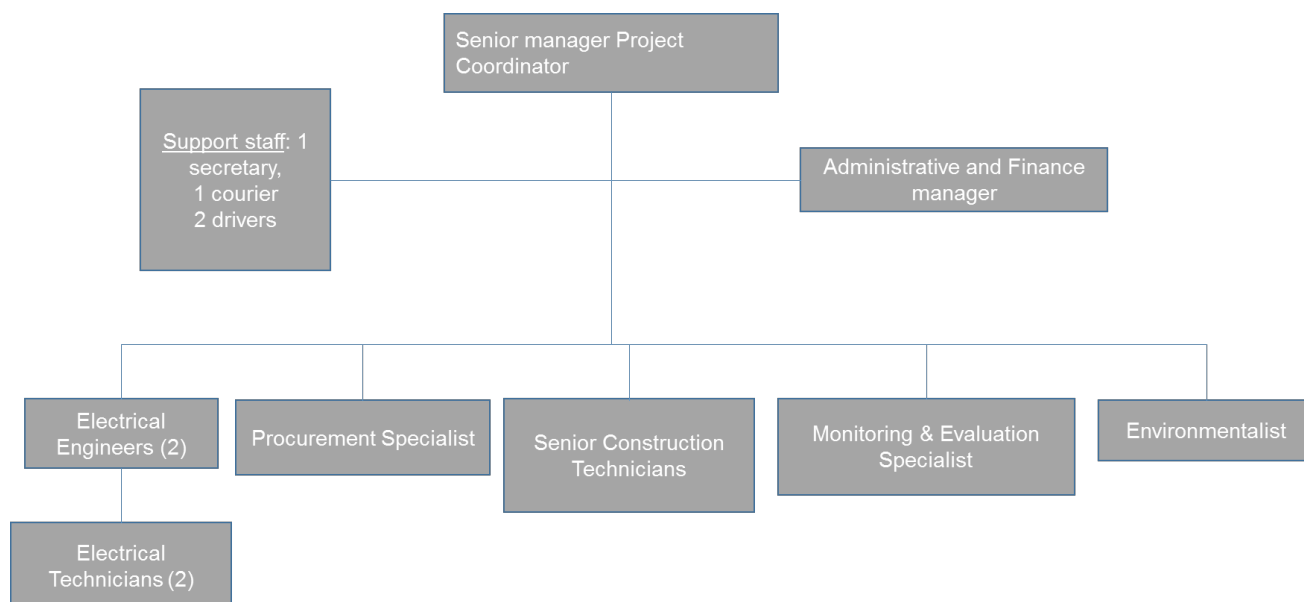


Figure14: PMU Flow chart

The PMU will coordinate the conduct of technical studies, the preparation of tender and consultation documents (in collaboration with the Consulting Engineer) as well as tender evaluation and procurement reports.

The PMU will also be assisted by a consulting engineering firm for the monitoring and inspection of works. This office will provide a monthly progress report.

The PMU staff could be reinforced through external recruitment. The project also provides for capacity building of the Unit through training.

This component takes into account the "Control and supervision of work" aspects. The work control and supervision services will include: (i) the technical and administrative follow-up of project execution; (ii) checking design notes and civil engineering reports and electromechanical equipment plans; (iii) the factory and on-site delivery of equipment, the validation of tests for commissioning and provisional acceptance of works; and (v) quality control of works and quantities implemented in accordance with Special and Technical Specifications.

This sub-component includes (i) feasibility studies as well as environmental and social impact assessment already carried

out; (ii) detailed technical engineering studies and environmental and social impact statements to be carried out for each locality; and (iii) the tender documents for construction works and the Request for Proposals for the control and supervision of construction works.

Control and supervision of the works will include: (i) the technical and administrative follow-up of the execution of the works; (ii) verification of calculation notes and plans for civil engineering and electromechanical equipment; (iii) factory and on-site equipment approvals, trial validation for commissioning and temporary acceptance of work; and (v) the quality control of the works and quantities implemented in accordance with the Particular Technical Prescriptions Manual.

The Bank will monitor the project from its headquarters, based on the Borrower's quarterly reports and field supervision mission's reports. A technical and financial audit of the project will be carried out by a consultant who, in accordance with the project specifications, will inspect works carried out and the services provided by the inspection team, check the books and accounting records.

Works will be carried out by companies. Work supervision and inspection as well as technical audit will be carried out by engineering firms. The technical and financial audit of the project will be conducted upon completion of works before final handing over. Services relating to raising awareness among the populations living in the project catchment zone will be conducted by local NGOs or associations with the support from the Local government.

Upon completion of works, a final report shall be prepared by the firm responsible for monitoring and control and submit to the Bank. The report shall provide detailed information on the technical progress of the project, its financial cost and the project execution timeline.

Project management and operation

AMADER is the regulator for rural electrification activities and will grant the concession agreement to the operators. There will be no legal agreement between AER and AMADER while AER will host the Project Management Unit (PMU).

The private operators will be in charge of assets management and operation. They will be selected on the basis of specifications defining the performance of the operation and preventive and curative maintenance, in accordance with the manufacturer's standards.

The operation and maintenance of the installations are carried out by the concessionaires, the rural electrification project management entities, and / or the associative structures of the rural communities of the localities concerned, according to the regulatory and legislative framework of the project. country. The policy of choosing a private operator aims to bring professionalism, skills and resources in some localities electrified and not connected to the network interconnected.

The dealer maintenance policy is characterized by the execution of a systematic maintenance program (preventive maintenance) and repair and repair interventions (curative maintenance). Maintenance by the operator is based on manufacturer's maintenance manuals, maintenance schedules developed by the programming unit, and statistics of breakdowns and incidents.

During the life of the contract, the operator must comply with a number of rules to ensure quality of service and customer satisfaction. In this context, he is responsible for the maintenance of the infrastructure entrusted in perfect working order, while ensuring regular maintenance. It also renews parts at the end of its life.

The operator manages the maintenance and upkeep of the infrastructures, on a daily basis, relying on light teams, present in each of the villages. Thus, three (03) types of intervention are retained:

- the daily maintenance is provided by a trained technician present in the locality through daily visits of the production facilities with information record (voltage, current, frequency, power, number of hours of operation, abnormal noise, leakage, etc.) and visual inspections of distribution lines;
- systematic, conditional or corrective maintenance interventions are provided by the regional private operator; and
- for breakdowns, overhauls or major repairs, the operator may call on the supplier if necessary.

Local authorities and national energy agencies control the quality of the facilities and service to ensure that the private operator manages the infrastructure.

A levy of EUR 0.02 will be collected from the private operators by the Rural Electrification Fund (FER).

Please provide a project/programme implementation timetable in [section I \(Annexes\)](#). The table below is for illustrative purposes. If the table format below is used, please refer to the activities as numbered in Section H. In the case of outputs, please mark when all the required activities will be completed.

[illegible]

D.1. Value Added for GCF Involvement

Energy access and energy savings are key to the development of Mali and in the West Africa region. Given that the energy mix in the target region already heavily depends on fossil-fuel based electricity, further adding high-emission diesel generators to power SMEs is not an option from a climate perspective. Low-carbon energy production can be achieved by scaling up the use of renewable energy resources. GCF for this programme is critically important and it provides significant value-added above the business as usual situation. Funding from government and development partner institutions is not sufficient to meet the vast demand for clean energy access. As a matter of fact, it remains a major challenge throughout West Africa to meet the installing, operating and maintaining cost requirement, not to mention the equally important costs for training and capacity building for local stakeholders, despite the obvious social and environmental benefits of renewable energy solutions.

A key reason to seek GCF financing is to help Mali scale up investment in renewable energy. Without such an incentive, the country would continue to suffer from the slow pace of investment in the renewable sector. GCF financing is critical to allow the successful structuring of finance for this project so as to address some of the barriers as faced by the private sector investors, such as by promoting the establishment of a conducive regulatory framework to encourage private sector involvement in renewable energy investment in Mali. The financing from GCF is expected to help reduce the risks of investment. Concessional financing from GCF can offset the high cost related to the barriers and risks as described in the project FSR.

The Government of Mali has been struggling to mobilize the necessary financial resources. Without GCF financing, the government will have to borrow from financial institutions at market rates, which are still very high considering their capacities, but above all faced with the urgency of the problems to be solved. Hence the need for external support becomes more urgent if community resilience is to be achieved.

Given the large investment needs in the sector, a significant volume of investments into the renewable energy markets is required. Currently, only the GCF is able to provide such finance within the given timeframe. BOAD's strategic objective is to increase its portfolio of investments in the renewable energy sectors, but so far this has been low because of the lack of alternative sources of public sector funding available. The involvement of the GCF is required as it will allow a combination of concessional lending and grant for technical assistance required to achieve the objectives of the project and the consequent paradigm shift in Mali.

D.2. Exit Strategy

The Renewable Energy Fund will be underpinned by a legal framework and an annual budget from Government and will be a real financial institution able to borrow, issue guarantees, build loans, and equity in order to coordinate the mobilization on the international market, of the financial resources necessary to the financing of the investments by the agencies in charge of promoting the energy programs renewable resources.

The project will build capacity of the national actors (public, private and financing institutions) engaged in rural electrification and renewable promotion. With enhanced understanding of the RE sectors and their profitability, MFIs are expected to continue providing finance to these businesses even without the project participation. It is expected that there will be a greater demand for financing energy demand projects and that MFIs will start using their own resources for climate lending.

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's [Investment Framework](#), should be addressed where relevant and applicable. This section should tie into any request for concessionality made in [section B.2](#).

E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation / adaptation impact potential

As mentioned before, traditionally, the growing energy access demand has been met through fossil fuels. This project however provides a total shift in energy production by promoting the use of renewable energy instead of continuous addition of fossil fuel based generation

The project will result in a real and visible paradigm shift in Mali Rural electrification towards low-carbon sustainable development. The project directly addresses the second largest-emitting sector in the country – i.e. energy generation – and implements actions specifically recommended in the Nationally Determined Contribution and the Second National Communication to the UNFCCC.

The project is expected to result in direct emission reductions of 41 089 t CO₂ eq/year by installing solar minigridd in 70 communities of which 32 871 t CO₂ eq/year for 50 communities for Tranche 1.

E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

GCF core indicators	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (Mitigation only)	Annual	41,089 t CO ₂ eq (32,871 tCO ₂ eq/year for Tranche 1)
		Lifetime	1,027,227 t CO ₂ eq (821,782 t CO ₂ eq for Tranche 1)
	- Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience); - Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Total	310,698 (283,777 for tranche 1) (50% women)
		Percentage (%)	

Other
relevant
indicators

Examples include:

- Expected increase in the number of households with access to low-emission energy: 31 000 (28,778 for tranche 1)

Describe the detailed methodology used for calculating the indicators above.

The methodology chosen is the AMS I.L.

(https://cdm.unfccc.int/methodologies/documentation/meth_booklet.pdf#AMS_I_L) as approved by UNFCCC CDM Executive Board for CDM projects. It is considered as the most suitable methodology with robust and conservative means to estimate the mitigation impact for this project in Mali, as it provides a solid method for the quantification of the emission reduction outcome and has been approved by a constituent body under the UNFCCC:

- This methodology is applicable to electrification of a community achieved through the installation of renewable electricity generation systems that displace fossil fuel use, such as in fuel-based lighting systems, stand-alone power generators, and fossil fuel based mini-grids.
- This methodology is applicable to greenfield individual, renewable energy system projects or mini-grid activities.
- This methodology is applicable in situations where consumers that were not connected to a national/regional grid prior to project implementation are supplied with electricity from the project activity. It is also applicable to situations where a fraction of consumers that are supplied with electricity from a fossil fuel based mini-grid prior to the implementation of the project are now supplied with electricity from the project activity.
- At least 75 per cent (by number) of the consumers connected to the project renewable electricity generation system(s) shall be households

Baseline calculation

Two parameters are required to be known to determine the baseline:

- The amount of renewable electricity utilized by the consumers served by the project renewable electricity generation systems;
- The number of consumers supplied with renewable electricity by the project activity.

Baseline emissions for the entire project activity are calculated as:

$$BE_y = BE_{55,y} + BE_{250,y} + BE_{250 \text{ plus},y}$$

Equation (1)

Where:

BE_y	=	Baseline emissions in year y (t CO ₂)
$BE_{55,y}$	=	Aggregate baseline emissions for consumers that consumed equal to or less than 55 kWh of renewable electricity from project renewable electricity systems in year y (t CO ₂)
$BE_{250,y}$	=	Aggregate baseline emissions for consumers that consumed more than 55 kWh but equal to or less than 250 kWh of renewable electricity from project renewable electricity systems in year y (t CO ₂)
$BE_{250 \text{ plus},y}$	=	Aggregate baseline emissions for consumers that consumed more than 250 kWh of renewable electricity from project renewable electricity systems in year y (t CO ₂)

Emission reduction calculation:

$$\text{Emission reductions (ER)} = \text{Baseline Emissions (BE)} - \text{Project Emissions (PE)}$$

Since project activity uses solar energy, PE= zero, therefore (ER) = BE.

Estimated Power Generation

Communities Type 1	919800 kWh
Communities Type 2	2299500 kWh
Communities Type 3	4730400 kWh
Communities Type 4	2628000 kWh
Total yearly generation	10 577 700 kWh
	10 578 MWh
Estimated Power Distributed	8 991 MWh

Therefore total emission reduction (ER) per year:

	31000	Energy (MWh)	Emission Factor
Total households			
Proportion of Households that consume less than 55 kWh a year	60%	5394,627	6,8
Proportion of Households that consume less than 250 kWh a year	30%	2697,3135	1,3
Proportion of productive use with consumption above 250 kWh per year	10%	899,1045	1,0

Emission Reduction = 41,089/year

Lifetime of the sub-projects is 25 years

Total emission reduction (ER) over the lifetime of the projects:

41,089 tCO₂ eq X 25 years = 1,027,227tCO₂ eq

For Tranche 1: ER = 32,871 tCO₂ eq/year, and 821,782 tCO₂ eq for the entire lifetime of the project.

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/programme investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

Mali has over 74% of the population living off-grid and without modern energy access. In the rural areas where this project targets, the proportion of the population without electricity access is 91%. The country has been making a calculated shift of attention to off-grid electrification strategies. In effect, numerous policies and regulations have been developed. Incentives such as customs rebates are in place. However, there has not been significant to-scale investments off-grid.

This GCF project seeks to remove these barriers: it has a dedicated component on institutional & technical capacity building and Technical Assistance. It will support the revision/updating of the mandates of agencies and institutions actively involved in RE. Having had a legacy of not-so-successful off-grid interventions, this project has the ambition to demonstrate working models that factor pertinent factors. The learning sub-component will ensure best practices and lessons are well-documented and communicated effectively to stakeholders in the emerging sub-sector.

The project will develop mini-grids that outnumber the existing installations and represent a significant ratio of the total Installed capacity of Mali. This way, the confidence in the community-run energy systems will rise and thereby permanently change the investment environment.

The project is founded on existing experiences from historic and ongoing projects and programmes. All stakeholders will be involved. It will therefore contribute sound and pragmatic additions and/or amendments to the existing regulatory framework and policies.

The promotion of demand to increase productive uses of renewable energy will demonstrate the competitiveness of isolated grids with respect to grid tariffs so as to influence the national tariff regime (for both on-grid and off-grid).

A successful scale up of Mali's rural electrification project combined with the greening of Mali's rural energy mix will provide lessons to be learned and best practices for rural energy agencies in the sub-region in the framework of the African club of rural energy agencies. BOAD is committed to developing a case study replicable in the 8 member countries and across the WAEMU countries.

In Mali, AMADER is the only government authority in charge of rural electrification regulation, setting of policies under the guidance of the Ministry of Energy.

At present the government of Mali is in the process of revising the current regulatory framework

The theory of change of the is described below, based on activities that support enabling environment through strengthening policy regulation and national institutions involved in rural electrification, private sector engagement in clean energy thanks to affordable financing that makes solar investments attractive and viable. The project intervention will result in a paradigm shift in terms of avoiding substantial GHG emissions for the country.

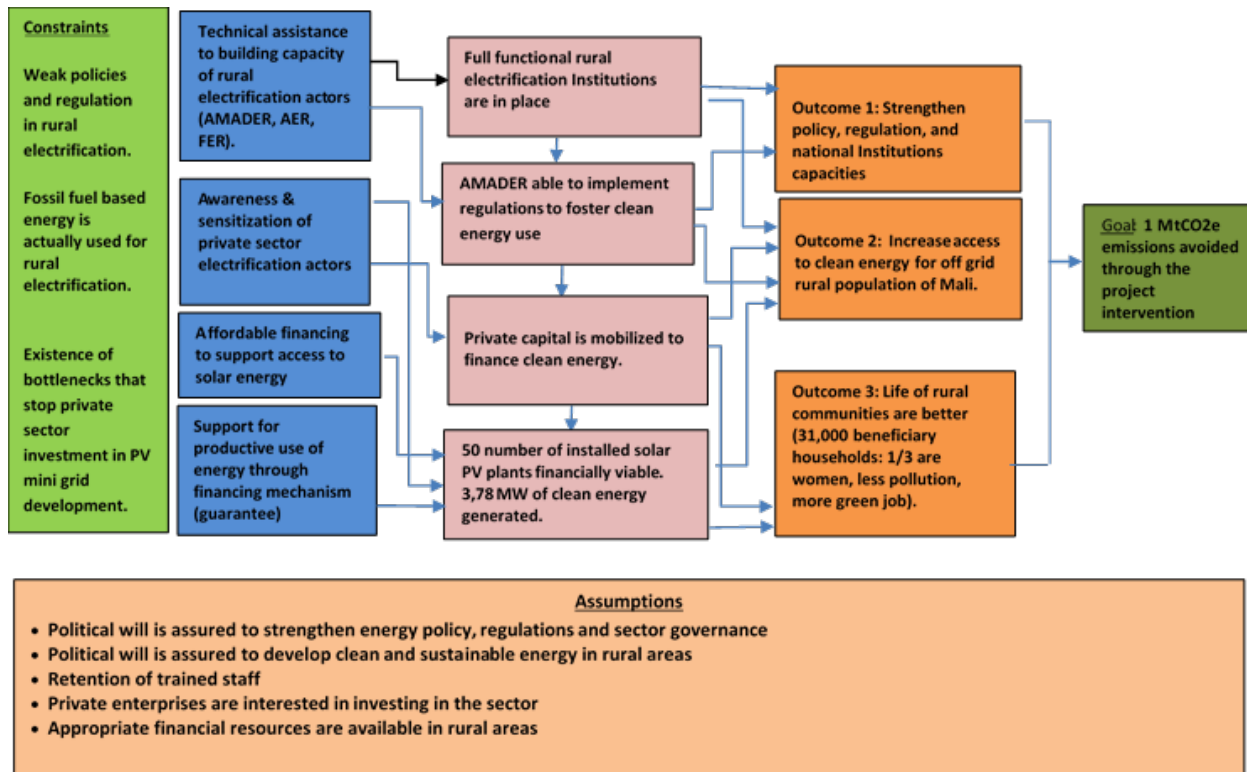


Figure15: Theory of change

E.2.2. Potential for knowledge and learning

Describe how the project/programme contributes to the creation or strengthening of knowledge, collective learning processes, or institutions.

The project is creating a unique partnership consisting of multiple stakeholders (public, private and NFI) who will work together in a collective learning process. Transfer of expertise and skills is a core objective of this project (component 1). It will mainly benefit local NFI, while potential clients active in promoting sustainable energy will also receive support. In this context, the following learning and know-how transfers and benefits will be achieved:

On the NFI's side, the project will enhance their understanding of evaluating RE investments and change the risk perceptions towards these new financing opportunities.

The project will deliver trainings to public institutions (AER, AMADER) staff on climate change issues and RE technologies development.

Potential MSME clients will also benefit from technical support on how to structure climate business plans to make them bankable.

E.2.3. Contribution to the creation of an enabling environment

Describe how proposed measures will create conditions that are conducive to effective and sustained participation of private and public sector actors in low-carbon and/or resilient development that go beyond the program.

The project will contribute to creating an enabling environment for RE investments through targeted capacity building, awareness raising activities and connecting a diverse range of private sector stakeholders under component 1. The technical assistance activities will achieve a framework in Mali that enables AMADER to unlock private sector resources on rural electrification activities while also having their participation on O&M activities. This will shift from current paradigm where public and private are competing and pave the way for a PPP model.

It will contribute to structuring the solar based rural electrification market through financial mechanism such as financial guarantee to remove the barrier of lack of a financial service adapted to support Productive Energy Users and de-risk loans demands for rural areas productive users of energy.

E.2.4. Contribution to regulatory framework and policies

Describe how the project/programme strengthens the national / local regulatory or legal frameworks to systematically drive investment in low-emission technologies or activities, promote development of additional low-emission policies, and/or improve climate-responsive planning and development.

GCF financing is critical to allow the successful structuring of finance for this project so as to address some of the barriers as faced by the private sector investors, such as identifying and establishing a conducive regulatory framework to encourage private sector involvement in renewable energy investment in Mali. The financing from GCF is expected to help reduce the risks of investment. Concessional financing from GCF can offset the initial high cost related to the barriers and risks as described in the project FSR.

The current regulatory framework allows concession agreements; however, this was never operationalized at scale. The standard O&M contracts will contribute to the regulatory environment by setting a framework that can be replicated and which operationalizes the regulation.

This project will contribute to having a fully functional rural electrification agency in place and produces contracts and supervision reports in line with industry standards.

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

The environmental and social benefits, including gender-sensitive development impacts are as follows:

Environmental cobenefits:

- reduction of air pollution due to fossil fuel consumption and improvement of air quality thanks to promotion of renewable energies;
- reduction of greenhouse gas emissions;
- reduction of pollution by waste oils at thermal power plants;

Socio-economic and gender-sensitive development impacts

- increased socio-economic activities,
- Improvement of the health and hygiene conditions of local populations;
- Improvement of the living conditions and comfort of local populations;
- access to public lighting and improvement of security conditions;
- increase in the duration of supply of electricity;
- increase of the service and the access rate for the safety of people and goods;
- better conservation of pharmaceutical and food products;
- improvement of the permanent drinking water supply of the populations;
- lightening women's work,
- improvement of household income levels and creation of income-generating activities;
- improvement of the literacy rate;
- development of the emergence of manufacturing and processing industrial units;
- opportunity for local people to benefit from capacity building actions.
- The focus on gender in ownership and management of the energy services centres will ensure inclusion of all gender and youth. So the improvement of livelihoods and other development benefits will be shared fairly.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

Describe the scale and intensity of vulnerability of the country and beneficiary groups, and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc).

In Mali, most of the energy supplied comes from biomass (78%). It represents the largest source of energy for the majority of the population, while fossil fuels contribute 21% of the energy supply and 1% hydroelectricity (excluding electricity trade). All of these factors are vulnerable to climate and economic shocks. Due to the impacts of climate change, such as drought and decreasing rainfall, wood energy is suffering from increasing cutting, soil degradation, etc., and the hydroelectricity of the energy deficit, water for its operation. As for fossil fuels, their fluctuations strongly impact Mali's trade balance, which imports all its consumption. This situation of poverty is accentuated in rural areas which use mainly biomass.

Mali is particularly vulnerable to climate change. The most plausible scenarios foresee a rise in temperatures and a decrease in precipitation in all the localities of the country, with possibly serious repercussions on agriculture, forests, health, energy and water, among others. According to the National Adaptation Program of Action (NAPA) Mali, energy is considered one of the sectors requiring urgent adaptation measures to climate change. In terms of vulnerability, climate variability particularly affects the hydroelectricity subsector. Under the climate scenario developed under the NAPA, a 1% drop in flow would result in a 1.3 million kWh reduction in electricity generation. In particular, the mitigation of the impacts of climate change on hydroelectric production and biomass production will be at the heart of any strategy for energy development and poverty reduction.

The majority of the population of Mali (about 64%) lives in rural areas but the urban population is growing at a constant rate of 5% per year. This demographic growth is a major concern for the future of the country because it implies a sharp increase in basic needs, in particular energy. Mali's economy is dominated by the primary sector, which contributes to 36.5% of GDP and employs almost 85% of the working population. The electricity and water subsector contributes to 1.91% of GDP. Mali imports all of its fossil fuel needs, making it highly sensitive to price volatility.

The results of the Survey of Consumption, Praetorship and Welfare of the National Institute of Statistics of Mali reveal an incidence of poverty estimated at 11.1% in Bamako against 46.6% and 52 respectively, 8% in other cities and rural areas.

This level of poverty accentuated the vulnerability of rural communities through the following factors:

- At the level of education, the literacy rate in relation to the 15-year poverty level in rural areas is 17.9% compared with 52.3% in Bamako and 44% in other cities;
- at the health level, because of the costs deemed too high by the poor (74.9%), they are consulted by traditional healers or community health centres that do not have sufficient resources,
- at the level of access to energy, 92% of poor rural households use wood as a source of cooking and only 23.3% have access to electricity compared to 74% of poor households in Bamako.

E.4.2. Financial, economic, social and institutional needs

Economic co-benefits

The project will create 600 permanent jobs when fully implemented and another 1,500 temporary jobs during the construction period. With the skills attained from the installation period, the temporary workers will be available for similar subsequent projects in the localities with the most need.

Assuming each site would be the equivalent of a 20kVA diesel generator, an all RE system would avoid 21,000 litres of diesel annually. This translates to 1,470,000 litres annually. This would save the country upwards of EUR 13,41 million annually. This would in turn improve the fiscal balance.

Social co-benefits

The absence of modern energy hinders community development; climate change complicates the situation. Poor lighting & indoor air pollution increases the prevalence of health complications including respiratory tract infections and heart complications (16,900 people die in Mali annually due to indoor air pollution, WHO, 2012), which dramatically reduce life expectancy and strain the national health budget.

The introduction of modern energy in the rural communities will lead to various social benefits including:

- better social services such as medical services and education
- increased safety in communities following the introduction of street/public lighting
- better health of especially women and girls following improved indoor lighting and decreased air pollution; as well as reduced distances to cover in search of energy services and/or resources
- better sanitation and increased availability of clean safe drinking water

Increased access to and improved use of basic social services, such as education and health care (lighting/heating/cooling of schools and health centers through solar panels), and increased access to water in rural areas (through solar PV pumping systems). Solar home systems also prevent health issues related to indoor use of firewood and improve household safety by decreasing domestic accidents caused by candles and kerosene lamps.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

Please describe how the project/programme contributes to country's identified priorities for low-emission and climate-resilient development, and the degree to which the activity is supported by a country's enabling policy and institutional framework, or includes policy or institutional changes.

The GoM's vision and targets have been formulated in key policy papers, namely: National Energy Policy (2006); National Strategy for the Development of Renewable Energies (2006); National Strategy for the Development of Biofuels (2006); and the National Energy Sector Policy Letter (2009-2012).

BOAD has a strict focus on ensuring that not only do countries own and lead all projects financed but also that the beneficiary community needs drive the choice of interventions. This project will take this approach ensuring that communities have full control of the systems created while at the same time supporting the country to have a proper policy environment for this and similar projects in future to thrive.

Mali's NAMA aims at reduction of GHG emissions by 1,285,034 tCO₂ per year through renewable energy production (e.g. with hydro, wind, biomass and solar PV). The UNFCCC lists of 19 priority projects in Mali contains 2 projects that are directly related to this GCF project: Contribution to barrier removal for the promotion of the use of solar energy in Mali, and promotion of income-generating activities and development of mutual assistance.

Mali's National Adaptation Programme of Action, completed in 2007, identifies Agriculture and Health as the most vulnerable sectors, followed by Fisheries, Energy and Water Resources.

Implementing these mitigation and adaptation activities are slower than desired because of various constraints include absence of capital and weak institutional capacity and experience. However, the country has instituted the agencies and structures required for actualisation of these climate action plans/strategies.

E.5.2. Capacity of accredited entities and executing entities to deliver

Please describe experience and track record of the accredited entity and executing entities with respect to the activities that they are expected to undertake in the proposed project/programme.

Please refer to section C4 for information about the Executing Entity and Responsible Parties

The West African Development Bank (BOAD) is the common development finance institution of the member countries of the West African Monetary Union (WAMU). It was established by an Agreement signed on 14 November 1973, and became operational in 1976. Member countries include Benin, Burkina, Côte d'Ivoire, Guinea Bissau, Mali, Niger, Senegal, and Togo.

BOAD is an international public institution whose purpose, as provided under Article 2 of its Articles of Association, is to promote the balanced development of its member countries and foster economic integration within West Africa by financing priority development projects

The Bank's financial support has made it possible not only to increase power generating capacities in the member countries, but also to erect more than 3,527 km of interconnection lines among them, and to reinforce as well power generation, transmission and distribution facilities on the interstate grids for a nominal power of over 1,350 MW.

As at 30 September 2016, the aggregate amount of funding provided by the Bank in the energy sector stood at XOF 863.9 billion excluding the Energy Development Fund

- BOAD provides financial assistance to WAEMU member countries to support their development efforts. Such assistance targets various areas, including the organization of workshops, conferences or seminars, the training of government officials and the implementation of social projects.
- The Bank also grants interest rate subsidy for non-commercial public sector loans to member countries. Also, BOAD participates in various roundtable meetings with a view to financing national development programmes in the member countries.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

During project preparation, several project stakeholders were consulted on the basis of the potential impacts analyzed in order to gather their points of view. The consultations took place from 02 to 17 May 2018 in the regions of Sikasso, Segou, Koulikoro and Kayes. They mainly concerned local elected representatives, civil society actors and local populations. The methodological approach adopted is the participative approach: meeting information, exchange and discussion around the project. The exhaustive list of the villages consulted and persons met is in Appendix.

Key points discussed

To collect the population opinions, the following points were discussed after presentation of the project by the consultant.

- Possible sites for the implementation of equipment;
- Advantages of such a solar equipment project for them;
- Special concerns or possible fears for a successful project;
- Experiences or have they heard about this kind of solar electrification projects;
- Project benefits for village women and youth;
- Training needs for equipment maintenance or other;
- Suggestions and recommendations for the project.

Furthermore, the national regulator for rural electrification (AMADER) and the the Private operators' organization were fully engaged. The 20% of investment cost as parallel financing demonstrates strong support from the private sector for the proposed project.

NDA was also engaged and provided no-objection letter (see appendix)

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers; providing the least concessionality; and without crowding out private and other public investment.

Component 1 (capacity building) is not revenue-generating in any meaningful sense.

Component 2 has revenue-generation aspects. Therefore, detailed financial and economic analyses have been conducted.

The GCF support to component 3 (financial guarantee mechanism) is designed to ensure that productive use for electricity is implemented and therefore there is enhancement of the consumers' ability to pay when generating income with electricity, and as a consequence there is an enhancement of the commercial viability of mini-grids, which as a consequence results in GCF investment being more sustainable

The GCF grants will be augmented by considerable loans provided as co-finance by BOAD. Therefore, the project is proposing a package for investors consisting of a mix of grants, BOAD loans.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

Please provide the co-financing ratio (total amount of co-financing divided by the Fund's investment in the project/programme) and/or the potential to catalyze indirect/long-term low emission investment.

Co-financing details are provided in Section B.2.

The co-financing ratio of the entire project is 3:1. The private sector's contribution to the investment for the assets is 20%. If further investments in renewable energy as a result of creating an enabling environment through institutional capacity building and technical assistance are taken into account, then the co-financing ratio, including leveraged funds from private investors for another 10.000 off-grid rural communities will be much higher.

E.6.3. Financial viability

FINANCIAL ANALYSIS OF THE PROJECT

Methodology and assumptions

The criteria used is the cost price per kWh and the valuation is made on constant basis. The study covers a period of 15 years of operation. It was considered a unique model for financial analysis, corresponding to the electrification of new localities by a system including a photovoltaic solar field and a low voltage mini grid. The methodology used is the computation of expected cost of operating the mini photovoltaic power plants compared to the use of thermal production source of isolated plants of private operators in rural areas. Thus, a cost per kWh was calculated taking into account the cost of the project's power plants and mini grids over their entire life cycle (construction, operation, maintenance and renewal) in relation to the net quantity of power produced over the same period.

Therefore, the financial projections of the model used were based on the following assumptions:

- that the facilities start operating from 2021 and their technical lifetime is 30 years for the photovoltaic solar field and the mini-grids;
- that battery and inverter life is 8 and 15 years, respectively;
- that depending on the size of villages, four (04) types of solar power plants shall be considered: 30 kWp, 50 kWp, 80 kWp and 150 kWp;
- the various photovoltaic solar fields have a total power of 4 830 kWp;
- total annual production is estimated at 14,103,600 kWh;

- that technical losses of production and distribution networks are estimated at 7.5% and 8%, respectively for photovoltaic fields and mini distribution grid;
- that the price of electricity is EUR 0.24/kWh (XOF156 FCFA/kWh) to which will be added a levy of EUR 0.02, leading to a tariff of 0.26 (XOF 174/kWh);
- that photovoltaic fields will run for 8 hours and batteries for 6 hours;
- that the operating costs of photovoltaic fields and mini distribution grid are 10% of sales turnover;
- that the maintenance cost of photovoltaic fields and mini distribution grid is estimated at 3.5% of the turnover of the power sold;
- that the cost of facilities is XOF10.8528 billion for the hybrid solar systems (photovoltaic solar field + mini grids + connections + batteries + inverters + spare parts + training);
- the average cost for the transmitting of power from isolated plants operated by private operators in rural Mali is about 350 CFA Francs/kWh;
- that the periodic expenses for replacing defective equipment are estimated at XOF5.423 billion for batteries;
- that the operating period of the project extends from 2021 to 2035 and the residual value of the investments amounts to XOF6.58 billion for the photovoltaic solar fields and the mini-grids.

Tariff per kWh

The above assumptions show that the tariff will be EUR 0.26 (XOF 174 /kWh) against an average (current) cost of XOF 235 /kWh for isolated groups (from private operators' generating park) supplying some villages far from the electricity grid.

Financial viability for operators

Based on these assumptions, and considering the private operators' contribution being 20% of the assets value, the project IRR at an electricity price of XOF 156/kWh would be 18%.

Economic analysis of the project

Methodology and assumptions

The criteria used for economic appraisal of the project is the Economic Rate of Return (ERR) determined according to the economic costs and benefits method. It has been retained that the economic costs are equal to 85% of the financial costs to take into account the transfers contained in the costs excluding taxes. The assumptions for computing the economic benefits are same as those set out in the project financial appraisal. However, for economic analysis, the cost used is 85% of the total investment cost.

Economic Rate of Return

Based on the above assumptions, the project's economic rate of return, presented in Annex Excel spread sheet, is

16.61%.

Sensitivity tests

The sensitivity test of the project is based on assumptions of a 10% increase in investment costs, a 10% decrease in benefits and a one-year delay in project execution. The results are as follows:

	Project	10% increase in investment costs	10% decrease in benefits	a one-year delay in project execution
ERR	16.61%	16.00%	15.82%	15.98%

The appraisal shows that the ERR is appreciably high, which confirms the economic viability of the project. However, the AER-Mali will have to ensure that deadlines are respected when executing the project and the Consulting Engineering Firm and companies are recruited on time in order to guarantee the beneficiary populations all the benefits thereof.

Financial viability for operators

The government is the borrower of GCF proceeds and will use the proceeds as 80% subsidies to the mini-grids, through construction. Private sectors operators will be selected based on 20% equity contribution. The financial analysis gives a return of 18% to the operators, based on their 20% expected contribution.

E.6.4. Application of best practices

A solar photovoltaic (PV) micro-plant comprises four large parts; namely: i) the solar field; ii) the storage system; (iii) the regulatory system; and iv) the conversion system.

Regarding the solar field, there are four (04) types of solar panels: flexible, amorphous, monocrystalline and polycrystalline.

- Flexible solar panels: These panels are designed for so-called nomadic applications: to charge mobile devices, camping, hiking, boating. The average yield is about 9%;
- The amorphous solar panels: These panels have the palm of the report quality / price; they work very well even with a weak luminosity. The average yield is about 7%;
- Monocrystalline solar panels: These panels are currently the most on the photovoltaic market, they are real generators of energy by any time. The average yield is between 18 and 21%;
- Polycrystalline solar panels: These panels are used when direct sunlight is guaranteed, they are generally used to create large photovoltaic plants. The average yield is about 15% to 18%.

The tender will look for crystalline panels.

As far as storage is concerned, there are several types of batteries, the main ones being acid lead, nickel-cadmium and Silicium batteries.

E.6.5. Key efficiency and effectiveness indicators

GCF core indicators	Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)
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(a) Total project financing	EUR 34.25 million
(b) Requested GCF amount	EUR 25.97 million
(c) Expected lifetime emission reductions overtime	821,782 tCO ₂ eq
(d) Estimated cost per tCO₂eq (d = a / c)	EUR 41.7 / tCO₂eq
(e) Estimated GCF cost per tCO₂eq removed (e = b / c)	EUR 31.6/ tCO₂eq

By the end of the project, 224,776 MWh will have been consumed, or 8 991 MWh per year over the expected life of 25 years of equipment. This scenario does not include any allocation for additional mini-grids that could be installed over the life of the project taking advantage of the momentum created by the project.

Thus, in the framework of the hypothesis of a strong interest aroused for the solar-based mini-grids for rural electrification in connection with the execution of the projects and taking into account the favorable environment for the investment that the project will have created, it is highly likely that more mini-grids of this type will be built during the 10-year period after the project, exceeding several times the number installed during the project implementation period. This is especially true given the donor's interest in promoting scaling up in the event of successful results under this project.

Thus, indirect emission reduction estimates after the project, based on the conservative scenario of the status quo and a causal factor of 80% considering 10,000 villages to be electrified in Mali, may be of the order of magnitude 117 397 359 tonnes of CO₂ avoided. In total, the project will have contributed to 118 424 586 tonnes of CO₂ avoided.

The project is considered to be highly cost-effective, providing 1 027 227 tCO₂eq of emission reductions and 821,782 tCO₂eq for tranche 1 at a cost to the GCF of EUR 31.6/tCO₂eq. This is considerably lower than the social cost of carbon estimated as approximately USD 40.

Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)

Component	Activity	Tranche 1				
		Indicative cost EUR	GCF funding		BOAD Co-financing	
			EUR	Financial instrument	EUR	Financial instrument
1. Capacity building of rural electrification institutions and Technical Assistance	Activity 1.1 Training, raising awareness of stakeholders and dissemination of project activities/results	164,645	164,645	Grant		-
	Activity 1.2 Communicating project results and lessons learned	164,645	164,645			
	Activity 1.3 Capacity building of the Rural Electrification Fund (Fonds d'Electrification Rural -FER)	98,787	98,787			
	Activity 1.4 Technical Assistance on regulatory framework and on Procurement and Contracting of minigrid construction and O&M companies	59,272	59,272			
2. Detailed Engineering and Installation of solar powered mini-grids	Activity 2.1 Preparation of detailed Engineering Studies	853,214	853,214	Grant		-
	Activity 2.2 Supply installation and commissioning of equipments	31,007,376	22,821,475	Loan	7,891,892	Loan
	Activity 2.3 Implementation of environmental and social measures	294,009	-	-	294,009	
3. Support to Productive Use of Electricity	Activity 3.1 Financial guarantee	1,500,000	1,500,000	Loan		-
Project Management Cost		404,269	306,575	Grant	97,694	Loan
Indicative total cost		34,252,208	25,968,612		8,283,595	

The private sector will provide additional EUR 8.5 M at the whole project level and EUR 6.2 M for Tranche 1 as parallel financing, which corresponds to 20% of the investment for the installation of the minigrids.

Actor	Amount (EUR)
GCF	25.9 M
BOAD	8.3 M
Private sector (Additional Parallel financing (20% of assets investment))	6.2 M

Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)

* The information can be drawn from the project/programme appraisal document.

F.1. Economic and Financial Analysis

Methodology and hypotheses

The criterion used for the economic analysis of the project is the economic rate of return (ERR) determined according to the cost-benefit method. It has been assumed that the economic costs are equal to 85% of the financial costs to take into account the transfers contained in the pre-tax costs. The assumptions used to calculate the economic benefits are the same as those described in the financial analysis of the project. However, for the economic analysis, the cost is 85% of the total cost of investments.

Economic rate of return

On the basis of the above assumptions, the economic rate of return of the project, presented in Annex 13, amounts to **16.61%**.

Financial viability for operators

The government is the borrower of GCF proceeds and will use the proceeds as 80% subsidies grant to the mini-grids, through construction. Private sectors operators will be selected based on 20% equity contribution. The financial analysis gives a return of 18% to the operators, based on their 20% expected contribution.

F.2. Technical Evaluation

Component 2: Detailed Engineering and Installation of solar powered mini-grids

For each of the localities selected, the acquisition, installation and commissioning of the following equipment: (i) autonomous solar power plants of 30, 50, 80 and 150 kWp power with storage systems; (ii) low and medium voltage networks of respective lengths 5, 8, 10 and 12 km; (iii) 14,189 prepaid meters; (iv) domestic electrical installations of households; (v) civil engineering of buildings; and (vi) 10 autonomous solar street lighting systems.

F.3. Environmental, Social Assessment, including Gender Considerations

The project is classified in Category B, according to the BOAD classification, as a project for installation and production of renewable energy. In practice, for projects with multiple sites and where the exact location of the sites is unknown, an ESMF is developed.

Locations and scope of the environmental and social impacts for each minigrid will be determined during the project execution based on a screening process will be undertaken according to the guidelines provided in the ESMF. For such small scale minigrids, there is no need for a fully detailed ESIA, since the overall project is categorized medium risk (B).

The sites of mini solar power plants will not affect protected areas and reserves because they will be located at the village level; and in general protected areas and reserves are far from the villages. Besides, the project sites will be set up so as not to displace populations or destroy economic assets. The beneficiary communities will provide small sites in villages for small scale solar plants as their contribution to the project. There is no need for resettlement plan since there will be no resettlement. So PS5 will not be triggered. PS 7 does not apply for there is no indigenous as defined by the World Bank Group's policy, on which those of BOAD and GCF are based, in the project area. Furthermore, PS8 does not apply too for activities related to such small isolated solar PV systems' construction will not need great excavation and there will be no potential risk for cultural heritage degradation in beneficiaries' communities.

A simplified E&S analysis is sufficient as per BOAD ESS. Required studies (ESIA/analysis) will be realized when specific site is identified before installation of the minigrids. The ESIA documents for subprojects will be disclosed locally in the local administrative buildings (townhall, and local chief offices).

Environment and Social Management Framework (ESMF) and the Gender Assessment and Action Plan documents were prepared and are attached to this Funding Proposal and also published in both English and French under the links provided bellow as a Category B project:

English:

- Parent link to the document on BOAD web site:
<https://www.boad.org/en/environmental-and-social-assessments/>
- Direct link to the document on BOAD website:
https://www.boad.org/wp-content/uploads/2016/10/ESMF_Report_Mali_EN.pdf

French:

- Parent link to the document on BOAD web site:
<https://www.boad.org/evaluations-environnementales-et-sociales/>
- Direct link to the document on BOAD website:
https://www.boad.org/wp-content/uploads/2016/10/Rapport_CGES-PERS-Mali_FR2.pdf

A certification letter has been issued following the national validation of the ESMF which stands for the environmental permit. (Please see letter attached).

Potential impacts and risks of the project

The major potential positive impacts of this type of project relate to i) the creation of jobs for the realization of the works; (ii) improvement of land-use planning; (iii) development of rural entrepreneurship through access to electricity; (iv) improving the conditions of care in social structures (health centers, education, etc.); (v) improving the security of goods and people through public lighting; (vi) reduction of greenhouse gas emissions per kWh produced; and (v) strengthening the resilience of populations to the adverse effects of climate change.

Regarding gender aspects, access to electricity in rural areas will save time on domestic chores to which women are subjected, eg. thanks to the installation of electric mills, and electric water pumps. This time-saving can be devoted to the development of income-generating activities (IGAs) and the schooling of the girl, which will contribute to strengthening women's empowerment and their participation in the country's development. In addition, the availability of electricity will improve the functioning of social services for women (eg ease of deliveries during the night), etc.

The main risks and potential negative impacts are related to the phase of the works: (A) on the biophysical environment, i) the deterioration of air quality and the pollution of soil and surface water; (ii) noise pollution caused by machinery traffic and site-level activities; (iii) the destruction of vegetation and wildlife habitats; and (iv) waste generation at worksites; (B) on the human environment; (i) the spread of STIs due to the arrival of foreigners in villages and unprotected sex; (ii) accidents during brushing, installation of equipment and drawing of power lines.

During the operation phase of the project, there are risks of: i) accidents by electrocution of workers responsible for the maintenance of equipment and populations; (ii) fire related to the use of hydrocarbons at the sites; and (iii) lack of management of waste electrical and electronic equipment (WEEE) from batteries and photovoltaic panels.

Mitigation measures for Project activities

For the most part, the significant negative impacts can be greatly reduced with the application of (i) appropriate planning measures (good implementation and landscape integration of equipment); (ii) environmental and social clauses relating to safety, hygiene, the management of solid and liquid waste (batteries and used oils) during the construction and operation phases.

The coordination of the implementation of these measures will be the responsibility of the AER. To this end, under its Environmental and Social Management System (ESMS), AER will develop an appropriate policy, environmental and social guidelines, and will have an Environmental Specialist to lead the planning and environmental monitoring of projects. rural electrification. In the context of this project, the Environment Specialist will be responsible, among other things, for (i) support for the selection of sub-project sites in accordance with BOAD's land acquisition policy; (ii) the development of Terms of Reference (TORs) and the conduct of the consultant recruitment process for specific environmental assessments, as appropriate; (iii) monitoring the integration of the environmental and social clauses in the DAO, and the examination of the E & S aspects of the companies' offers; and (iv) coordinating the development of EHS manuals and organizing OSD training sessions in environmental monitoring and collaboration and monitoring the implementation of mitigation measures at the construction phase.

As part of the environmental monitoring of the project, an agreement will be signed with the the National Directorate for National Directorate for Sanitation and Pollution Control (DNACPN), and capacity building sessions will be organized for the different actors involved in the project.

Gender considerations

The project is linked to national and international framework documents and policies for the promotion of gender equality. the national gender policy and the SDGs provide for an important place for the implementation of projects in the field of energy and solar energy as a means of promoting gender equality. as well, gender is mainstreamed in particular project context analysis, objectives and results definition, activities and indicators determination. In this case, women and men are part of the project implementation. In the impact analysis, women and men are equally addressed and it shows that the project will have positive impacts for both women and men. Among the positive impacts, there are (i) the increase in socio-economic activities, the improvement of the health and hygiene conditions of local populations; (ii) the improvement of living conditions and the comfort of local populations; (iii) improving the permanent supply of drinking water to the population, (iv) the reduction of women's labour, (v) the improvement of the literacy rate, etc. To alleviate potential negative impacts, measures have been taken to benefit men and women. Above all, a gender action plan has been draft to highlight gender consideration in the project. The purpose of this gender action plan is to mitigate gender risks and constraints and to operationalize opportunities for women and men identified during the gender analysis. Base on a participative diagnostic, the plan includes (i) gender-responsive actions that address and strengthen the voice and agency of vulnerable women and men in climate action; (ii) gender performance indicators and sex-disaggregated targets that can be incorporated into a results framework; and (iii) presentation of gender-responsive development impacts. Consultations to date with stakeholders such as the EAR, AMADER, local authorities and beneficiaries have revealed the needs and priorities of men and women in the renewable energy sector, and the way in which whose project will respond to it. These elements are taken in consideration in priority as actions and presented in the gender analysis; and it is on this that actions have been proposed jointly with beneficiaries in view of a greater relevance of these actions proposed in this plan. These actions are concretely related to (i) Stakeholders' Institutional and organizational governance capacity strengthening, (ii) Improving rural women accessibility to off-grid/mini grid mode based on renewable energies (solar PV, wind), (iii) Supporting beneficiary communities, and (iv) Communication and capitalization on gender best practices and approaches in gender mainstreaming and scaling up.

At the end of the project implementation, a capitalization study will show results, changes, impacts, good approaches and best pratics in gender promoting in energy and solar domain. The gender focal point of AMADER is chosen as the gender staff to be part of the implementing staff to ensure gender mainstreaming during the implementation, reporting, evaluation, and to take the lead on all the works. This gender staff will have the technical assistance of the AE gender Expert in this work.

The gender action plan in annex gives more information on gender considerations in the project.

F.4. Financial Management and Procurement

The provision of the BOAD loan will be made either by direct payment to the suppliers (BOAD / I procedure), according to the schedule of payment provided for in the contract and at the express request of the Borrower, or on reimbursement of the expenses made according to the BOAD / II procedure.

Goods, and services financed from the GCF loans and the BOAD Loan will be acquired by:

- limited consultation after expression of interest for sub-components "Training, awareness raising and dissemination of knowledge", "RMF capacity building" and the "Financial mechanism to support the rural electrification market", component
- international call for tenders for the components "setting up an equipment control system", "supplies, installation and commissioning of equipment";
- open consultation on a UEMOA scale of specialized design offices for the realization of the sub-components "control and inspection of works", "Technical and financial audit of the project;
- National call for tenders for the "Environmental and Social Measures" components and the "Procurement of 03 vehicles" for the "Project Management" component;
- direct agreement with AMADER for monitoring the implementation of the ESMP of the "Environmental and Social Measures" sub-component;

Procurement

To ensure that financing is applied in ways that adequately secure the BOAD's mandate while maximizing development effectiveness, the Bank encourages and promotes sound, fair, transparent and well performing procurement systems.

BOAD's Guidelines for procurement of consultancy services financed by a loan or advance of funds (2016); and Guidelines for the award of works contracts, goods and services (other than consultancy services) (2016) will be applied to the project's procurement.

The bank will conduct an analysis of the procurement files and give its no objection.

AML/FT Due diligence

As part of the prevention and fight against money laundering, BOAD has developed a Financial Security Policy (2016). This internal document constitutes the general framework of the control system covering all of BOAD's activities relating to the prevention, surveillance and management of money laundering and terrorist financing risks.

BOAD's Financial Security Policy is inspired by the international standards defined by the United Nations through its specialized structures, the Organization for Economic Cooperation and Development (OECD) whose recommendations are issued by the Groupement d ' International Financial Action (FATF) and provisions of Directive 02/2015 / CM / UEMOA of 02 July 2015 on the fight against money laundering and terrorist financing in the Member States of the Economic and Monetary Union. West African currency ("WAEMU Directive").

The policy requires to disclose information about its clients' transactions to the relevant authorities in cases where international rules and local law require regulated financial institutions to do so, including with respect to against money laundering.

Under its due diligence, the Bank will assess civil / criminal and regulatory antecedents and sanctions lists. The Bank conducts also administrative investigations into corruption, fraud, coercion, collisions and inconvenient practices, and make use of the relevant national authorities for the necessary criminal investigations.

The Bank integrates measures to combat illicit financial flows, the fight against money laundering and terrorist financing in the internal operations of the Bank Group.

In the context of this project, concerning ALM/FT, the concerned stakeholder will:

- ensure that the funds financing the Project are not of illicit origin and in particular are not related to fraud against Mali's financial interests, corruption, organized criminal activities, terrorism or drug trafficking; and
- forward to the Bank without delay any information raising suspicions as to the unlawfulness of the sums invested in the company and in the Project;
- notify the Bank without delay if it has known at any time of any information indicating the illicit origin of all or part of the funds of the structure;
- will not enter into a business relationship, directly or indirectly, with persons or entities on the lists established by the United Nations Security Council or its committees pursuant to Security Council resolutions, by the Council of the European Union in application of its Common Positions and / or by the African Union as well as any other relative or complementary resolution and any act of implementation thereof in connection with the fight against the Laundering of Capital and the financing of terrorism.

AUDIT

Audits will be undertaken in accordance with BOAD Guidelines for Financial Reporting and Auditing of Projects. The Financial Agreement with Mali will require the submission of Audited Financial Statements to BOAD within six months after each year-end. An independent external auditor will be recruited based on Terms of Reference acceptable to the Bank (not later than four months after effectiveness) for the entire duration of the project. The Financial Statements will be audited in accordance with international auditing standards.

BOAD will prepare a Management Letter to provide observations, comments, and recommendations for improvements in accounting records, systems, controls and compliance with financial covenants in the Financial Agreements. The cost of the audit will be met from project resources.

BOAD will ensure KYC standard due diligence process including anti-money laundering and other evaluations of sponsors is followed thoroughly.

G.1. Risk Assessment Summary

The main risks identified are related to the failure that could occur in the realization, project management and operation of equipment to be acquired under the project.

Risks in a project could be determined at a planning stage or during project implementation. However so, the foresee risks seems modest and could be managed through preventive measures and complementary BOAD programs such as

1. Construction risk: Failure in the installation of equipments of the mini-grid plants. This will be mitigated by applying best practices in the selection process.
2. Environmental and social risk (site-specific): This is mitigated as described in the ESS report.
3. Operating and maintenance Risk: This will be mitigated by the choice of quality equipments at construction
4. Regulatory Risk: This is mitigated by implementation of component 1 Technical Assistance activities
5. Currency exchange Risk: this is mitigated since by allowing 20% deposit in local currencies
6. Country security Risk

G.2. Risk Factors and Mitigation Measures

Please describe financial, technical and operational, social and environmental and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures.

Selected Risk Factor 1

Description	Risk category	Level of impact	Probability of risk occurring
Construction risk: Failure in the installation of equipments of the mini-grid plants	Technical and operational	Low (<5% of project value)	Low

Mitigation Measure(s)

This risk will be mitigated by applying best practices in the selection process. Indeed, this risk is mitigated by the arrangements made for the choice of high-performance companies and the quality of the technical teams of the AER, supported by the Consulting Engineers' offices, which have proven references for the control and supervision of similar works.

Selected Risk Factor 2

Description	Risk category	Level of impact	Probability of risk occurring
Environmental and social risk (site-specific)	Social and environmental	Low (<5% of project value)	Low

Mitigation Measure(s)

The Environmental and Social Management Framework (ESMF) for the project has been developed and submitted. All sub-projects will be assessed according to the relevant BOAD policies, and an environmental and social action plan (ESAP) will be developed during the implementation of the project. The project ESAP will be an integral part of the required agreement with the project developers. BOAD will apply its highest standards in the screening and

managing of E&S related issues. The framework targets only small-scale projects and the sites will be multi-location, which will reduce the overall E&S risk.

Selected Risk Factor 3

Description	Risk category	Level of impact	Probability of risk occurring
Operating and maintenance Risk	Technical and operational	Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

The operation and maintenance of a solar based mini-grid systems is relatively simple as it consists of cleaning, regular inspections, minor repairs and measurements, data verification, reporting and site security. The Project benefits from a 15-year Operation & Maintenance (O&M) Agreement. A performance bond will be made available by the operator for failure of the plant to achieve performance guarantee. Furthermore, this risk will be mitigated by the choice of quality equipments at construction.

Selected Risk Factor 4

Description	Risk category	Level of impact	Probability of risk occurring
Regulatory Risk	Technical and operational	Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

This is mitigated by implementation of component 1 Technical Assistance activities.

Selected Risk Factor 5

Description	Risk category	Level of impact	Probability of risk occurring
Currency exchange Risk:	Financial	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

Project financing will be in EUR. This risk is therefore mitigated given the stowage of the EUR to the XOF.

Selected Risk Factor 6

Description	Risk category	Level of impact	Probability of risk occurring
Country security Risk	Social and environmental	Low (<5% of project value)	Medium

Mitigation Measure(s)

The security situation which used to be fragile has improved. Despite a difficult security past, medium term macroeconomic prospects are good with overall growth forecast of 5% in 2017. These important factors can positively contribute to a consolidation of Mali political situation and secure stability in future. In fact it is expected that with the secure and wider access to clean energy services, economic activities will further emerge and the society feeling much safer.

Other Potential Risks in the Horizon

Please describe other potential issues which will be monitored as “emerging risks” during the life of the projects (i.e., issues that have not yet raised to the level of “risk factor” but which will need monitoring). This could include issues related to external stakeholders such as project beneficiaries or the pool of potential contractors.

H.1. Logic Framework.

Please specify the logic framework in accordance with the GCF's [Performance Measurement Framework](#) under the [Results Management Framework](#).

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level⁶

Paradigm shift objectives

<i>Shift to low-emission sustainable development pathways</i>	The project intends to contribute to the reduction of Mali's chronic electricity deficit which will result in reducing energy-related CO2 emissions by introducing photovoltaics (PV). At the same time, it will enhance the access to clean energy in rural and remote households, specifically in the 70 villages targeted. In addition, the project will develop local capacities to identify technical and financing options and to formulate the regulatory, financial and marketing instruments necessary to demonstrate the technical, economic, and financial viability of using the private sector as a vehicle to deliver basic electricity services.					
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Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	

Fund-level impacts

<i>M1.0 Reduced emissions through increased low-emission energy access and power generation</i>	<i>Tonnes of carbon dioxide equivalent (t CO2eq) reduced or avoided as a result of the funding of Projects/programs (main indicator)</i>	<i>Annual CO2 performance monitoring reporting and validation</i>	<i>0</i>	<i>821,782 tCO2eq</i>	<i>821,782 tCO2eq</i>	CO2 emission reduction was based on the most conservative calculations and data using international default values. CO2 emission reduction could be increased if local data will be available.
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⁶ Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that [some indicators are under refinement](#)):

[http://www.greenclimate.fund/documents/20182/239759/5.3 -
Performance Measurement Frameworks PMF .pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4](http://www.greenclimate.fund/documents/20182/239759/5.3_-_Performance_Measurement_Frameworks_PMF_.pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4)

	<i>Volume of finance leveraged by Fund funding (funding leverage factor)</i>	<i>Project management report</i>	0	EUR 5,98 M	EUR 19,92 M	We assume that private sector will participate in the project
	Cost per tCO ₂ eq decreased	Annual CO ₂ performance monitoring reporting and validation	0	EUR 31.6/t CO₂eq	EUR 31.6/t CO₂eq	

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level

Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/program outcomes	Outcomes that contribute to Fund-level impacts					
M6.0 Increased number of small, medium and large low-emission power suppliers	6.2 Number of Households and individuals (males and females) with improved access to low- emission energy sources	Project management Report, household surveys, Malian Agency for the Development of Domestic Energy and Rural Electrification (AMADER) report	0	At least 283,777 people directly benefit [of which 50% are female]	At least 283,777 people directly benefit [of which 50% are female]	Households are aware of the climate change; Households have the capacity to pay; households are sensitized about using clean energy
	6.3 MWs of low-emission energy capacity installed, generated and/or rehabilitated as a result of GCF support	Utility company (EDM) report; AMADER report	0	3,78 MWp	3,78 MWp	Private enterprises are interested in investing in the sector
M5.0 Strengthened policy and regulatory framework, regulation of tariffs	5.1 Institutional and regulatory systems that improve incentives for low emission planning and development and their effective implementation	National Energy Strategy and policy documents, EDM and AMADER annual reports	National Energy Sector Policy 2006 does not take into account regulation regarding climate change challenges	Reinforcement of existing 2006 policy taking into account climate change issues	Climate change issues are addressed in all energy policies and enforced in projects implemented in Mali	Political willingness to implement policies and regulations

Project/program outputs	Outputs that contribute to outcomes					
1. Strengthened Rural Energy Regulatory Agency	Functional rural electrification Agency with trained agents in legal matters (concession contracts), technical aspects (tariff methodology), supervision of the sector (monitoring and evaluating the project)	AMADER report, project report.	Regulatory agency exists but need capacity reinforcement	Legislation and regulations to encourage clean energy use and facilitate private sector involvement in place	Fully functional rural electrification agency in place and produces contracts and supervision reports in line with industry standards	Investment targets are achieved, less turnover and higher retention rate of trained agents
2. Rural localities electrified with solar PV power plant	Number of Isolated solar PV installed Number of Localities benefiting from PV installations in the project site MW produced by installed solar plants	Household surveys, Installation project reports and end of Project execution	0	50 number of installed solar PV plants installed 50 localities benefiting from power supply from project financed solar plants 3,78 MW clean energy generated by project financed solar plants	50 number of installed solar PV plants installed 50 localities benefiting from power supply from project financed solar plants 3,78 MW clean energy generated by project financed solar plants	Investment targets are achieved, Government willingness to electrify rural localities with clean source energy
3. Increased access to clean energy by households and for productive use	Number of households connected to the clean solar energy Proportion of connected households benefiting	Household surveys Evaluation of development result, audit report, private sector structures reports	0	28,378 households connected of which 10% of connected households have access to financial	28,378 households connected of which 10% of connected households have access to financial	Mali government willingness to reduce GHG emissions in the energy sector

	due to access to credit under financial mechanism (guarantee scheme) for productive use of clean energy			mechanism (guarantee scheme)	mechanism (guarantee scheme)	
Activities	Description	Inputs		Description		
1.1 Training, raising awareness of stakeholders and dissemination of project activities/results	1.1.1 - organizing dissemination workshops on potential actions to be implemented in order to achieve the objectives of the NDC in the field of energy 1.1.2 - organizing specialized training sessions for designated executives of DNE, AMADER and AER-Mali 1.1.3 -developing and disseminating public awareness material on climate change and NDC in the energy sector.	Consultancy services		Consultancy services will be required to organize sensitization and information dissemination workshops and train target beneficiaries		
1.2 Communicating project results and lessons learned	1.2.1: - Developing a communication and raising awareness/promotion plan for national (and international) investors and production of materials (including videos) 1.2.2 - Organizing information sessions with investors who are already active in the energy/renewable energy sector 1.2.3 - organizing mid-term and end-of-project workshops for national institutions (DNE, AMADER, AER-Mali, FER) to ensure monitoring and document the project experience	Consultancy services		Consultancy services will be required to develop communication and awareness raising plan, produce related materials and organize sensitization sessions as well as project workshops with relevant stakeholders and agencies		
1.3 Capacity building of the Rural Electrification Fund (Fonds	1.3.1: - Drafting of organisational and operational legal provisions as part of a reform of the FER 1.3.2 - TA for recruitment of a dedicated staff	Consultancy services		Consultancy services will be required to build and train staff, draft organizational and operational legal provisions, etc		

d'Electrification Rural -FER)	1.3.3 - Capacity building of the FER staff		
1.4 Providing Technical Assistance on procurement and contracting of mini-grid construction and O&M companies	1.4.1 - TA - writing of standard contract for 15 years O&M and the procedures that set the relations between AMADER and O&M companies 1.4.2 - Tender processes and selection of construction companies and O&M companies that will operate the mini-grids.	Consultancy services PMU Staff time	Consultancy services and staff time will be required to support the procurement and contracting construction and O&M companies
2.1 Preparation of detailed Engineering Studies	2.1.1 - Site selection, engineering studies & Specific environmental and social analysis for each site	Consultancy services	Consultancy services will be required to support site section, preparation of engineering and environmental studies and analysis for each site
2.2: Supply, installation and commissioning of equipment	2.2.1 - Solar PV power plants construction and minigrid deployment	Consultancy services Goods Works	Consultancy services, equipment and construction works will be needed to install Solar Mini-grids
2.3: Implementation of Environmental and social measures	2.3.1 - Project impact mitigation and improvement activities as per the ESMF	Consultancy services PMU Staff time	Consultancy services and works and PMU staff will be needed to mitigate and ensure compliance with ESMS standards
3.1 Operationalisation of Financial guarantee facility	3.1.1- Implementation of Financial guarantee activities	staff time	Staff time will be needed

Most of mid-term data are not applicable to the project because if we refer to the timetable c.8. the project will be completed in the fourth year. Therefore, the key deliverables will be available at the end. However, there will be a mid-term evaluation that will allow to monitor the progress of the implementation. In the mid-term, other important activities will be implemented such as tender & selection of construction companies. But there will not be PV power plant installed yet.

H.2. Arrangements for Monitoring, Reporting and Evaluation

Monitoring of the project

As with all projects supported by BOAD, this intervention will be monitored by BOAD project management team as per the relevant internal policies and procedures. As the accredited executing agency, BOAD will be responsible for supervising the monitoring activities, implementation conditions and reporting periodically to the GCF under the terms to be defined by GCF and BOAD. The project progress will be monitored by the responsible of M&E of the Project Management Team based in Mali.

The present project will comply with the BOAD appraisal, approval, monitoring and supervision standards and procedures

involving all relevant teams (engineer, financial analyst, procurement expert, E&S specialist, climate finance specialist, financial management officer, and monitoring specialist). The implementation and monitoring of each stage of the project will be guided and managed by the BOAD project cycle management framework. The Division in charge of Climate finance will perform due diligence, implementation monitoring, risk monitoring and mitigation. For that purpose, the Division in charge of finance climate assisted by the Department in charge of Monitoring and Evaluation will be working closely with the project management team based in Mali in order to monitor the progress of each component of the Project.

Reporting

Reporting of project management team based in Mali to BOAD will be in line with the standard loan agreement, and the BOAD will conduct a by-annual supervision.

Reporting of the BOAD to GCF: The BOAD will comply with the relevant GCF policies in reporting and evaluation arrangements for this framework. The BOAD will provide the annual performance report (APR) to the GCF during the four-year implementation period. In addition, during the sub-loan lifetime, semi- annual activity report on the status of the GCF-financed individual sub-projects will be provided.

In addition, following the arrangement under the AMA, inception report, mid-term and final evaluation reports, and financial information reports (semi-annually throughout the life of the loan) will be submitted.

Evaluation

The evaluation arrangements for this framework will comply with the related BOAD and GCF policies. A formative evaluation will be carried out by the BOAD's project management team that is composed of (engineers, lawyers, project finance specialists, procurement experts, E&S specialists, climate finance officers, supervision and monitoring specialists). The final evaluation is conducted 06 months after the reception of the project achievement report.

After two to five years of closing, the project will undergo ex-post evaluation that is guided by internationally accepted principles for the evaluation of development assistance, in particular, the Organization for Economic Co-operation, and Development Assistance Committee (OECD DAC) evaluation guiding principles, and the good-practice standards.

Budget

The monitoring and evaluation budget, estimated at EUR 41 161, will be used to conduct monitoring and evaluation activities related to indicator collection and impact evaluation activities.

I. Supporting Documents for Funding Proposal

- ☒ NDA No-objection Letter
- ☒ Feasibility Study
- ☒ Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable)
- ☒ Confirmation letter or letter of commitment for co-financing commitment (If applicable)
- ☒ Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – *see the Accreditation Master Agreement, Annex I*
- ☒ Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan (If applicable)
- ☐ Appraisal Report or Due Diligence Report with recommendations (If applicable)
- ☐ Evaluation Report of the baseline project (If applicable)
- ☒ Map indicating the location of the project/programme
- ☒ Timetable of project/programme implementation

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter issued by the national designated authority(ies) or focal point(s)

MINISTRE DE L'ENVIRONNEMENT,
DE L'ASSAINISSEMENT ET DU
DEVELOPPEMENT DURABLE

REPUBLIQUE DU MALI
Un Peuple - Un But - Une Foi

Agence de l'Environnement et du
Développement Durable (AEDD)

Bamako, le 13.0 NOV 2018



*The Director General of the Environment
and Sustainable Development Agency*

To

Mr Howard Bamsey
Director Executive of Green Climate Funds
G-Tower, 24-4 Songodo dong Yeonsu gu
Incheon City Republic of Korea

N° 2463 /MEADD/AEDD-DMRF

Re: Funding proposal for the GCF by The West African Bank of Development (BOAD) regarding the project « Mali solar rural electrification project».

Dear Sir,

We refer to the project « **Mali solar rural electrification project** » in Mali as included in the funding proposal submitted by The **West African Bank of Development (BOAD)** to us on 26th of November 2018.

The undersigned is the duly authorized representative of Boureïma CAMARA, the National Designated Authority/focal point of Mali.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Mali has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Mali's national priorities, strategies and plans ;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



The Director general,

Boureïma CAMARA
Chevalier de l'Ordre national

Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Mali solar rural electrification project
Accredited entity	Banque Ouest Africaine de Développement (BOAD)
Environmental and social safeguards (ESS) category	Category B
Specific project location or target country/ies of the programme, as the case may be	Mali (Kayes, Koulikoro, Sikasso, Segou, Mopti and Tombouctou)
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	2019-01-22
Language(s) of disclosure	English and French
Link to disclosure	English: https://www.boad.org/wp-content/uploads/2016/10/ESMF_Report_Mali_EN.pdf French: https://www.boad.org/wp-content/uploads/2016/10/Rapport_CGES-PERS-Mali_FR2.pdf The Environmental and Social Management Framework (ESMF) contains a preliminary characterization of environmental and social risks and impacts which is consistent with the requirements for category B projects.
Other link(s)	English: https://www.boad.org/en/environmental-and-social-assessments/ French: https://www.boad.org/evaluations-environnementales-et-sociales/
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	2019-01-22
Language(s) of disclosure	English and French
Link to disclosure	English: https://www.boad.org/wp-content/uploads/2016/10/ESMF_Report_Mali_EN.pdf French: https://www.boad.org/wp-content/uploads/2016/10/Rapport_CGES-PERS-Mali_FR2.pdf The ESMF contains a preliminary ESMP which is consistent with the requirements of Category B projects.

Other link(s)	English: https://www.boad.org/en/environmental-and-social-assessments/ French: https://www.boad.org/evaluations-environnementales-et-sociales/
Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	N/A
Language(s) of disclosure	N/A
Link to disclosure	N/A
Other link(s)	N/A
Disclosure in locations convenient to affected peoples	
Date	2019-01-22
Place	The documentation is published on BOAD website, and made available in hard copies at BOAD HQ in Togo, and also in Mali in locations that are accessible for the affected people, especially BOAD Representative Office, Agence des Energies Renouvelables (AER-Mali), Direction Nationale de l'Assainissement du Contrôle des Pollutions et des Nuisances (DNACPN-Mali) in Bamako and at regional level within Regional Directorate for environmental affairs (Direction Régionale de l'Assainissement du Contrôle des Pollutions et des Nuisances-DRACPN)

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP102

Proposal name:	Mali solar rural electrification project
Accredited entity:	Banque Ouest Africaine de Développement (BOAD)
Country/(ies):	Mali
Project/programme size:	Medium

I. Overall assessment of the Secretariat

- The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project proposes an innovative approach, public-private partnerships, to address the barriers for renewable-based, mini-grid investments in Mali. If successful, this could be duplicated across the whole country and many other least developed countries	The productive use guarantee facility for local microloans will be established from sovereign loan proceeds, leading to a repayment obligation by the country. While this model is more stringent for the country than a direct guarantee from GCF, the model is requested by the accredited entity and builds on its own experience in the region for similar schemes
The project targets a least developed country, Mali, that is vulnerable to climate change and has one of the lowest rural electricity access rates in Africa	The private sector will invest 20% of their own equity and will have the operation and maintenance rights and revenue collection rights over 15-year concessions while the ownership of the assets remains with the government. This dual arrangement usually results in reduced quality of service towards the end of the concession arrangements. The Government of Mali, through the regulatory agency, will need to be more vigilant on technical regulation, especially towards the end of the concession agreements
Banque Ouest Africaine de Développement is a direct access entity that has never accessed GCF project resources before despite abundant experience in the renewable sector in the region	
The project unlocks private sector investment with a private sector project equity contribution (20%) while enhancing sustainability with operation and	

maintenance also being covered by private sector operators	
Revenue collection on the mini-grids will take a levy to be paid to the Rural Electrification Fund on each kilowatt hour delivered. The proceeds from this levy will be used for other renewable energy mini-grids or refurbishments of the assets, which will remain under government ownership	
The end users tariff is capped at 174 African Financial Community (CFA) francs per kilowatt hour and remains within the communities' ability to pay while country ownership is strongly demonstrated with capital expenditure financing being based on a sovereign debt	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the respective term sheet and addendum XVI, titled "List of proposed conditions and recommendations".

II. Summary of the Secretariat's assessment

2.1 Project background

3. Mali is a landlocked least developed country (LDC) in the Sahel belt of West Africa. The rate of access to electricity in rural areas is estimated at only 19 per cent. It is very unlikely that a significant proportion of the low-income population living in isolated areas will be connected to the national electricity network in the next decade. Even if they could be connected, about 80 per cent of Mali's main grid power generation relies on fossil fuel while the majority of the rest is based on hydro, which is declining due to drying rivers. At the moment, 96 per cent of the operational mini-grids in rural areas are diesel based, thus highly dependent on imported fossil fuels and global price volatility. Despite the immense potential of solar resources in Mali, they represent only 3 per cent of electricity production. Their development remains slow due to various financial and technical barriers, among which is the high upfront investment cost, inadequate financial incentives to attract the private sector, and weak institutional planning capacity.

4. The main objective of this project is to promote rural electrification through isolated solar photovoltaic (PV) green mini-grid systems as a low-carbon and resilient solution to the effects of climate change in the energy sector of Mali. This would be achieved by strengthening the capacity of relevant public institutions engaged in rural electrification; (ii) increasing the rural populations' access to electricity in 70 communities (50 communities in tranche 1) through installation of isolated solar PV mini-grid systems; and (iii) a guarantee facility to promote increased access to financial services for productive energy users in rural areas.

5. The full project cost is EUR 46.2 million, comprising EUR 34 million as a loan and EUR 1.6 million in grants requested from GCF. Banque Ouest Africaine de Développement (BOAD), the accredited entity (AE), will also contribute EUR 10.6 million in the form of loan. The AE has

proposed a two-tranche approach to finance this project to adapt the commitment authority of GCF. For tranche 1, 50 out of the 70 identified localities would be financed to install the PV mini-grids, of which EUR 24.3 million would be a loan and 1.6 million a grant from GCF. The impact of tranche 1 will be 80 per cent of the full project's impact.

6. The environmental and social risk of this project falls under category B of the GCF accreditation framework. An environmental and social management framework has been published by the AE 30 days prior to GCF Board consideration as required for category B projects.

2.2 Component-by-component analysis

Component 1: Capacity-building of rural electrification institutions and technical assistance (total cost: EUR 0.4 million, GCF cost: EUR 0.4 million)

7. This component aims to create an enabling environment for the promotion of rural electrification from solar PV with the chosen model that is inclusive of the private sector for investment and operation and maintenance (O&M) while allowing the public sector to provide an adequate subsidy to keep the tariffs affordable. It includes three activities: (i) capacity-building for relevant institutions including the Malian Agency for the Development of Household Energy and Rural Electrification (AMADER), Agence des Energies Renouvelables (AER), and the Rural Electrification Fund; (ii) raising investor awareness and disseminating project activities and results; and (iii) providing technical assistance for the tender process for the selection of O&M companies that will operate the mini-grids and project implementation controls.

8. The proposed activities are key to addressing weak institutional capacity and the implementation of this project. The full cost will be financed by a GCF grant of EUR 0.4 million.

Component 2: Detailed engineering and installation of 70 solar powered mini-grids (50 mini-grids for tranche 1) (total cost: EUR 43.9 million (total tranche 1: EUR 31.9 million), GCF cost: EUR 33.4 million (GCF contribution to tranche 1: EUR 23.7 million))

9. This component will finance the installation of solar-based mini-grids (power plants with battery storage and distribution infrastructure) in 70 selected localities (50 localities in tranche 1). The GCF concessional loan to the Government of Mali will finance the installation of mini-grid systems, which will remain under ownership of the Government of Mali while O&M contracts will be awarded to eligible private sector operators. With this model, the Government is subsidizing the infrastructure at construction to achieve tariff levels that are within the rural areas' ability to pay. The GCF loan to the Mali Government will be blended with the BOAD loan whose proceeds are obtained by BOAD at market rates. The level of concessionality, which is applied by GCF, is mainly justified by the specific vulnerability situation of Mali as an LDC, the use of a debt instrument instead of a grant, and the objectives of the project to keep the electricity tariff to affordable levels at EUR 0.26 per kilowatt hour (kWh), or 174 African Financial Community (CFA) francs (XAF) per kilowatt hour.

10. The project is designed to be a public-led investment due to the immature market for private sector involvement and the need to keep the level of affordability for the end users. The financial analysis shows that at the targeted affordable tariff of 174 XAF, a subsidy of 80 per cent is required to make a positive return, thus the requirement for a 20 per cent contribution from the private sector. While the Government keeps the ownership of the built assets in the proposed model, concession agreements to eligible private operators for operation and maintenance will last 15 years and could be renewed in the event of satisfactory service. This model is deemed appropriate in the specific case of Mali, where the electricity market requires a soft transition from purely public service to more involvement from the private sector in the operations.

11. Fifty out of the planned 70 localities would be covered in the first tranche, achieving about 80 per cent of the impact. The carbon dioxide (CO₂) abatement calculation used is a clean development mechanism (CDM) approach and the assumptions on the percentage of users that will be on productive use, middle tranche or basic use are realistic.

Component 3: Support to productive use of electricity (total cost: EUR 1.5 million, GCF cost: EUR 1.5 million)

12. A guarantee facility is proposed to address the issue of limited financial services for productive energy users in rural areas. The productive users will usually have more demands and their loads coincide more with solar generation, hence they will serve as immediate new income for mini-grid businesses and will be critical for the long-term viability of mini-grids. However, productive-use appliances, like grain mills, irrigation and cold storage often come with high upfront cost. Surveys revealed that although some producers' cooperatives are eligible for credit from financial institutions, they have trouble accessing credits due to the absence of adequate collateral. Financial service providers often request guarantees for lending to small institutions. This would be addressed by this project through setting up a financial guarantee mechanism that covers part of the risk taken by the micro-finance institutions towards productive users, hence achieving greater sustainability for the GCF investment.

13. The GCF concessional loan is requested to finance this facility. This means that this credit risk would be borne by the Government of Mali, a key detail that shows the ownership and commitment of the government in developing this sector.

Project Management Cost (total cost: EUR 0.4 million, GCF cost: EUR 0.3 million)

III. Assessment of performance against investment criteria

14. The GCF investment is key to helping Mali scale up investment in renewable energy, providing successful structuring of finance for this project to address some of the barriers faced by private sector investors. Overall, the Secretariat deems the anticipated performance of the project to be satisfactory in aligning with GCF investment criteria, especially in terms of the paradigm shift potential and cost effectiveness.

3.1 Impact potential

Scale: Medium/High

15. The project is expected to result in direct emissions reduction of 41,089 tonnes of carbon dioxide equivalent (tCO₂eq) annually (tranche 1 will result in reductions of 32,871 tCO₂ per year), which works out as 1,027,227 tCO₂eq in total over the project's lifetime. This is achieved through the switch in energy demand from diesel generators, kerosene lamps, paraffin candles and other emitting sources to solar mini-grids in the selected communities. This figure does not take into account the potential of opening up the market of renewable energy mini-grids, which is vast in Mali.

3.2 Paradigm shift potential

Scale: High

16. The paradigm shift potential of this project is deemed high as it is an attempt to support a new model that includes both public and private efforts in the mini-grid market in Mali. GCF financing is critical for allowing the successful structuring of finance for this project while addressing some of the barriers faced by private sector investors. If successful, the relevant institutions, including the renewable energy agency, AER, will be underpinned by an enabling environment that is equipped with a strengthened capacity of national players, private sector awareness and interest in rural electrification and renewable energy promotion. With the

enhanced understanding of the renewable energy sector and potential profitability for lending to rural electrification operations, microfinance institutions are expected to continue providing finance to this sector. This is especially true in a country with vast mini-grid market potential such as Mali.

3.3 Sustainable development potential

Scale: High

17. By improving access to energy and microfinance, the localities will benefit from increased socio-economic activities, job creation, improvement of household income and creation of income-generating activities. The environmental benefits include reduction of air pollution, through the replacement of diesel generators, and greenhouse gases. The project design also includes gender considerations and makes efforts to provide equal access to GCF resources to all people, regardless of gender, especially in O&M, thus ensuring gender mainstreaming in the energy development plans.

3.4 Needs of the recipient

Scale: High

18. Mali is an LDC country with a high poverty rate and low electrification rate in rural areas, which further exacerbates the climate vulnerability faced by the localities. Mali's energy consumption is heavily dependent on the importation of fossil fuels, which is impacted by price volatility. Even with its determination to grow its renewable energy sector, the Government has been struggling to mobilize the necessary financial resources. With a high level of indebtedness and under serious fiscal stress, the Government is not able to borrow from financial institutions at market rates. GCF finance is critical for an attempt to develop the country's immature solar mini-grid and allow vulnerable communities to access affordable energy.

3.5 Country ownership

Scale: High

19. With the Ministry of Energy and Water as the executing entity (EE) and BOAD as the key partner, this integrated proposal is fully aligned with the national priorities and supports Mali's goals detailed in its national adaptation programme of action and nationally appropriate mitigation action, namely, reducing greenhouse gas emissions by 1,285,034 tCO₂eq per year through renewable energy production, and fisheries, energy and water resources as the most vulnerable sectors. The proposed project also contributes to several Sustainable Development Goals (SDG), especially SDG 7, affordable and clean energy, and SDG 11, sustainable cities and communities. The country ownership is further evident in the strong commitment by the Government in this sector and in their taking the full credit risk of the producers' cooperatives.

20. BOAD, as a direct access AE to GCF, has a wealth of experience working in the energy sector in West African countries. The project design is a result of inputs from various stakeholders, including the concerned local elected representatives, civil society actors, local populations and the national designated authority. Finally, almost the entire financing package is a loan to be repaid by the country.

3.6 Efficiency and effectiveness

Scale: Medium/High

21. A GCF high concessional loan with a sovereign guarantee is essential for ensuring financial adequacy and minimum concessionality. Operating costs will be covered through revenue collection from power distribution while the infrastructure will be financed through the concessional loan and repaid by the Government of Mali. Given the Government's ongoing economic and financial programme with the International Monetary Fund, Mali is required to

carefully manage its debt and budget, and the country has requested high concessionality to stay within these constraints. If lower concessionality were sought, the Government would not be able to take the loan, threatening the viability of the project.

22. The project demonstrates high cost effectiveness. The project has a cost of EUR 41.7 per tCO₂eq, which is lower than many global estimates of the social cost of carbon. The project also has an economic rate of return of 16.61 per cent, which a sensitivity analysis shows to be robust to variations in costs and benefits.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

23. **Environmental and social risk category:** BOAD has assigned environmental and social risk category B to the project in accordance with the GCF environmental and social safeguards standards. The risk category is consistent with the BOAD environmental and social safeguards accreditation for carrying out funded activities that fall within the medium risk, including lower risk, category. The project's impact is expected to derive from the installation and operation of off-grid solar PV mini-grid systems. The sites for the mini-grid solar PV power plants will not be situated in or near protected areas and reserves since they will be located in villages and are not expected to have biodiversity concerns. The establishment of project sites will also avoid involuntary physical displacement of populations or negative impacts on economic activities. The environmental and/or social risks and impacts of the activities under the project are predicted to have potentially limited adverse impacts, which are site-specific, largely reversible, and can be readily addressed through the implementation of generally accepted, good industry practices.

24. **Safeguard instrument and disclosure:** an environmental and social management framework (ESMF) has been prepared and submitted as the safeguards instrument for the project. The project will be implemented in multiple sites and the exact locations are as yet unknown. Once the scope and location of the mini-grids are identified, a fit-for-purpose environmental and social impact assessment (ESIA) in the form of a simplified environmental and social analysis will be conducted as per Mali's national requirements, the BOAD environmental and social safeguards requirement and GCF environmental and social safeguards standards. The site-specific ESIA documentation will be made available on the BOAD website, in various government offices, and in places that are accessible to the affected people, such as local administrative buildings (e.g. the townhalls and local chief offices) in areas where the solar PV plants are situated. BOAD has disclosed the ESMF on its website in English and in French, at least 30 days in advance of the BOAD or GCF Board decision, whichever is earlier, to adhere to GCF disclosure requirements.

25. **Environmental and social risks and impacts and mitigation measures:** the project is expected to have positive impacts on socio-economic activities, an improvement on the health of the rural villagers, an increase in income sources and improvement of learning conditions for pupils. The project would also result in avoidance of traditional fossil fuel-based energy systems resulting in reduced greenhouse gas emissions and reduced cutting of timber for fuel-wood, among others.

26. The main environmental and social impacts and risks of the project include those in relation to labour and working conditions, resource efficiency and pollution prevention, and community health, safety and security. No involuntary land acquisition and resettlement is expected since the project facilities will be installed in existing rights of way and in voluntarily provided small parcels of land (generally less than a hectare). The project by design will also

not be located in environmentally sensitive areas. Construction of the facilities will involve only minor excavation for installation of the solar PV plant and of the distribution lines.

27. Construction related impacts on communities include potential generation of dust and nuisance noise. Workers may also be exposed to minor occupational safety hazards and risks that could result in injuries. These include physical impacts and hazards such as noise, vibration, electrical and eye hazards, working at heights, as well as potential chemical hazards as regards handling the battery and its components. Appropriate planning, safety training and use of personal protective equipment will greatly minimize, if not avoid, such potential risks and impacts.

28. Risks and impacts during the operations phase include potential injury (electrocution or shock) to workers and the community as well as pollution caused by batteries and end-of-life solar PV panels. The project will largely mitigate these potential risks and impacts by: conducting appropriate occupational health and safety training of personnel; avoiding incidents and accidents; setting up limited public access and security protocol at the solar PV plants; raising public awareness in the community on safety of the solar PV plants; and the development of an effective waste management plan for the battery and solar panels.

29. No involuntary resettlement is expected to occur since beneficiary communities are expected to voluntarily provide the sites (of an area less than one hectare) in villages as a donation for the mini-grid solar PV power plant installation. Consultation with communities in selecting a consensual site free from disputes and conflicts will be undertaken and agreed upon and will exclude areas that would result in involuntary land acquisition and resettlement. The project provides for the conduct of an inclusive stakeholder engagement, meaningful consultation and information disclosure for all members of the communities in an inclusive, culturally appropriate manner including vulnerable and marginalized groups and individuals.

30. **Institutional arrangements:** the project will comply with BOAD standards and procedures for appraisal, approval, monitoring and supervision. It will conduct supervision monitoring and reporting on project activities and ensure the implementation of environmental measures in accordance with national requirements, as well as those of BOAD and of GCF.

31. AER-Mali will carry out environmental and social screening, coordination and supervision at the national level of implementation and will ensure that the selected operators comply with environmental and social obligations. AER-Mali will coordinate with Agence Malienne de Développement de l'Energie Rurale (AMADER) and will forge an agreement as regards E&S matters. Project-level monitoring will be conducted by the E&S personnel of the solar PV plants. Additional monitoring and supervision of the project will be carried out by the Environmental and Social Safeguard Unit of AER-Mali while external monitoring will be provided by DRACPN and DNACPN. Institutional strengthening measures on environmental expertise of AER-Mali, AMADER and of private operators as well as in assisting in the conduct of environmental and social assessments and in the development of maintenance manuals, including those that relate to occupational health and safety will be carried out with support from the Environment and Sustainable Development Agency. An indicative training programme with the corresponding budget is included in the ESMF.

32. **Stakeholder engagement:** a participatory approach was implemented in the consultation with all stakeholders and partners involved in the project. The consultations provided a venue for the engagement and involvement of different actors and stakeholders in identifying the overall environmental and social impacts of project activities. Further stakeholder engagement at the site level will take place during the conduct of the environmental impact assessment. Such meetings will bring together the local authorities, residents, and concerned organizations to identify the main issues and concerns and determine how these will be taken into account in conducting the assessment and in developing the management actions and plans.

33. **Grievance redress mechanism (GRM):** the ESMF provides for a mechanism for complaint management and grievance settlement. At the plant level, a Complaint Management Unit will be set up and at the AER-Mali level, a Complaint Management Committee will be established. The mechanism also provides for procedures for filing and duration for resolution of concerns. Local communities and other interested stakeholders may likewise lodge their grievances with BOAD through its Compliance Division.

4.2 Gender policy

34. The proposal contains a gender assessment and the funding proposal therefore complies with the operational guidelines related to the GCF gender policy and action plan. The assessment recognizes women as the main users of energy in households for tasks such as cooking. Energy is mainly derived from biomass sources, and rural areas have less access to electricity than urban areas.

35. Consultations conducted on the project by the AE identified the energy needs of women, some of which include access to a continuous supply of energy and the use of electrical equipment for income-generating activities. The project includes a component for supporting the productive use of electricity. This involves the provision of microfinance to access electricity-driven equipment to address some of the identified needs. It is encouraging to note that the project will provide microfinance to local communities for the productive use of energy and to contribute to income-generating activities and capacity-building in business management. The AE is recommended to reduce loan requirements or interest rates for women, if further assessments indicate that they have less access to credit and finance.

36. The proposal also contains a gender action plan with activities aligned with the project's components. Costs are also included in the gender action plan. The AE is advised to include its role in implementation of the gender action plan, in particular on monitoring and reporting. Key performance indicators have been included against each activity and some sex-disaggregated targets. Baseline information, which will be collected in the first quarter before the project commences implementation, will be used to update the gender analysis. It is advised that the AE should also use this information to rationalize sex-disaggregated targets in the gender action plan. Furthermore, targets should be for both men and women to ensure that both benefit from the project.

37. Targets are expected to be achieved by the estimated end date of the project. Additionally, the timelines for the implementation of activities in the gender action plan are not mentioned. Whilst targets are expected to be achieved at the end of the project, some activity timelines are too stretched and can be achieved before the end of the project to allow for reporting on achievements as the project is implemented.

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timelines are too stretched and can be achieved before the end of the project to allow for reporting on achievements as the project is implemented.

4.3 Risks

40. **Overall proposal assessment (medium risk):**

- (a) The funding proposal is to increase the access to the electricity in 70 rural communities in Mali through installation of a solar-powered mini-grid with aggregated capacity estimated to 4.83 megawatts (MW). GCF is requested to provide a sovereign loan of EUR 33.8 million and a grant of EUR 1.6 million. The AE is co-financing a loan of EUR 10.6 million. The GCF loan will be blended with AE resources and used for installation of a mini-grid system at a lower interest rate. The Government of Mali is the borrower and an owner of the assets. The concession agreements will be signed with the government and private sector operators for O&M;
- (b) Mali is an LDC and the key reason to seek GCF financing is to help the country to scale up investment in renewable energy. GCF financing is expected to help reduce the high cost related to the development of renewable energy projects; and
- (c) The success of the project depends on the capacity of the AE and EE to select construction companies and O&M companies. In addition, the tariff (estimated around EUR 0.24/kWh) needs to be affordable to the user communities. The project includes the training and technical assistance for the tender and procurement process under component 1 and support for the productive use of electricity for communities' affordability under component 3.

41. **AE/EE capability to execute the current programme (medium risk):**

- (a) BOAD, as a direct access AE, has been operating in the West African region for 40 years and has experience in financing solar-based rural electrification projects in Senegal, the Niger and Togo over the last three years. The AE has a team of engineers and specialists with relevant expertise for the implementation of the project and has been building a relevant track record in energy sector by managing an energy development fund. Although the AE doesn't have a track record of investment in mini-grid projects in Mali, the experience the AE has gained from other countries in the region is expected to support the implementation of this project; and
- (b) AER is the EE for this project. It will lead the construction (installation) of the mini-grids and carry out all associated procurement following BOAD procurement policies. It will also closely coordinate with AMADER, the government regulatory authority in charge of rural electrification. The EE has managed an average annual budget of USD 518,000.

42. **Programme-specific risks (medium risk):**

- (a) Construction and technical risk: the funding proposal states that the risk of failure or delay in the installation of equipment of the mini-grid power plants can be mitigated by the engineering, procurement and construction (EPC) contract with fixed-price and timebound conditions. In terms of the technical risk, the procurement of quality materials and using a proven technology will be critical. The funding proposal indicates that these risks will be mitigated through a joint agreement between BOAD and an engineering firm in charge of implementing the project for the use of products that are compliant with a certain level of quality/certificate (e.g. Conformite EURpeenne). The AE and EE are important for overseeing procedures and providing the technical assurance of the connections to the end users;

- (b) O&M risk: AMADER will enter into a 15-year concession agreement with private operators for the O&M. The private operators are expected to provide 20 per cent equity contribution. The funding proposal states that the current regulatory framework allows concession agreements; however, this was never operationalized at scale. A component of technical assistance in component 1 includes the writing of the standard concession agreement that will set the contractual relations between AMADER and O&M operators;
- (c) Use of transmission network: the project includes the construction of 124 kilometres of electricity distribution network in addition to the installation of solar PV. The network may be used for the transmission of non-renewable power generation along with the transmission of solar power. The funding proposal states that AMADER, the regulatory authority, will ensure that the distribution is used for renewable energy sources;
- (d) Credit risk and foreign exchange risk: the loan to the Government will be provided and repaid in EUR. The funding proposal states that the source of the repayment of the loan is the Government treasury and not from the cash flow to be generated from the project. Therefore, the Government will be partly exposed to currency fluctuation risk. Mali is an LDC; however, the long tenor and concessional pricing of GCF financing and the preferred creditor status of the AE are expected to partially mitigate the credit risk;
- (e) Tariff levels and ability to pay: the funding proposal mentions that a framework for the development of renewable energy, including setting up the feed-in tariffs and rural electricity tariff methodology, is not fully developed. The AE states that the Government is in the process of revising its regulatory framework. The tariff for the project will be around EUR 0.24/kWh and the end users will benefit from the concessionality of GCF through a lower electricity tariff. The funding proposal indicates that there is a willingness to pay despite the low level of incomes in the project target area. However, a study on the affordability of the suggested tariff is not provided; and
- (f) Security concerns: the AE has identified security concerns as a risk due to the outbreaks of political violence. The funding proposal states that Mali's membership in the West Africa Economic Monetary Union provides for relatively low exchange rates and transfer risk, and these can positively contribute to a consolidation of Mali's political situation and secure stability in the future. An insurance cover (if available) can potentially minimize the loss of the equipment due to fragile security situation. The Government will play an important role in providing security support for the project.

43. **Project viability and concessionality**

- (a) The viability of the project depends on the ability-to-pay of the end users and the possibility of attracting more private investment into the sector. Without GCF financing, the Government will have to take loans at market rates, which are costly considering it has a high fiscal deficit and has been struggling to mobilize the necessary financial resources. The AE and the Government are requested to pass on GCF concessionality to end beneficiaries through a reduced tariff.
- (b) The economic analysis is provided under the same assumptions for the financial analysis but with 85 per cent of the financial costs, considering the transfers contained in the costs exclude taxes. The analysis provided an internal rate of return of 16.61 per cent. A sensitivity analysis has been carried out under different scenarios, a 10 per cent increase in investment costs, a 10 per cent decrease in benefits and a one-year delay in project execution. The analysis shows the economic viability of the project resulting in an internal rate of return of 16 per cent, 15.82 per cent and 15.98 per cent, respectively.

44. **Compliance risk (medium risk):**

- (a) This project presents an elevated level of money-laundering/terrorist financing (ML/TF) risk particularly in the context of component 3 in evaluation of private sector

investors, microfinance institutions and end-user loan applications alike. In line with the AE anti-ML/TF due diligence and risk management obligations, under the accreditation master agreement (AMA), the AE is expected to ensure that GCF resources are robustly safeguarded against these risks. Accordingly, as per the funding proposal, the AE shall undertake ML/TF due diligence as a mandatory element of the funding eligibility criteria and take into account ML/TF and/or sanctions risk considerations throughout the project cycle. In addition, the funding proposal contains a description of the project-specific measures proposed by the AE to evaluate and effectively mitigate these risks as prescribed by its internal policies and procedures;

- (b) A combination of the highlighted controls is expected to reduce the likelihood of ML/TF exposure; and
- (c) Furthermore, compliance has conducted in-house enhanced due diligence on the key project stakeholders with satisfactory results.

45. **GCF portfolio concentration risk (low risk):**

- (a) In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results areas and single proposal is not material.

46. **Conclusion:**

- (a) It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall programme	Medium	- GCF is requested to provide a sovereign loan to the Government of Mali - The concessionality of GCF financing is required to be passed on to the end beneficiaries - BOAD is the accredited entity for the funding proposal and has preferred creditor status with the Government of Mali
Accredited entity/executing entity capability to implement this programme	Medium	
Project-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	Medium	

4.4 Fiduciary

47. The Ministry of Energy and Water will be the EE of the project. It has delegated execution of component 2 to the renewable energy agency, AER. AER will manage the project and shall see to the programming, coordination, monitoring and supervision of all project activities.

48. The AE, BOAD, has conducted technical and financial capacity assessments of the entities involved in the project as per the requirements of the AMA and in line with BOAD policies and procedures. BOAD deems the national entities involved in the project to be capable of carrying out the execution and operation of the project component activities, under the assistance and support of BOAD.

49. A Project Management Unit shall be set up within AER-Mali and managed by a project coordinator, who shall be supported by nine other experts. The Project Management Unit will coordinate the conduct of technical studies, the preparation of tender and consultation documents, tender evaluations and procurement reports.

50. The Government of Mali will be the borrower and will take the credit risk through a sovereign guarantee. BOAD will monitor the progress implementation based on quarterly

reports submitted by the borrower, and field supervision mission's reports. An implementation team for this project will be constituted within BOAD and will be in charge of due diligence. It will coordinate the preparation, design, financing and implementation as well as the monitoring and evaluation aspect of the project.

51. The project's audits will be undertaken in accordance with the BOAD guidelines for financial reporting and auditing of projects. An independent external auditor will be recruited based on terms of reference acceptable to the bank for the entire duration of the project. The financial statements will be audited in accordance with international auditing standards. The AE will also ensure a "know-your-customer" standard due diligence process, including anti-money-laundering and other evaluations of sponsors, is followed thoroughly.

52. The BOAD guidelines for procurement will be applied to the project's procurement. The bank will conduct an analysis of the procurement files and provide no objection.

4.5 Results monitoring and reporting

53. This project proposes to promote rural electrification through isolated solar PV systems as a low-carbon and resilience solution to the effects of climate change in the energy sector in Mali. The project aims to achieve this by: (i) strengthening the capacity of public institutions engaged in rural electrification to accelerate solar-powered electrification; and (ii) increasing the rural population's access to electricity in 70 communities through installation of isolated solar PV mini-grid systems for a total capacity of 4.83 MW. The project hopes to catalyse the development of an efficient solar market in Mali and help the country to meet its renewable energy and greenhouse gas reduction targets. The project will: help avoid 41,089 tCO₂ per year, and 1,027,227 tCO₂ for the entire life of the equipment (25 years); benefit an estimated 310,698 direct beneficiaries (50 per cent of whom are women) at the project site; and help around 31,000 new rural households to access solar-powered electricity. In its first tranche, the project is expected to achieve 80 per cent of the impact (821,782 tCO₂eq avoided) and benefit 50 localities.

54. The logical framework (section H.1) conforms substantially to GCF standards and requirements and has been cleared through Secretariat review.

55. The implementation timetable (section C.8) includes proper sequencing of proposed activities and broadly conforms with the Secretariat format.

56. The funding proposal's theory of change is considered adequate as is the monitoring and evaluation section, which clearly outlines the frequency of reporting and refers to the other reporting requirements to GCF as per the AMA, for example, annual performance reporting, midterm and terminal evaluations, and project completion reports.

4.6 Legal assessment

57. The AMA was signed with the AE on 3 April 2017 and became effective on 31 August 2017.

58. The AE has not provided a certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project. It is recommended that, prior to submission of the funding proposal to the Board: (a) the AE has obtained all its internal approvals; and (b) GCF has received a certificate or legal opinion from the AE in form and substance satisfactory to GCF confirming that all final internal approvals by the AE have been obtained, and that the AE has the authority and capacity to implement the project.

59. The proposed project will be implemented in Mali, a country in which GCF is not granted privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, whose risks need to be further assessed. The Secretariat submitted a draft privileges and immunities agreement to the Government of Mali on 7 December 2015 and a meeting with the National Designated Authority was held on 9 October 2018 during the Global National Designated Authority Conference. No response on the draft agreement has yet been received.

60. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF are made only after GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

61. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Delivery by the AE to GCF of a certificate or legal opinion within 120 days of the Board approval confirming that it has obtained all its internal approvals and that the AE has the authority and capacity to implement the project;
- (b) Signature of the funded activity agreement in form and substance satisfactory to the Secretariat within 180 days from the date of Board approval or the date the AE has provided a certificate or legal opinion confirming that it has obtained all its internal approvals and has the capacity and authority to implement the project; and
- (c) Completion of legal due diligence to the satisfaction of the Secretariat.

Independent Technical Advisory Panel's assessment of FP102

Proposal name:	Mali Solar Rural Electrification Project
Accredited entity:	Banque Ouest Africaine de Développement (BOAD)
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential *Scale: High*

Brief Description of the programme concept, goals and objectives

1. In order to understand the rationale of the conclusions reached by the independent Technical Advisory Panel (TAP) in this review, the concept of the project proposed in this submission will need to be summarized as concisely as possible. The summary is presented as follows:
 - (a) Mali's energy situation is characterized by a biomass-dominated balance sheet that accounts for 76 per cent of primary energy, followed by 20 per cent for hydrocarbon imports and 4 per cent for electricity;
 - (b) Energy consumption in Mali is dominated by the residential sector, which basically relies on biomass. Wood and charcoal are mainly used as cooking or heating fuels, a situation that results in strong pressure on the country's forest resources;
 - (c) According to national statistics for 2017, national average for access to electricity was 42 per cent. This national average disguises the fact that a sizeable portion of Malians who live in the rural areas do not have access to electricity, the figure for this section of society being reported as 19 per cent in 2017;
 - (d) Improving access to electricity in rural areas in Mali has been a key pillar of the sustainable development goals of the country. Strategies that have been employed in recent past include:
 - (i) The gradual extension of the Malian grid to rural areas – the Malian grid is owned and operated by the Malian National Electricity Company (EDM-SA);
 - (ii) Supplying some rural consumers with electricity through isolated EDM-SA off-grid facilities (isolated EDM-SA Centres); and
 - (iii) Also supplying electricity through off-grid supply facilities operated by contractors engaged by the Government of Mali under its rural electrification programme;
 - (e) According to EDM-SA data, energy security in Mali suffers from a chronic gap even on the national grid. Moreover, it has been concluded that it is very unlikely that over the next ten years that the expected expansion of the national electricity network will be able to ensure that a significant proportion of the low-income population living in isolated areas will be connected to the network. This situation is expected to create an important market for off-network electrification schemes;

- (f) The supply of electricity to off-network consumers is mainly ensured by local energy service providers, which operate mini-grids in isolated rural areas. In 2015, the country had 160 isolated mini-networks. To this day, the production technologies used in these mini-grids is basically diesel-run units (in 2015, 96 per cent of the electricity produced off-network was through diesel);
 - (g) The negative impact of high and perennially increasing fossil fuel prices, in this case diesel, which must be imported into Mali, with its negative balance of payment impact, may not allow electricity access in these rural areas to increase much beyond what it has been for some time. This calls for another strategy for rural energy access in this least developed country (LDC);
 - (h) With the decreasing cost of delivery of electricity from solar power technologies (photovoltaic (PV) and concentrated solar power (CSP)), locally sourced solar power projects are needed to achieve a significant increase in electricity access in the rural areas of Mali; and
 - (i) This current intervention is planned for consumers in rural Mali without access to the main grid, including those both currently served and not currently served by off-grid mini-grids.
2. The main objective of this project is to promote rural electrification through isolated solar PV systems as a low-carbon and resilience solution to the effects of climate change in the energy sector in Mali. This project intervention has been designed to achieve the following goals:
- (a) Strengthen the capacity of public institutions, such as Malian Agency for the Development of Household Energy and Rural Electrification (AMADER), the Agence des Energies Renouvelables (AER), and the Rural Electrification Fund (FER), which are normally engaged in rural electrification to accelerate solar-powered electrification;
 - (b) Increase the rural population's access to electricity in 70 communities through the installation of isolated solar PV mini-grid systems for a total capacity of 4.83 megawatts (MW);
 - (c) Catalyse the development of an efficient solar market that will enable Mali to meet its renewable energy and greenhouse gas (GHG) reduction targets;
 - (d) Provide a total shift in energy production by replacing fossil fuel additions with renewable energy solutions;
 - (e) Contribute to the regulatory framework through capacity-building, demonstrations of technologies in areas where not available, putting in place an institutional framework for FER to strengthen its capacity to further promote the deployment of renewable energy; and
 - (f) Catalyse the inclusion of the private sector in the programme through the incentivization of workable and long-term operation and maintenance (O&M) concessions for them for the mini-grids.
3. The key issues that characterize this project intervention in Mali can be summarized as follows:
- (a) A key United Nations Sustainable Development Goal in many nations of the world, Mali included, is the need to increase energy access. In Mali, a key access goal is to improve the national average electricity access from the current 42 per cent to a much higher level that will promote development. More importantly is the recognition that electricity access in rural Mali is abysmally low at 19 per cent;

- (b) Efforts geared towards improving this figure in the rural areas of Mali include the following:
- (i) Extending the EDM-SA national grid to rural areas;
 - (ii) Building more EDM-SA mini-grids to serve rural areas without access to the main grid;
 - (iii) Encouraging the involvement of more private sector entities, which have operations in rural areas, build onsite generation facilities and supply excess power to rural consumers; and
 - (iv) Encouraging mini-grid facilities that are operated under contract with the Malian Government to deliver electricity to consumers in rural area;
- (c) Many of the facilities (more than 90 per cent) that supply electricity to rural consumers are powered by fossil fuels (mostly diesel), with their high GHG emissions and negative impact on the country's balance of payments, especially in a country like Mali where all the fossil fuel consumed is imported;
- (d) Key strategies that are incorporated in the proposed GCF/West African Development Bank (BOAD) intervention include the following:
- (i) The plan to utilize the very abundant solar irradiation in Mali for electricity generation, which will displace the status quo fossil-fuel-based generation in the current way power is generated to supply consumers in rural areas;
 - (ii) The acceptance of the fact that rural electrification will continue to be funded from public resources and, as such, the Government of Mali will still be responsible for the way finance is accumulated in this intervention;
 - (iii) The Government of Mali intends to use this project to incentivize the continued participation of rural energy service providers, which in the status quo ante operate the mini-grids fueled by fossil fuel power plants, in the O&M of rural mini-grids that will be developed as part of this programme;
 - (iv) A portion of the fund for this project will be targeted at developing (and where it already exists – strengthening) the productive use of electricity. A key reason for the failure of rural electrification programmes is the inability of programmes to attract consumer interest and thus develop consumer demand beyond the residential to small business and commercial users, who are referred to as productive users of power. In this project, a programme to incentivize the development of these productive users will be funded;
 - (v) To this end, local financial institutions (microfinances) will be provided with funding to put in place credit facilities for productive users to purchase “equipment or investment credit”. The types of energy equipment for productive use will include: electric tools for carpentry, woodworking and small industry; sewing machines; welding machines; battery charging equipment; refrigeration and ice cube production; seed oil presses, grain mills, crop drying equipment and other equipment for the value-added processing of local agricultural produce (honey, mango, cashew nuts, etc.);
 - (vi) By supporting the acquisition of equipment for productive use by rural populations, the project will contribute to increasing the use of energy for income-generating activities, which will stimulate the sale of electricity and economic development in the beneficiary localities;
 - (vii) Another important component of the strategy will be the use of part of the fund (grant fund) to strengthen the enabling environment for the development of

solar-powered mini-grids in rural areas (70 communities identified). The focus will include the following:

1. Strengthening the regulatory framework for renewable power development;
 2. Strengthening the policy environment, including tariff policy, power purchase agreement development and negotiations etc.; and
 3. Building of local capacity in the technology and engineering of solar power systems that will be adopted.
4. Greenfield mini-grid power facilities will be developed for each of the 70 rural communities already identified. Projects in these communities will be made up of the following:
- (a) 14 localities in category 1 (population less than or equal to 2,000 inhabitants) will each be supplied by a 30 kilowatt peak (kWp) PV solar power plant and a 5 km-long three-phase low voltage (LV) grid;
 - (b) 21 localities in category 2 (population between 2,000 and 4,000 inhabitants) will each be supplied by a 50 kWp PV solar power plant and an 8 km-long three-phase LV grid;
 - (c) 27 localities in category 3 (population between 4,000 and 7,000 inhabitants) will each be supplied by an 80 kWp PV solar power plant and a 4.5 km-long three-phase LV grid; and
 - (d) 8 localities in category 3 (population of 7,000 inhabitants or more) will each be supplied by a 150 kWp PV solar power plant and a 12 km-long three-phase LV grid.
5. The total installed capacity of solar power capacities that will be installed under this intervention has been estimated to be approximately 4.83 MW. A summary of the expected funding is presented as follows:
- (a) The total funding that will be provided by both GCF and BOAD is estimated to be about EUR 46.2 million;
 - (b) This total fund will be contributed as follows:
 - (i) GCF loan will be EUR 33.7 million as senior debt;
 - (ii) GCF will also provide a grant of EUR 1.9 million; and
 - (iii) BOAD will provide a loan of EUR 10.6 million as senior debt;
 - (c) At the request of GCF, the programme will be implemented in two tranches. Tranche 1 will cover 50 communities and will be funded by a total of EUR 34.2 million, including PMC, and the expected GCF funding is EUR 25.9 million. BOAD will provide a co-financing in loan for EUR 8.3 million. In addition, it will submit a request for Tranche 2 funding after the midterm evaluation, after which the balance EUR 12 million will be made available.
6. The structure of the programme as proposed in this funding proposal can be summarized as follows:
- (a) Component 1: capacity-building of rural electrification institutions and technical assistance;
 - (b) Component 2: detailed engineering and installation of solar-powered mini-grids; and
 - (c) Component 3: support to the productive use of electricity.

Climate impact potential of this project

7. The primary impact potential of this programme will be the reduction of GHG emissions through the displacement of fossil fuel electricity generation by the solar power facilities that will be implemented as a result of this programme.
8. It has been estimated that the solar mini-grid that will be implemented in the four category 1–4 communities will deliver a peak capacity for all categories totalling 4.83 MW of new solar power mini-grid, with a peak generation of about 10.578 gigawatt hours (GWh) of clean electricity annually.
9. Electricity (peak) expected to be delivered by this project at each of the four community categories¹ can be summarized as follows:
 - (a) Community type 1: 0.9198 GWh;
 - (b) Community type 2: 2.2995 GWh;
 - (c) Community type 3: 4.7304 GWh; and
 - (d) Community type 4: 2.6280 GWh;
10. It has been estimated that the project will directly avoid GHG emissions amounting to about 41,089 tonnes of carbon dioxide equivalent (tCO₂eq) per year, equivalent to 1.02723 MtCO₂eq over 25 years, that would have been emitted by the status quo fossil fuel system.
11. The GHG emissions reduction methodology that was utilized in this calculation can be aptly summarized as follows:
 - (a) Methodology AMS I.L as approved for CDM projects² by the Clean Development Mechanism (CDM) Executive Board of the the United Nations Framework Convention on Climate Change (UNFCCC) has been used in the estimation of the GHG emissions reduction implications of this project;
 - (b) This methodology is applicable to the electrification of a community achieved through the installation of renewable electricity generation systems that displace fossil fuel use, such as in fuel-based lighting systems, stand-alone power generators and fossil fuel based mini-grids. This methodology is applicable to implementation of individual, renewable-energy systems or mini-grid activities;
 - (c) This methodology is applicable in situations where consumers that were not connected to a national/regional grid prior to project implementation are supplied with electricity from the project activity. It is also applicable to situations where a fraction of consumers that are supplied with electricity from a fossil-fuel-based mini-grid prior to the implementation of the project are now supplied with electricity from the project activity;
 - (d) At least 75 per cent (by number) of the consumers connected to the project's renewable electricity generation system(s) shall be members of households;
 - (e) For each category of consumer, the emission factors developed in the UNFCCC methodology has been utilized in this calculation; and
 - (f) In addition to the GHG emissions reduction impact of the planned programme, the expected numbers of consumers to whom the programme will deliver clean electricity has been estimated at approximately 310,698 people. This number was arrived at by using the estimate of consumers that will be connected to the mini-grids.
12. Given these impacts, the independent TAP assessed the scale of the impact potential metrics of this programme as "high".

¹ A description of the characteristics of a community category is presented in section 1(d).

² Available at <https://cdm.unfccc.int/methodologies/documentation/meth_booklet.pdf#AMS_I_L>.

1.2 Paradigm shift potential

Scale: High

General overview of paradigm shift

13. The projects that will be covered by this GCF/BOAD funding intervention will involve the development of off-grid solar power systems that will supply power to the 70 mini-grids. In terms of the technology that will be implemented, it can be argued that these technologies (solar PV and CSP) are not new as they have been implemented successfully in many parts of the world. However, when one compares this technology intervention with what has happened off the country's national network, especially in existing community-based power mini-grids, the truth is that the adoption of solar technology has not happened at a rate that is consistent with the impressive primary resources in the country. Barriers to solar power technologies' delivery of power to the main grid as well as off-grid networks in Mali in the status quo ante are well documented in the funding proposal and include: technical, financial, regulatory and policy issues. A key goal of the proposed project funding is not only to develop and implement the solar power facilities but also to reduce and, where possible, completely eliminate these barriers. For example, a key component of the proposed intervention is the utilization of concessional financing from GCF to deliver solar power facilities at the mini-grids already conceptualized to serve 70 rural communities. The concessional loan available from GCF is expected to render these projects financially feasible, and to deliver electricity to these consumers with minimal public subsidy. The success of this intervention may have sound paradigm shift potential in that it will demonstrate a workable strategy by which public financing can deliver rural electrification projects. The metrics under which such paradigm shift potentials are enhanced by the proposed intervention are discussed in the following subsections.

Potential for scaling up and replication

14. As stated in the earlier section of this report, the solar power technologies that will be employed by this programme are well-developed internationally and have been deployed in many countries in the past. Since the focus of the programme in this intervention is to put in place activities that will reduce or completely eliminate barriers that have so far prevented the expanded deployment of these mini-grid technologies in Mali, it can be safely concluded that once these barriers are effectively mitigated by the programme, then the potential for scaling up of the activities, and replicating the solar power projects in other regions of Mali and even in other countries in the francophone subregion to develop workable mini-grid projects will be enhanced. Lessons learned from the technical assistance that is utilized to enhance the technical, regulatory, policy and market capacities for solar mini-grid development in Mali should be available if the decision to scale up or replicate the underlying projects is taken especially after this intervention. It is expected that the effect of the concessional funding from GCF will not only be that the underlying solar powered mini-grids for rural communities can be feasibly implemented, but that local rural energy service providers, who will be given opportunities to take on the O&M function responsibilities, would have learned so much from the process to be able to replicate and scale up their O&M functions in other similar projects (rural mini-grids) that will be replicated in the country in the future. Replication of the successful project in other countries in the francophone subregion of West Africa will also provide lessons and best practices for rural energy agencies in the subregion under the framework of the CLUB-ER (Club of African National Agencies and Structures in charge of Rural Electrification). BOAD, according to the submission reviewed by the independent TAP, has stated its commitment to developing a case study replicable in the eight francophone countries in Africa and across the West African Economic and Monetary Union (WAEMU) countries.

Potential for knowledge-sharing and learning

15. The potential for knowledge-sharing and learning in this proposed GCF/BOAD funded rural solar power mini-grid will be enhanced by thoroughly targeted capacity-building, awareness raising activities and the connection of a diverse range of relevant sector stakeholders under component 1. The activities in component 1 have been designed to encompass the following knowledge-sharing and learning characteristics:
- (a) Multiple stakeholders (public, private and microfinance institutions(MFIs)) will work together in a collective learning process;
 - (b) The project will deliver training and knowledge to public institution (AER, AMADER) staff members on climate change issues and renewable energy technology development;
 - (c) The productive use of the electricity component will transfer tremendous knowledge to the MFIs, whose knowledge will be enhanced on the evaluation of financing investments in energy equipment needed by rural artisans, who will be able to elevate electricity demand in the new mini-grid areas. This will surely change the risk perceptions towards these new financing opportunities; and
 - (d) Potential micro, small and medium-sized enterprise (MSME) clients will also benefit from technical support on how to structure climate business plans to make them bankable.
16. These activities will create a high potential for knowledge-sharing and learning.

Contribution to the creation of an enabling environment

17. The activities that will be carried out in this project will contribute to the creation of an enabling environment for the development of rural renewable energy mini-grid infrastructure in Mali as follows:
- (a) Rural electricity supply infrastructure is usually financed by public funds, which more than often is usually a budget item. With the increasing scarcity of public funds, rural electricity supply has been negatively impacted. The Government of Mali will be responsible for the debt that will be provided for this project and will take responsibility for servicing the debt. This will create an enabling environment for financing these infrastructures outside of bilateral and multilateral grants that was used in the status quo ante;
 - (b) The technical assistance activities will achieve a framework in Mali that will enable AMADER (the only government authority in charge of rural electrification regulation, setting of policies under the guidance of the Ministry of Energy) to unlock private sector resources for rural electrification activities while also including their participation in O&M activities. This will cause a shift from the current paradigm, where the public and private sectors are competing, and pave the way for a public-private partnership model; and
 - (c) It will contribute to structuring the solar based rural electrification market through financial mechanisms such as financial guarantee to remove the barrier of a lack of a financial service adapted to support productive energy users and de-risk loan demands for rural area productive users of energy.
18. The project will therefore contribute to the creation of an enabling environment for the development of solar-based mini-grids in other parts of Mali and offer good lessons for the development of rural power projects in other francophone nations. This will enhance electricity access in Mali and the region.

Contribution to regulatory framework and policies

19. GCF financing is critical to the successful structuring of finance for this project so as to address some of the barriers faced by private sector investors. One of the key barriers that has been identified is the weak regulatory framework in place that has not encouraged private sector involvement in renewable energy investment, especially in the development of mini-grid infrastructure to serve rural consumers in Mali. The financing from GCF is expected to help to reduce the risks of investment for these private entities. Concessional financing from GCF can offset the initial high cost related to the barriers and risks as described in the project feasibility study report. Component 1 of the proposed project will focus on building the capacity of national institutions such as AMADER, which will definitely assist in strengthening the legal, technical, policy and regulatory capacities of such institutions. It is expected that at the successful completion of this programme, the capacities of institutions such as AMADER and AER will be strengthened. The existing policy and regulatory environment allows for the participation of entities in the O&M of rural mini-grid facilities. However, because existing policies and regulatory frameworks were weak, the activities of private sector entities were not at scale. It is expected that activities planned in this intervention through component 1 will build the capacities of these public institutions to be able to create regulatory framework and policies that will enable the development of rural solar mini-grids at scale.

20. The independent TAP therefore concluded that based on the expected performance of the proposed project as discussed above, the programme will deliver a paradigm shift that will ensure the successful development of the planned 70 solar mini-grids and also catalyse the replication of the idea in other areas of Mali and other countries in the francophone subregion. This will catalyse a deep and accelerated departure from use of fossil fuels to the delivery of electricity to these rural communities through increased utilization of low-carbon solar power options (PV technologies). The independent TAP has therefore scored the paradigm shift metric as “high”.

1.3 Sustainable development potential

Scale: High

21. The major development contribution of this project’s efforts is expected to be the utilization of cleaner solar power technology at the rural mini-grid level in Mali through the enabling of a good financing, regulatory and project development environment in the country. The project is expected to deliver environmental and socio-economic co-benefits as well as gender-sensitive development impacts. These are summarized as follows:

- (a) Environmental co-benefits:
 - (i) Reduction of air pollution due to fossil fuel consumption and improvement of air quality thanks to the promotion of renewable energies;
 - (ii) Reduction of GHG emissions; and
 - (iii) Reduction of pollution by waste oils at thermal power plants; and
- (b) Socio-economic and gender-sensitive co-benefits:
 - (i) Increased socio-economic activities;
 - (ii) Improvement of the health and hygiene conditions of local populations;
 - (iii) Improvement of the living conditions and comfort of local populations;
 - (iv) Access to public lighting and improvement of security conditions;
 - (v) Increase in the duration of the supply of electricity;
 - (vi) Increase of the service and the access rate for the safety of people and goods;
 - (vii) Better conservation of pharmaceutical and food products;

- (viii) Improvement of the permanent drinking water supply to the populations;
- (ix) Better lighting for the work environment of women;
- (x) Improvement of household income levels through the creation of income-generating activities;
- (xi) Improvement of the literacy rate;
- (xii) Development of the emergence of manufacturing and processing industrial units;
- (xiii) Opportunity for local people to benefit from capacity-building actions; and
- (xiv) The focus on gender in ownership and management of the energy services centres will ensure the inclusion of all gender and youth. So the improvement of livelihoods and other development benefits will be shared fairly.

22. Based on these co-benefits that are expected to be generated by this intervention, the independent TAP concluded that the proposed programme has a “high” sustainable development potential.

1.4 Needs of the recipient

Scale: High

23. The intervention will reduce the vulnerability of the country and beneficiary group.

24. Mali is classified by the United Nations as an LDC³ and according to the Notre Dame Global Adaptation Initiative Country Index,⁴ it is among the most vulnerable countries to climate change. Key challenges faced by Malians that arise as direct and indirect consequences of climate change include low access to electricity, high cost of electricity and overdependence on imported fossil fuels for electricity generation (grid and off-grid). Low access of electricity has continued in the rural areas as power on mini-grids comes mostly from imported fossil fuels. Only 1.8 per cent of rural Mali’s population has access to electricity. As such, the majority of rural inhabitants who do not have access to modern electricity utilize mostly fuelwood and other biomass products to meet their energy consumption needs. Climate change effects, such as drought and decreasing rainfall, have a negative impact on biomass productivity, thus extending the time spent by rural households in fetching wood and other biomass products to meet energy needs. This will continue in the absence of this intervention, making normal living very difficult in these rural areas of Mali. The inadequate availability of public funds to invest in energy infrastructure to meet the growing demand for electricity in these rural areas will mean that Mali will continue to be vulnerable to these challenges. So, in the absence of this intervention, rural consumers will continue to face limited access to electricity. Therefore, the proposed solar-powered mini-grid projects will address the needs of these consumers by improving their access to clean and reliable energy.

Economic, social and institutional needs

25. The intervention will strengthen the economic development of Mali and especially that of the affected rural population in Mali.

26. Although Mali, like most of the seven other francophone countries in West Africa, has witnessed economic growth in recent years, which averaged more than 5 per cent of gross domestic product growth per annum, this impressive economic performance has not translated to reduction in poverty in the countries. In Mali, more than half of the population still live below

³ United Nations, Committee for Development Policy, List of Least Developed Countries (as of December 2018). Available at: <www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc_list.pdf>.

⁴ Available at <<https://gain-new.crc.nd.edu/ranking>>.

the national poverty line, with a significant proportion of the population living in the rural areas of Mali (approximately 98 per cent) still not having access to modern energy, including electricity. It is recognized that poor access to electricity is one of the key barriers to better economic growth and social development in the rural areas of the country. Therefore, the improvement of access to electricity in both the rural and urban areas of the country has long been recognized as a good development strategy in Mali. There is definitely a need for a sharp improvement in electricity access for rural Mali's inhabitants and the current intervention not only fulfils this need but it is also doing so through the use of a low-carbon system. The good news is that Mali is well endowed with solar energy resources, and, as such, solar power offers a credible alternative pathway to improved electricity access and hence a credible way of meeting the social and economic development needs in the country, especially in rural areas. For this to happen, the already identified barriers – financial, technology, regulatory and others – must be properly mitigated. The proposed project intervention will initiate a systemic mitigation of these key barriers and therefore contribute to unleashing a paradigm shift to a low-emission development pathway through the mobilization of climate finance at scale for the solar market. It will accelerate private sector involvement in the O&M of rural solar power based mini-grid (well above the level of the status quo ante), which will enable market transformation and contribute to the economic and social needs of a large segment of the population residing in the rural areas of Mali. The funding structure of the proposed programme will also contribute to strengthening the institutional capacity required to back up the development of mini-grids that will be essential for ramping up rural electrification in Mali.

27. The intervention will strengthen the social development of Mali and especially that of the affected rural population.

28. The absence of modern energy hinders community development; and climate change complicates the situation. Poor lighting and indoor air pollution increases the prevalence of health complications, including respiratory tract infections and heart complications (16,900 people die in Mali annually due to indoor air pollution, WHO, 2012⁵), which dramatically reduces life expectancy and puts a strain on the national health budget.

29. The introduction of modern energy in rural communities will lead to various social benefits, including the following:

- (a) Better social services, such as medical services and education;
- (b) Increased safety in communities following the introduction of street/public lighting;
- (c) Better health especially of women and girls following improved indoor lighting and decreased air pollution as well as reduced distances to cover in search of energy services and/or resources;
- (d) Better sanitation and increased availability of clean, safe drinking water;
- (e) Increased access to and improved use of basic social services, such as education and healthcare (lighting/heating/cooling of schools and health centres through solar panels);
- (f) Increased access to water in rural areas (through solar PV pumping systems); and
- (g) Solar home systems will also prevent health issues related to the indoor use of firewood and improve household safety by decreasing domestic accidents caused by candles and kerosene lamps.

30. The intervention will provide more robust alternative sources of financing of solar powered mini-grids in rural Mali.

⁵ Funding proposal, page 38.

31. A key barrier to a low carbon path in these nations as an approach to improving energy access in these nations has been the lack of the funds (local equity, loans/debt – with medium to long maturity and lower interest rates) needed to implement these clean power projects. BOAD as an important regional financing institution has a strategic objective to increase its portfolio of investments in solar with a reasonable participation of private sector actors. This has not materialized as BOAD has been unable to mobilize adequate funds in the status quo ante period, at the level needed and at terms that will enhance the participation of the private sector in this endeavour. The proposed project will create an enabling environment for a better understanding of how to mobilize support for public funding of mini-grid projects that are powered by renewable energy systems, especially given the dwindling public financing of the fossil fuel alternative. This path will also deliver GHG emissions reductions as a result of the displacement of fossil fuel-based generation, thus contributing to the achievement of the nationally determined contribution (NDC) of Mali.

32. The intervention will strengthen institutions and implementation capacities in the targeted countries.

33. Apart from the financial barrier discussed in the earlier section, weaknesses in institutional, regulatory, legal and policy frameworks (e.g. tariff structure, policy on the concessional participation of private actors versus public utilities in the O&M of mini-grids in rural areas, etc.) are known to be barriers to the implementation of solar power mini-grid facilities in Mali. The proposed GCF/BOAD programme grant funding, will provide technical assistance (TA) to relevant institutions in these countries so as to reduce or completely eliminate the negative impacts of regulatory and policy barriers. For example, as part of this grant funding, BOAD will carry out training and awareness-raising activities on the project for entities in Mali responsible for rural electrification policies and regulations. These entities will include NED, AMADER and AER-Mali. The rationale of component 1, which has been designed to address these institutional weaknesses and disseminate information to the public and other pertinent stakeholders in the country is aimed at supporting the implementation of regulatory and policy regimes that provides clear and predictable rules for solar mini-grid project development and replication in Mali. The TA under this programme is expected to stimulate the environment and the need in Mali for a robust takeoff and the successful implementation of the current projects and replication of the success in other parts of Mali. This will partly be achieved through institutional strengthening that will be provided by the TA component of this intervention.

34. Given the various country needs that will be satisfied by this GCF/BOAD intervention in the development of a sound rural mini-grid in Mali, the independent TAP has concluded that the country need of Mali and especially that of rural electricity consumers can be rated as “high”.

1.5 Country ownership

Scale: High

35. Country ownership of the proposed GCF/BOAD intervention is discussed in this review along the following submetrics.

Alignment of the proposed programme with national climate strategies

36. Key documents where the national climate strategies of Mali have been articulated in line with the goals of the proposed intervention are the intended nationally determined commitment (INDC) and the nationally appropriate mitigation action (NAMA). Both documents, which have been submitted as national documents to the UNFCCC Secretariat aim to reduce GHG emissions. For example, the NAMA aims to reduce GHG emissions by 1,285,034 tCO₂ per year through renewable energy production (e.g. with hydro, wind, biomass and solar PV). Mali's INDC pointed to the need to implement solar power facilities to the grid and off-grid in these

nations as a key strategy to national contribution to the international goal of global GHG emissions reductions. A recent UNFCCC publication⁶ listed 19 priority climate change projects that are likely to be pursued in Mali. This list included the following projects that are directly related to this GCF/BOAD intervention:

- (a) Contribution to barrier removal for the promotion of the use of solar energy in Mali; and
- (b) Promotion of income-generating activities and the development of mutual assistance.

37. Measures spelt out in these documents, especially for the energy sector, included GHG emissions mitigation through the implementation of solar power systems, especially to the power grids and rural mini-grids, as a means of increasing access to electricity in both the rural and urban areas of Mali. Therefore, it can be concluded that the proposed GCF/BOAD programme is aligned to the country's climate strategies.

Capacities of accredited and executing entities to successfully deliver this programme

38. BOAD will serve as both the accredited and executing entities for this project. Some of the relevant attributes of BOAD that points to the organization's capacity to deliver on this programme are summarized as follows:

- (a) BOAD is the common development finance institution of the member countries of the WAEMU. It was established by an agreement signed on 14 November 1973 and became operational in 1976. Member countries include Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, the Niger, Senegal and Togo;
- (b) BOAD is an international public institution the purpose of which, as provided under Article 2 of its Articles of Association, is to promote the balanced development of its member countries and foster economic integration within West Africa by financing priority development projects. Key examples of investment projects that have been funded through the support of BOAD in the francophone subregion of West Africa include the following:
 - (i) The bank's financial support has made it possible not only to increase the power generating capacities in member countries, but also to erect more than 3,527 km of interconnection lines between them, and in addition to reinforce power generation, transmission and distribution facilities on the interstate grids for a nominal power of over 1,350 MW;
 - (ii) As at 30 September 2016, the aggregate amount of funding provided by the bank in the energy sector stood at 863.9 billion African Financial Community (CFA) francs (XAF), excluding the Energy Development Fund:
 1. BOAD provides financial assistance to WAEMU member countries to support their development efforts. Such assistance targets various areas, including the organization of workshops, conferences or seminars, the training of government officials and the implementation of social projects; and
 2. The bank also grants an interest rate subsidy for non-commercial public-sector loans to member countries. Also, BOAD participates in various round-table meetings with a view to financing national development programmes in member countries; and

⁶ Funding proposal, page 39.

- (iii) The discussion above points to the fact that BOAD, which is a GCF accredited entity, is fully prepared to serve as the executing entity for this proposed programme.

Engagement with national designated authorities, civil society organizations and other relevant stakeholders

39. During project preparation, several project stakeholders were consulted, and the potential impacts of the project were discussed and jointly analysed in order to gather their points of view. Key characteristics of these engagements can be summarized as follows:

- (a) The consultations took place from 2 to 17 May 2018 in the regions of Sikasso, Segou, Koulikoro and Kayes;
- (b) The consulted stakeholders included:
 - (i) Elected representatives in government;
 - (ii) Civil society actors; and
 - (iii) Selected members of the rural communities;
- (c) The methodological approach adopted is the participative approach, which covered meeting information, exchange and discussion around the project;
- (d) Discussions with these stakeholders focused on the projects as much as possible so as to obtain the views of these stakeholders on the project. Key issues discussed included:
 - (i) Possible sites for the implementation of equipment;
 - (ii) Advantages of such a solar equipment project for them;
 - (iii) Special concerns or possible fears for a successful project;
 - (iv) Previous experiences or stories they have heard about this kind of solar electrification projects;
 - (v) Project benefits for village women and youth;
 - (vi) Training needs for equipment maintenance, among others; and
 - (vii) Suggestions and recommendations for the project;
- (e) BOAD also engaged the national regulator for rural electrification (AMADER) and the private operators' organization, the participation of which is crucial for the project's success as they are expected to contribute 20 per cent of the investment cost as parallel financing as a condition for winning the O&M concession for the facility that will be built; and
- (f) The engagement of the GCF NDA of Mali – Agence pour l'Environnement et le Développement Durable (AEDD) – also resulted in the issuing of a letter of no objection to the project.

40. Given the discussion above, the independent TAP has concluded that the country ownership is very strong and as such the project's country ownership has been ranked as "high".

1.6 Efficiency and effectiveness

Scale: High

41. Efficiency and effectiveness of the proposed programme will be measured by how much GHGs are reduced, which itself depends upon how much the intervention is able to release financing for the implementation of the solar-powered mini-grids to deliver power to the end

users, will displace fossil-fuel-based power generation that is prevalent in the absence of this programme for the supply of power to the few rural consumers who had access to power. The economic and the financial soundness of the proposed programme are discussed below.

Cost-effectiveness and efficiency

42. In the baseline, the Government of Mali utilizes public funding to implement rural mini-grids to serve the few consumers who have access in rural areas. This it does through its publicly owned power utility (AEM-Mali) and through grant funded (from bilateral and multilateral partners) mini-grid facilities with private investors selected as concessionaires for the O&M of these facilities. The status quo ante strategy of expanding electricity access in rural Mali cannot be said to be successful as electricity access to the rural population in Mali is still about 1.8 per cent. Also, the few facilities that were built were mostly fossil fuelled (mostly diesel) with high carbon emissions and an ineffective collaborative structure between public and private entities involved in this sector. Therefore, the efficiency and effectiveness measure of this intervention will be measured by a combination of how effective the intervention is to dismantle the inhibitive barriers existing before the intervention and the financial efficiency of the funds mobilized as debt. The financial disbursement for the intervention can be described as following:

- (a) Component 1 (capacity-building) is not revenue-generating in any meaningful sense;
- (b) Component 2 has revenue-generation aspects. Therefore, detailed financial and economic analyses have been conducted; and
- (c) Component 3 has GCF support in the form of a financial guarantee mechanism designed to ensure that productive use of electricity is implemented and therefore the GCF investment is sustainable with the enhancement of the commercial viability of mini-grids. The GCF grants will be augmented by considerable loans provided as co-finance by BOAD.

43. According to the information included in the funding proposal:

- (a) The economic rate of return of the project has been estimated to be about 16.6 per cent;
- (b) The financial viability of the private sector O&M concessionaires that are expected to contribute 20 per cent of the capital exchange of the micro-grid has an internal rate of return of approximately 18 per cent; and
- (c) Sensitivity analyses of these performance metrics were also carried out on: a 10 per cent increase in investment costs; a 10 per cent decrease in benefits; and a one-year delay in project execution. The appraisal shows that the economic rate of return is appreciably high, which confirms the economic viability of the project. However, AER-Mali will have to ensure that deadlines are respected when executing the project and the consulting engineering firm and companies are recruited on time in order to guarantee that the beneficiary populations receive all the benefits thereof.

Application of best practices

44. Equipment and technology will be selected from the best options available on the market. During the appraisal process of the projects to be funded under the programme, BOAD will support the application of the best available technologies by requiring projects to demonstrate that they use solar technologies that are:

- (a) Effective in addressing climate-related goals;
- (b) Market-tested, technically performing and commercially viable; and

- (c) Contribute to market transformation in terms of demonstrating the viability and commercial application of new technologies.

Key efficiency and effectiveness indicators

45. These can be summarized as follows:
- (a) Total project financing = EUR 46.2 million, of which:
- (i) GCF loan = EUR 33.7 million;
 - (ii) GCF grant = EUR 1.9 million; and
 - (iii) BOAD loan = EUR 10.6 million.
46. At the request of GCF, the intervention has been broken into two tranches:
- (a) Tranche 1: total project financing = EUR 34.2 million, of which
- (i) GCF funding = EUR 25.9 Mmillion;
- (b) Expected lifetime (25 years) emissions = 1,027,227 tCO₂eq; and
- (c) Key efficiency and effectiveness metrics:
- (i) Estimated cost per tCO₂eq (for total project financing – tranche 1) = EUR 26/tCO₂eq; and
 - (ii) Estimated GCF cost per tCO₂eq removed for (GCF financing – tranche 1) = EUR 12/tCO₂eq.
47. Based on the discussions presented above (parafraph 17 of this assessment), the independent TAP concluded that the programme will deliver results that will be adjudged to have a “high” efficiency and effectiveness measure.

II. Overall remarks from the independent Technical Advisory Panel

48. This proposed project meets GCF performance metrics at a high level for: impact; paradigm shift; sustainable development potential; need of the recipient; country ownership; and efficiency/effectiveness. As such, the independent TAP recommends the approval of this programme to the Board for funding.

Response of the accredited entity to the Independent Technical Advisory Panel's assessment (FP102)

Proposal name: Mali solar rural electrification project
Accredited entity: Banque Ouest Africaine de Développement (BOAD)

Impact potential
No comment.
Paradigm shift potential
No comment.
Sustainable development potential
No comment.
Needs of the recipient
No comment.
Country ownership
No comment.
Efficiency and effectiveness
No comment.
<i>Overall remarks from the independent Technical Advisory Panel:</i>
No comment.

REPUBLIC OF MALI

One People-One Goal-One Faith

MINISTRY OF ENERGY AND WATER RESOURCES

GENERAL SECRETARIAT

National Agency for Renewable Energy, Mali (AER-MALI)

MALI SOLAR RURAL ELECTRIFICATION PROJECT

GENDER MAINSTREAMING ACTION PLAN

January 2019

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INTRODUCTION

The gender action plan is based on both data collected in the project targeted regions and a literature review on gender and energy sector in Mali. The overall objective is to improve gender mainstreaming in the whole energy project, and to ensure its alignment with the national and international gender frameworks (National gender policy, back donors Gender policies, etc.). The plan also presents a relevant gender context analysis, gender challenges and opportunities identified with the other stakeholder to combat gender inequalities. This gender action plan formulation is also based on two specific levers: (i) the strong favourable political will, and the (ii) gender mainstreaming principles and guidance presented in the main stakeholder's reference documents (BOAD, GCF, State of Mali). Indeed, gender issues are actually priorities and need to be taken into consideration in all areas of socio-economic development including energy sector. The exercise of gender mainstreaming involves ensuring that gender is integrated across all levels of the project cycle by defining of various activities to implement (training, continuous gender-based analysis and assessment, etc.). Gender-related activities and outcomes within the project to meet the challenges constitute a whole and coherent broad. They are grouped together as follows: (i) capacity-building of the various stakeholders, (ii) establishment of a Baseline of gender data, (iii) establishment of a performance monitoring and evaluation system. The purpose of this gender action plan is to operationalize the constraints and opportunities for women and men identified during the gender analysis towards fully integrating them into the project design. The plan includes (i) gender-responsive actions that address and strengthen agency of vulnerable women and men in climate action; (ii) gender performance indicators and sex-disaggregated targets that can be incorporated into a results framework; and (iii) presentation of gender-responsive development impacts. The gender action plan is structured as follows:

- Gender context analysis of the project,
- Objectives and expected impacts of gender actions
- Gender performance indicators;
- Strategies for implementation of the plan
- Logical frame for gender mainstreaming.

I. GENDER CONTEXT ANALYSIS OF THE PROJECT

1. *National Policy et institutions promoting women access to energy*

The country of Mali is an agro-sylvo-pastoral land with about 15.8 million inhabitants (51% are women). 80% of the population lives in rural areas where the national electrification rate in 2016 is 19.39% (39% national, 86% in urban areas). The energy balance in total energy consumption in 2016 established 77% of biomass use mostly by women, the more vulnerable group. So the Government has adopted the National Gender Policy that includes an important component to improve women's access to appropriate technologies that are less costly and less wood energy use. This national policy sets the country's vision and strategic directions in terms of priorities and strategic directions for strengthening gender equality and equity. So, all interventions and back donors at the national level must be aligned in order to optimise the results and changes in this area. This gender policy developed in 2011 defines the national vision of gender promotion as follows: *"A democratic society that guarantees the development of women and men through the full exercise of their fundamental equal rights, active and participatory citizenship and equitable access to resources, with a view to making Mali an emerging*

country strong in its growth and proud of its values of justice, peace, solidarity and social cohesion". Six
 (6) Strategic directions are defined there and they are:

1. Consolidation of democracy and the rule of law in Mali through equal access and full enjoyment of fundamental rights for women and men.
2. Development of human capital (women and men) capable of meeting the challenges of the country's socio-economic development, poverty reduction and integration of Africans and Africa in the global village.
3. Recognition of women's contribution to economic development through their integration into productive channels and equal access to employment opportunities and factors of production.
4. Making use of the potential of women and men in national development through equal participation in the decision-making process.
5. Promotion, awareness-raising and communication for the entrenchment of egalitarian values and behaviours in Malian society while rallying the traditions of equality to the imperatives of modernity and openness to the world.
6. Gender mainstreaming as a guiding principle of good governance in public policies and reforms most conducive to change and in budgets, considering the context of deconcentration and decentralization.

The below analysis indicates that this project is also consistent with this national action plan as well as the Sustainable Development Goals (SDGs). Stimulating women's access to energy effectively would help ensure that the country achieves Sustainable Development Goals - SDGs. Overall, equitable access to energy is a goal targeted by the SDG Goal 7 aims to "ensure access for all to affordable, safe, renewable and modern energy".

In this project, AER has designated a Gender Focal Point, an actual staff of Malian Agency for the Development of Domestic Energy and Rural Electrification (AMADER). She will ensure and follow gender action plan implementation and to ensure that gender dimensions are taken into account during the whole energy project implementation, and to ensure gender-energy aspects integration into the existing monitoring-evaluation mechanism.

2. Women in energy challenges in rural Mali

The country economy is mainly based on primary sector (agriculture, livestock, fishing), which employs about 80% of the active population and mainly women. Poverty is more pronounced for women in rural areas than in urban ones. Studies have sufficiently demonstrated that the day-to-day agenda of rural women is heavy and that prevents them to develop income-generating. Their daily agenda contains activities as varied as they are painful (pounding cereal grains, looking for firewood with rudimentary technologies). Added to this is the family expenses they face, such as the cost of renting the cart to collect wood. It is to these multiple problems of women that the national energy policy and the national gender policy want to tackle, so that equal access to energy is a solution to create opportunities for women's entrepreneurship in rural areas. To date, one of the remarkable responses has been the concept of Units for the Transformation and Marketing of Local Products with use of Energy-Efficient Equipment operating with renewable energy sources. This low access to energy (mainly for women) is a national challenge and particularly for improving the quality of women life (lightening household work and increasing income). Other challenges are the reduction of the rate of

biomass use in the energy balance; the increase of the rural and urban electrification rate; the increase of the quota of renewable energy in the production of electric power; the improvement of women's access to modern energy services and technologies or equipment including professional training for a best use of that equipment; and the systematization of gender mainstreaming in all energy programs and projects.

3. General gender issues analysis in Mali climate context

The climate issues determine the quality of men and women life, and globally the future of the planet. In this case, Mali is currently facing significant environmental problems such as desertification, floods, and rapid deforestation. Women are more victims of the adverse effects of climate change, but on the other, certain women's activities, such as the artisanal transformations of agricultural products, the sale of wood and coal have negative impacts on The environment. So, concrete measures should accompany the ongoing awareness-raising campaigns: development of collection pits for dyeing wastewater, development of soaps workshops, etc. The national priority is to strengthen efforts on the extension of appropriate technologies that are less costly and less energy-consuming for all men and women in rural areas.

But, Mali is one of those countries where the energy paradox is great. There is more sunshine, but the people have less access to energy. Analysis shows that more than 10% of households is run by a woman in rural area. And women have lower access to energy than men. The energy balance reveals that biomass (fuelwood and charcoal) accounts for about 80% of national energy consumption, petroleum products 16%, electricity 3%, and energy Renewable 1%. Wood remains the main source of energy in rural. The strong use of firewood and charcoal as domestic fuel has significant negative impacts, the main ones being deforestation of about 400 000 ha per year, and the emission of greenhouse gases (15.45 MT of carbon dioxide), the dependence on biomass and the increase in the workload of rural women because the collection of firewood is getting farther away from homes. Finally, pollution inside the households has a negative impact on the health. So, the country energy policy has several strategies and actions, supported by multiple actors both state and non-state and back donors. It is expected that this present project will improve the living conditions of households, in particular of women, such as the reduction of domestic corvee related to water and firewood, and cooking, and will allow to more save time to carry out productive activities and income opportunities. This project will also make a decisive contribution to the poverty reduction and sustainable development.

The gender diagnosis has revealed that poverty is a global problem in Mali and more access to energy could be a solution. However, as it has been said, this access is actually more limited for women than men. The diagnosis has also shown that women have more energy access needs because of the type of their activities. The weak level of efficient energy equipment reduces men and women potential and makes women more vulnerable socially and economically. Rural women generally have not enough information on opportunities of energy to modernize their income-generating activities. Concretely, it is necessary to provide solar energy equipment to women such as motor pumps for gardening, dryers for agro sylvo pastoral products transformation, solar fridge, slicer, grinder, etc. and provide more training to them for a best sufficient et relevant use of these equipment.

Considering that mainstreaming gender into energy projects can increase the benefits of electrification for rural communities, the project proposes to contribute in reducing economic and social vulnerability

of rural women through increased access to energy facilities and economic opportunities. As precised in previous session, the interviews of beneficiaries reveal that productive activities requiring modern energy equipment are many and include all economic domain.

Up to now, projects and programs have been developed with AMADER, and other local NGOs in Mali funded by the State or other donors (UN-Women, the World Bank, etc.) for a good recovery of renewable energies. Diagnosis revealed that to solve the specific problems of men and women, and meet the practical and strategic needs of women and youth, solar equipment must be provided. They also need to be trained for a best use equipment or technologies (component 2 and 3 in this project). The experiences generated interesting results against poverty in which women are more victims. The more productive and less arduous to use energy equipment acquisition can significantly improve conditions, especially for women and young people in the implementing localities. Providing local communities with efficient technologies and equipment will enable them to take better advantage of the abundant available natural resources, increase income from fruit trees, better transform, conserve, and sell livestock products. Based on these achievements and on the positive impacts of women's access to energy services, the present plan will allow to expand the intervention area and develop business opportunities for women and young people. The activities of components 2 and 3 of the project will be reinforced by the activities proposed in axes 1, 2 and 3 in this actual gender action plan.

4. Gender challenges analysis in energy sector in implementing communities

The weak economic power of women and weak access to electricity, particularly in rural areas, has important consequences for their access to factors of production, their economic empowerment, etc. Rural women's capacities are severely constrained by low access to energy technologies and services and they use traditional biomass (wood, agricultural residues, animal excrement) for cooking food and small artisanal processing of products. They generally have little information on the opportunities offered by the use of energy sources to modernize their income-generating activities. The reported access to energy difficulties of women are many and include the cost of energy and equipment maintenance, the insufficient coverage of the public energy network in the villages, payment problems. The new energy policy aims to increase production and diversify energy sources by promoting new and renewable energies in order to support economic activity and ensure access to electricity for the greatest number of people. The concrete results are to install solar kits mainly in households, schools and health centres, solar pumps for water pumping, off-grid lighting installations and mini-grid installations.

Rural women and girls are especially affected and the major gender challenges include (i) advocacy for a favourable institutional environment ; (ii) masculinity perspective (Men's awareness and involment for Women's Access to opportunity) ; iii) women's access to project decision-making during implementation ; iv) women's entrepreneurship capacity building (access to renewable energy technologies) ; and v) taking discriminatory measures in favour of women to ensure equity in the equipment endowment

So, to combat detrimental to economic and social emergence of women factors, joint initiatives have to be implemented in the renewable energy sector. It involves the provision to women of various equipment, such as mills, freezers and dryers that operate with solar energy, and mixers and gas lamps. For most of these women, renewable energy is will be the appropriate source of energy and economic

empowerment. After training, they will manage to use this equipment to develop income generated activities. for marketing purposes. Therefore, the project will have to provide for gender equality, bring innovative technologies to women, and strengthen their trade capacity. The project will also provide capacity strengthening of rural women to increase women's income and decision-making autonomy.

5. Project rationale

Climate change has a profound impact on our planet and its inhabitants. New climate records show alarming frequency with multiple climate impacts. For instance, we note the increase in the frequency and intensity of droughts or floods; the rise in extreme temperatures, with the drying up of our water bodies. These changes have major indirect impacts such as decrease in available agricultural or fishery yields, increase in food prices and decrease in food self-sufficiency opportunities leading to food insecurity, conflicts over land tenure rights due to pressure on land, water, wildlife and among others.

However, it should be noted that in this project, access to land is not an important and decisive issue. As a result, the access problems mentioned above represent a general situation and not a specific issue of women's involvement in the project.

Climate change impacts have more effects on the poorest and most rural populations in Mali because of the very low rate of rural electrification (15% in rural villages against 55% in urban centres). 77% (16.5 million inhabitants in 2013) of these rural populations reside in rural areas. 78% of households' energy needs are therefore met by biomass resources (firewood and coal), which, in addition to creating health problems linked to indoor air pollution in rural areas, aggravate deforestation and soil degradation by releasing carbon stored in trees and soils.

It is worth noting that research findings have estimated very significant environmental and social consequences of the energy situation, namely: (i) deforestation of about 400,000 ha/year; (ii) the emission of 15.45 megatons of carbon dioxide, representing 94.33% of Mali's total greenhouse gas (GHG) emissions in carbon equivalent; and (iii) decrease in the purchasing power of the population in the face of rising oil prices. Women and youth are particularly vulnerable and more exposed to the risks of natural disasters with low adaptation capacity to the effects of climate change.

It is against this backdrop that Mali, through the "Green Energy Services for Communities in Rural Mali" Project, is working to "electrify 300 localities in off-grid/mini-grid mode using autonomous renewable energy systems (solar PV, wind)".

This project is in perfect coherence with the country's national orientations on gender promotion and is part of guideline 3 of National Gender Plan "Recognition of women's contribution to economic development through their integration into productive circuits and equal access to employment opportunities and factors of production" and will allow women and men to have equitable access to energy services in rural areas.

6. Relevance of the targeted project implementation localities

As in many parts of the country, the localities chosen for the implementation of the project, women have a great need for energy services than men. For the moment, women's financial resources come mainly from the sale of wood and coal. They have diversified activities that require greater access to

renewable energy in rural areas. The needs are the use of solar equipment, the less use of wood, solar fridges, solar dryers, solar equipment in the kitchen, gardening, culinary training, solar panels. This climate project integrates gender perspectives because it integrates men and women resilience needs, allows women to abandon traditional activities (selling wood and coal) and allows them to improve their economic autonomy.

The project implementation targeted areas have potential for innovative women activities. Both women and men have been involved in diagnostic and they all expressed the problems, challenges, and needs presented in developing profitable economic activities due to lack or insufficiency of modern energy sources and appropriate and efficient processing equipment. They were also involved in the definition of needs and priorities as part of the project. Localities will be easily accessible for interventions. This participatory approach is a factor favorable to their adhesion to the project. Local authorities (mayors, prefects, etc.) have undertaken to provide substantial support to women to facilitate the installation of equipment. The local operators of rural electrification intervening in the localities expressed a real will to collaborate with the local populations for the smooth running of the activities.

II. OBJECTIVES AND EXPECTED OUTCOMES OF GENDER ACTION PLAN

1. Energy needs of project implementation localities

The diagnostic study done by the consultants indicates that the specific energy needs of women are mainly related to the nature of their activities in communities. The energy needs identified for women in localities can be summarized as follows:

- the provision of electricity on a continuous basis and at an affordable price.
- the establishment of modern and efficient infrastructures and equipment for processing and marketing for the valorization of local products and the increase of the income of the populations;
- the gardening equipment mainly using renewable energy for women;
- the technical supervision of beneficiaries in the use and the maintenance of infrastructures and equipment;
- women's access to markets (sales kiosks, drop off points, motorbike, etc. for the transportation of products in fairgrounds etc.); because women are heads of households

For training, special attention must be paid to girls and women to enable them to ensure the maintenance of energy equipment.

2. General objective of gender action plan

Equality between women and men is a condition for the effectiveness of public policies and the sustainable and equitable human development. This gender action plan is designed to contribute to better mainstream gender in the "Green Energy Services for Communities in Rural Mali" project, in particular the "Rural Electrification of localities in off-grid/mini-grid mode from autonomous renewable energy systems.

3. Specific objectives of gender action plan

This project aims to promote the productive use of energy by rural men and women and the reduction of time in domestic chores in order to contribute to the economic empowerment of rural women in a resilient energy system in climate change. The specific objectives of the plan are to:

- Improve women, and youth participation in decision-making and project implementation at all levels;
- Improve women participation of project benefits (access to equipment);
- Enhance collaboration between stakeholders to ensure appropriation and sustainability of the project;
- Improve women and youth as well as others' access to learning and training opportunities offered by the project;
- Promote information and communication in mainstreaming gender throughout the process
- Promote the acquisition of gender knowledge by project team members and other local stakeholders;
- Encouraging the sharing of experiences, capitalisation and dissemination of good practices
- Set up monitoring and evaluation system to follow results and change achievement;
- Determine project indicators to help results and change measuring and follow up.

4. Expected outputs and outcomes

The expected outcome at the end of the project is the increased access of rural men and women to economic opportunities and women's economic empowerment in the country with the development of adaptation and mitigation of climate change effects energy services that preserve the environment and improve the economic situation of women. Specifically, for gender mainstreaming in the project, these are:

- Rural women have greater access to adapted energy technology equipment;
- Participation of women, youth and others in decision-making and implementation within the framework of the project has improved at all levels;
- Participation of women of project benefits (access to equipment);
- Collaboration between stakeholders for the appropriation and sustainability of project activities is encouraged;
- Access of women, young and others to learning and training opportunities offered by the project has improved;
- Information and communication for mainstreaming gender throughout the project have been promoted;

- The acquisition of gender knowledge by project team members and other local stakeholders is enhanced;
- Experience sharing, capitalisation and dissemination of good practices are enhanced;
- Monitoring and evaluation system for achieving the indicators has been set up.

III. ACTIONS FOR A BETTER GENDER MAINSTREAMING IN THE PROJECT

The AER and AMADER provided additional information to the consultants on the gender analysis of the sector; this reassures that the proposed interventions respond to the needs and priorities of men and women. Consultations to date with stakeholders such as the AER, AMADER, local authorities and beneficiaries have revealed the needs and priorities of men and women in the renewable energy sector, and the way in which whose project will respond to it. These elements are taken in consideration in priority as actions and presented in the gender analysis; and it is on this that actions have been proposed jointly with all the actors in view of a greater relevance of the actions proposed in this plan.

The orientation of the project's implementation strategy will be based on the achievements, lessons learned, and gender best practices. The aim of this plan is also to carry out additional concrete actions in terms of optimising gender-responsiveness in the new sustainable adaptation initiatives to climate change, through operational principles, linked to different Elements of governance. Consequently, it is important to note that most actions will not be budgeted in a specific way. It will be for the management team to ensure that planned activities respect the principles of gender mainstreaming as presented here in the gender action plan. Indicators are defined (see the logical framework below) are defined to serve as baseline for monitoring actions to strengthen the gender-taking in the project. The four major strategic pillars and actions for gender mainstreaming in the project implementation are presented in below.

1. Component 1: Capacity building of rural electrification institutions and technical assistance

The output is to Strengthen Rural Energy Regulatory Agencies. In fact, many stakeholders will be involved in this project with different and complementary roles. Mainly, it is AER (the implementing organisation), the Gender Focal Point of AMADER (AMADER is the organisation responsible of rural Electrification), local NGOs and women civil society organizations such as cooperatives, BOAD the financial organisation who will provide funds and technical assistance and control of the implementation, Private sector, the GCF, the Government represented by the ministry and the direction of Energy, and the local administrative board, etc. Furthermore, the national regulator for rural electrification (AMADER) and the Private operators' organization were fully engaged. Capacity building actions will be implement to enable all stakeholders to play their respective roles. The project management team must opt to follow the principles in Gender mainstreaming. Thus, it is important to take gender into account in all actions.

So, the management team including gender staff must have strong gender capacities. These principles must not only apply to management bodies but also to stakeholders for a better follow up. The following activities will be carried out : (i) Capacity building and raising awareness of AMADER, AER, the Rural Electrification Fund, DNE, etc. ; (ii) Train women to make optimal use of solar energy

technologies ; (iii) Strengthen collaboration between actors for ownership and sustainability of equipment ; (iv) Carry out actions (awareness-raising actions) of communications on the project at community level ; (v) Advocate at the institutional level to strengthen gender mainstreaming in the national regulatory framework for solar energy ; (vi) Strengthen the participation of women and youth in decision making in the project at community level.

2. Component 2: Detailed engineering and installation of solar powered minigrids

The project main output is Rural localities electrified with solar PV power plant. To improve the accessibility of women in the project and in line with above gender analysis, the actions should be carried out at two levels and in line with the constraints and needs identified by men and women at the level of the analysis. in the renewable energy sector. It is about the involvement of women and vulnerable groups in the definition of the equipment to be provided. Priority criteria should be based on gender and diversity analysis to ensure that the most marginalized benefits from the actions. these criteria must also be focused on the level of vulnerability, the economic empowerment of women. there is a need to ensure the participation of women, other vulnerable groups such as young people in decision-making and in the implementation of project actions at all levels. In fact, the gender diagnosis showed that women and men have different needs and different levels of access to renewable energies, and also different needs for equipment. in the context of the project, the actions to be carried out must take into account these gender-specific needs, these differences in the level of access of men and women; in order to promote equality and gender equity.

Among the actions to be implemented in this component to support the mainstreaming of gender, (i) Improve the accessibility of rural households to solar energy; (ii) Promote access to information for men and women on the opportunities of the solar project; (iii) Advocate to reduce women's access to solar equipment.

3. Component 3: Support for productive use of electricity

The project main output is the Increased access to clean energy by households and for productive use, in particular, to set up a Financial guarantee mechanism to compensate microfinance institutions. So the gender analysis has shown the relevance of the choice of localities of implementation of the project. The project will aim to fight against poverty, to promote women and men access to renewable energy services. This requires the realization of actions favouring the involvement, the mobilization and the adhesion of the communities, the actions of sensitization, and the concrete actions of creation of incomes as well for the men as for the women.

Among the actions to be implemented in this component, there are (i) Improve access to financing for energy efficiency among small and medium-sized enterprises, including those run by women; (ii) Create and support income generating activities based on micro-credits; (iii) Train women for better use of solar equipment; (iv) Women's capacity building in business management, including the processing and proper maintenance of accounting documents, as well as marketing techniques.

These planned actions will improve the implementation, monitoring of gender mainstreaming in the project. In the context of this project, the participation and involvement of all stakeholders (project staffs, communities and authorities) is necessary. Participation must be mainly at three levels (project staff, implementation, and monitoring and evaluation). At Project implementation level the staff must

be recruited and trained in order to have strong gender operational capacities. At the implementation level, it is a matter of strengthening partnership with other stakeholders which would encourage the mobilization and participation of all to achieving the expected results. Supporting beneficiary communities also means strengthening the level of community participation in the actions and benefits of development in an equitable way for men and women.

4. *Management audit and reporting*

Planned activities her include: i) Operationalize the mechanism for monitoring and evaluating actions and results of gender promotion in the project; ii) Organize balance sheet and results monitoring and evaluation workshops; (see logical framework), etc.

As it has been noted in the component one, at the project implementation level, communication is an important and essential ingredient. Accessibility to information implies the right to seek, obtain and communicate information on social and environment matters. The project must ensure that the stakeholders involved are sufficiently informed about their rights but also about their duties in the project. Access to information implies the right to seek, obtain and communicate information on gender issues. The project must ensure that stakeholders are sufficiently informed of their rights, but also of obligations under the project. Communication aims at to make aware of the impact of the climate change on daily life of rural populations. It is also to make aware of the opportunities for active participation in the development and implementation of appropriate technical and technological innovations to adapt to the adverse effects of climate change.

At local communities - beneficiaries - of the interventions women and men, including those from the most vulnerable groups, are consulted. And their specific needs and priorities for change are taken in consideration. They must participate equitably in the diagnosis, design, implementation, monitoring and sharing of benefits generated through project intervention. At this level, synergies will be developed with local authorities, technical services and other gender sensitive stakeholders to boost the benefits of community interventions. Cooperation with local women's and other groups, such as young people and others, will be strengthened in order to encourage the participation of the whole community in the promotion, planning and implementation of the project. This level of involvement is a determining factor in the level of ownership and sustainability of the project's actions and results by beneficiaries.

IV. GENDER MAINSTREAMING STRATEGY

The gender mainstreaming strategy is based on, (i) the willingness and adherence of project stakeholders to the principles of gender equality and equity in development actions, institutional and operational capacity building, Development of tools for monitoring the gender and capitalization actions. The institutional framework of the strategy comprises three levels, namely the coordination of the Strategy, the management of implementation and the evaluation and capitalization of actions and good practices in gender.

1. Coordination of the strategy

The project team will assist in planning the operationalization of the gender-based strategy at all levels. It will also ensure that the results are achieved by the members with the support of the various national partners (making available to human and financial resources), and their capitalization in the processes.

2. The management of the implementation

In the operational phase, the piloting of the strategy will be carried out by a staff which will take charge of the orientation, facilitation of capacity-building sessions, periodic follow-up and the Monitoring mechanism assessment and implementation of the strategy. The implementation of project gender actions will be carried out by the project team, which shall benefit from stakeholders support to mainstream gender dimensions in line with the donor's gender strategies.

3. Information and actors' capacity building

This involves providing specific access to informing the actors involved in the project implementation and strengthening their capacities in the domain of gender at national and local level. Information and training on the themes of the project is an essential prerequisite for the success of implementation of this gender action plan. The project will use technical capacity building in advocacy and lobbying for gender equality as a gender mainstreaming strategy. Stakeholders that will have specific roles in the project are: AER, the Gender Focal Point of AMADER, local NGOs and women civil society organizations, BOAD, GCF, Private, Government, etc.

4. Assessment and capitalization of actions and good practices in gender

A monitoring and evaluation mechanism will be put in place and be active throughout the implementation process of the strategy in order to assess and measure progress made in terms of gender mainstreaming. A participatory approach will be adopted to assess the progress of activities and the level of achievement of outcome. The monitoring and evaluation process will therefore involve stakeholders directly concerned by gender mainstreaming at the project level. At the end of the intervention, a study of the capitalization of the actions and good practices in gender will make it possible to highlight the taking into account of the gender.

5. Gender mainstreaming M&E

The project staff is responsible for measuring the outcomes and impacts of project activities on women and men's resilience to climate change through gender responsive M&E. The objective is to assess the outcomes and impacts of interventions on women and men, support the equal participation of women and men in M&E activities and decision-making processes, and collect sex-disaggregated data to track gender impacts and benefits for women and men. To conduct gender-sensitive M&E, the foundation is a results framework (such as in the bellow table) with gender-responsive indicators and targets (sex-disaggregated). The collection of sex-disaggregated data is essential for M&E and for reporting on gender outcomes so that lessons learned and best practices are captured and disseminated, and fed back into project design.

Tools are proposed to measure how gender is mainstreamed during and after the project implementation. So the project staff will use these tools for that purpose.

6. *Reporting on Gender Outcomes*

Based on gender-responsive approaches and gender-sensitive data collection that is built into project design, implementation and M&E, reporting on gender outcomes is a component of all project reporting processes:

- Include mechanisms to monitor and report on gender impacts in project design;
- Document and disseminate gender-relevant best practices and lessons learned throughout the project cycle; and
- Ensure that progress reports, implementation status reports, aide memoires, management information systems and completion reports present gender equality results and lessons learned.

Reporting on gender outcomes should be robust enough to inform future project identification and development as a means of deepening and increasing development co-benefits between gender equality and climate action. An optimal result would be the growing capacity to influence national climate change plans and strategies so that they become more gender responsive.

7. *The sustainability of the project*

The sustainability of the project will be ensured through the technical capacities building for women and youth to take over the repairs of obtained equipment. These skills in maintenance and repair of energy equipment will help reduce the rate of failure in terms of equipment installed. The commitment of the authorities and the involvement of community leaders will help to ensure the continuation of the activities.

V. GENDER MAINSTREAMING ACTION PLAN FRAMEWORK

COMPONENTS	PROJECT OUTPUTS	ACTIVITIES	INDICATORS	BASELINE 2019	TARGET IN 2023	TIMELINE	RESPONSIBLE	BUDGET USD
The implementation of the "Environmental and Social Management Framework (ESMF)" will contribute in the long term to the off-grid/mini grid electrification of 70 villages, nearly 50% of which are women and men, using renewable energy systems (solar photovoltaic and wind) with the aim of moving towards low carbon solutions in the energy sector and reducing the sector's vulnerability to climate change, linked to fossil fuels. The intermediate and immediate results of the project will generate multiplier effects including: the improvement of in health of more than 50% of women, men, young people and children by reducing greenhouse gases, the economic empowerment of nearly 50% women explained by the improvement of their purchasing power, access to new economic and environmental opportunities, the reduction of deforestation by about 400 000 ha/year ; the reduction of emission carbon dioxide, representing 94.33% of Mali's total greenhouse gas (GHG) emissions in carbon equivalent.								
GENDER OUT PUT: • The participation of women, youth and others in decision-making within the project at all levels is effective; • The accessibility of nearly 50% women, young people and others to the learning and training opportunities offered by the project is improved; • Information and communication to mainstream gender throughout the process is real • The acquisition of gender knowledge by project team members and other local actors is effective • Sharing of experiences, capitalisation and dissemination of good practices are encouraged; • The monitoring and evaluation system for achieving the indicators has been set up.								
▪ 1 : Capacity building of rural electrification institutions and technical assistance	▪ Strengthened Rural Energy Regulatory Agency	▪ Capacity building and raising awareness of AMADER, AER, the Rural Electrification Fund, DNE, etc.	▪ Number of organized sessions of Capacity Building ▪ Number of structures formed	▪ TBD ▪ TBD	▪ 5 ▪ 5	▪ During the implementation of the project	▪ Private sector operators ▪ Project staff ▪ Private providers	100 000
		▪ Train women to make optimal use of solar energy technologies	▪ Number of people trained and % of women ▪ Number of new income-generating activities initiated by women	▪ TBD ▪ TBD	▪ 310 698 and 50% of women ▪ 155 349	▪ During the implementation of the project	▪ Private sector operators ▪ Project staff	
		▪ Strengthen collaboration between actors for ownership and sustainability of equipment	▪ Number of actors involved ▪ Number of women and men involved	▪ TBD ▪ TBD	▪ 05 ▪ 15	▪ During the implementation of the project	▪ Project staff	
		▪ Carry out actions (awareness-raising actions) of communications on the project at community level	▪ Number of people affected by the sensitization ▪ Women's rate	▪ TBD ▪ TBD	▪ 310 698 and 50% of women ▪ 155 349	▪ During the implementation of the project	▪ Media ▪ Project staff	
		▪ Advocate at the institutional level to strengthen gender mainstreaming in the national regulatory framework for solar energy	▪ Gender mainstreaming in policies ▪ Measures taken in favor of women in the solar field	▪ 02 ▪ TBD	▪ All public policies ▪ At least 50%	▪ During the implementation of the project	▪ Private sector operators ▪ Ministry of Energy	

		Strengthen the participation of women and youth in decision making in the project at community level	- Number of women members of decision-making bodies - Number of young members of decision-making bodies	- TBD - TBD	- At least 2 in each decision board - At least 1 in each decision board	During the implementation of the project	- Project staff - Community leaders	
2 : Detailed engineering and installation of solar powered minigrids	Rural localities electrified with solar PV power plant : Acquisition, installation and commissioning of solar powered minigrids equipment ; Implementation of Environment & Social measures	Improve the accessibility of rural households to solar energy	- Number of Households with improved access to low-emission energy sources - Number of female lead Households with improved access to low-emission energy sources	- TBD - TBD	44 385 50% (22 193)	During the implementation of the project	Private sector operators	46 000
		Promote access to information for men and women on the opportunities of the solar project	- Number of people affected by information actions - Women's rate	- TBD - TBD	310 698 50% (55 349)	During the implementation of the project	- Private sector operators - Media	
		Advocate to reduce women's access to solar equipment	- Number of beneficiaries with access to solar equipment - Women's rate	- TBD - TBD	310 698 50% (55 349)	During the implementation of the project	- Private sector operators - Project staff	
3 : Support for productive use of electricity	Increased access to clean energy by households and for productive use : Setting up a Financial guarantee mechanism to compensate microfinance institutions	Improve access to financing for energy efficiency among small and medium-sized enterprises, including those run by women	Number of small and medium-sized enterprises, including women-led enterprises, receiving energy efficiency funding provided through the project	TBD	50 000	During the implementation of the project	- Project staff - Micro-credit institutions	150 000
		Create and support income generating activities based on micro-credits	- Part of households benefiting from financial mechanism (guarantee scheme) to access credit for productive use of clean energy - Additional jobs created for men and women in the solar field - Number of beneficiaries with access to credit - Women's rate	- TBD - TBD - TBD - TBD	44 385 50 000 310 698 50%	During the implementation of the project	- Project staff - Micro-credit institutions	
		Train women for better use of solar equipment	Number of women trained	TBD	155 349	During the implementation of the project	- Project staff - Private sector operators	
		Women's capacity building in business management, including the processing and proper maintenance	Number of women trained	TBD	155 349	During the implementation of the project	- Project staff - Private providers	

Comments of the above table

It is important to specify that the target persons or group in charge of actions defined in this table are link to the stakeholders in the implementation of the mentioned activities. The project team is responsible to follow up and monitor the performance indicators progress during the project implementation phase. The project promoter should ensure that the first quarter of the year has been used by the project team to collect and complete the analysis of gender information and data on the project baseline. This collection of information will enable the updating of gender analysis during the implementation of the project. The project team will benefit from the support of other stakeholders involved to regularly inform SMART indicators formulated as part of the monitoring of gender equality and equity in the project. It's also important to have upon this framework a M&E system that will show exactly periods and frequency of data collection and gender analysis with the purpose to ensure a real gender consideration in the project.

CONCLUSION

From all the above, it is undeniably established that taking gender issues into account in any development process constitutes Mali's response both to public policies and international commitments in a context of globalization, and even as a democratic requirement. Finally, it is still useful to remind that this gender action plan is an integrated part of the project's operational framework and that the actions identified are aimed at improving gender mainstreaming in the project. It is therefore appropriate for actors to be involved at all levels to ensure and ensure that the principles of equality and gender equity are respected.

Annex : 3 R Gender analysis sheet : This below matrix of analysis is relatively simple and allows to ask some essential questions about an intervention (before or during its realization) and to study an intervention according to gender relations. The first two lines correspond mainly to a collection of information specified by gender: the involvement of women and men in the intervention and the distribution of resources related to the intervention. The analysis begins by questioning both the causes and consequences of the representation of women and men and the distribution of resources between these two groups. The study of the causes and consequences encourages the designers of an intervention to measure objectively the taking into account of gender relations. If the consequences for the intervention are likely to have negative effects, it may be necessary to rethink the design of the intervention and to carry out this analysis again.

Representation	• How many women and how many men benefit from the intervention? • How many women and how many men participate in the intervention?
Resources	• How are the resources used and/or made available through intervention between men and women (money, time, etc.)? • Analysis of cash flows by gender • What are the direct and indirect gains of men and women in the intervention? How do women and men use time?
Achievements	• What is the distribution of benefits between men and women? • Are there inequalities in the distribution? • What are the reasons of this distribution? • What are the actions to be taken to restore equality?

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